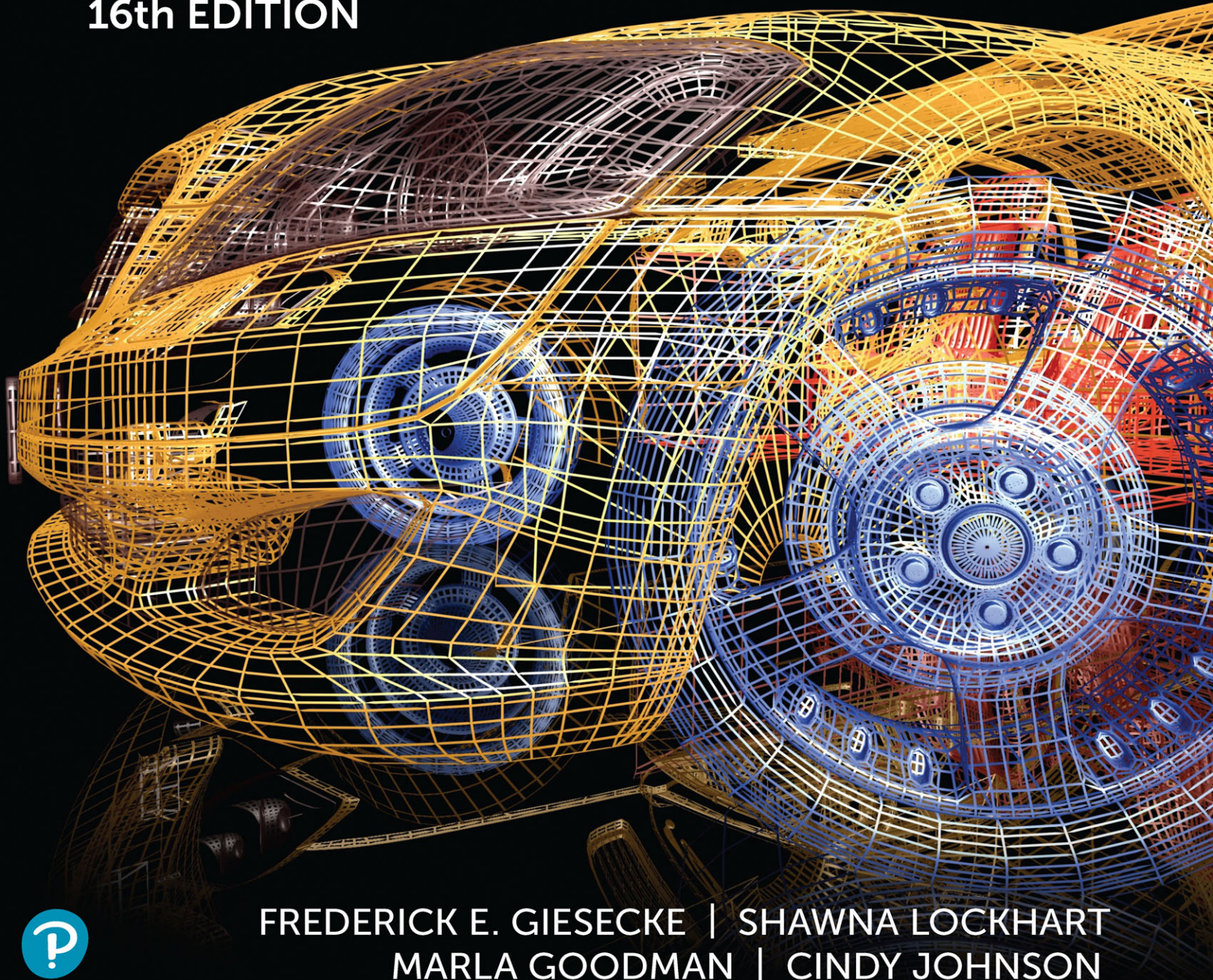


TECHNICAL DRAWING WITH ENGINEERING GRAPHICS

16th EDITION



FREDERICK E. GIESECKE | SHAWNA LOCKHART
MARLA GOODMAN | CINDY JOHNSON

SIXTEENTH EDITION

TECHNICAL DRAWING WITH ENGINEERING GRAPHICS

FREDERICK E. GIESECKE

*Late Professor Emeritus of Drawing
Texas A&M University*

SHAWNA LOCKHART

*Formerly Adjunct Professor, Engineering Graphics
Department of Industrial and Mechanical Engineering
Montana State University*

MARLA GOODMAN

CINDY M. JOHNSON



Pearson

Editor in Chief: Mark Taub
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SIXTEENTH EDITION

TECHNICAL DRAWING WITH ENGINEERING GRAPHICS

ABOUT THIS BOOK

The sixteenth edition of Giesecke's *Technical Drawing with Engineering Graphics* is a comprehensive introduction and detailed reference for creating 3D models and 2D documentation drawings.

Continuing its reputation as a trusted reference, this edition is updated to convey recent standards for documenting 2D drawings and 3D CAD models. It provides excellent integration of its hallmark illustrations with text and contemporary examples, and consistent

navigational features make it easy to find important information.

This edition illustrates the application of both 3D and 2D modeling and technical drawing skills to real-world work practice and integrates drawing and CAD skills in a variety of disciplines. Reviewers advised us on how to make *Technical Drawing with Engineering Graphics* a superb guide and resource for today's students.

Updated Content

- Coverage of 3D design and modeling techniques
- Updated for current ASME standards, particularly for GD&T and surface finish symbology
- Updated examples of rapid prototyping and direct printing
- Updated software examples
- Thoroughly checked for accuracy
- Web chapters available for axonometric projection and perspective drawing

Teaching/Learning Features

Visually oriented students and busy professionals will quickly locate content by navigating these consistent chapter features.

- **Splash Spread** An attention-getting chapter opener interests readers and provides context for chapter content.
- **References and Web Links** Applicable references to standards and links to handy websites are at the beginning of each chapter.
- **Foundations Section** An introductory section, set off by a topic heading tab at the top of the page for easy navigation, covers the topic's usage and importance, visualization tips, and theory related to the drawing techniques.

- **Detail Section** This is the “brass tacks” part of the book, where detailed explanations of drawing and modeling techniques, variations, and examples are organized into quick-read sections, each numbered for quick reference in the detailed table of contents.
- **CAD at Work** This breakout page includes tips related to using the 2D or 3D CAD model to generate drawings.
- **Industry Case** 3D modeling practitioners share their best practices for modeling and documenting design.
- **Portfolio** Examples of finished drawings wrap up the chapter by showing real-world application of topics presented.
- **Key Words** Set in bold italics on first reference, key words are summarized at the end of the chapter.
- **Chapter Summary**
- **Review Questions**
- **Chapter Exercises** The excellent Giesecke problem sets feature updated exercises, including plastic and sheet metal parts, modeling exercises, assembly drawings from CAD models, and sketching problems.

The following features were designed to provide easy navigation and quick reference for students and professionals who look to Giesecke both as a helpfully-organized teaching text and a lasting reference.

CHAPTER OPENER

THE WORLDWIDE GRAPHIC LANGUAGE FOR DESIGN

1

CHAPTER ONE

THE WORLDWIDE GRAPHIC LANGUAGE FOR DESIGN

OBJECTIVES

After studying the material in this chapter, you should be able to:

1. Describe the role of graphics in the design process.

2. Identify stages in the design process.

3. Contrast concurrent versus traditional design processes.

4. List five professions that use technical drawings.

5. Describe four creativity techniques.

6. Explain why standards are important.

7. Identify three purposes for technical drawings.

Refer to the following standard:

• ANSI/ASME Y14.100 Engineering Drawing Practices

THE WORLDWIDE GRAPHIC LANGUAGE FOR DESIGN

3



Conceptual Sketches. Exploring many design options through quick sketches is one method that Lunar, recently named one of the top 10 award-winning American product design firms by BusinessWeek magazine, uses to create beautiful products and successful brands. (Images courtesy of LUNAR.)

OVERVIEW

Regardless of the language they speak, people all over the world use technical drawings to communicate their ideas. Graphic representation is a basic, natural form of communication that isn't tied to a particular time or place. It is, in a sense, a universal language.

Accomplishing ideas, from the simplest to the most elaborate, requires teamwork. A new product, machine, structure, or system may exist in the mind of the engineer or designer, but before it can become a reality, the idea must be communicated to many different people. The ability to communicate design concepts quickly and accurately through technical drawings is key to meeting project budgets and time constraints. Effective graphic communication is also an advantage in the global marketplace, where team members may not always share a common language.

Like carpenters who learn to use the tools of their trade, engineers, architects, drafters, designers, manufacturers, and technicians learn the tools of technical drawing. They learn specific methods to represent ideas, designs, and specifications in a consistent way that others can understand. Being an effective graphic communicator ensures that the product, system, or structure that you envision is produced as you specified.

Topics that you can expect to learn about in this chapter are listed here.

A large illustration and an interesting overview give you a real-world context for what this chapter is about.

Drawing standards that apply to this chapter are shown here.

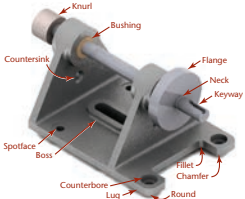
"SPOTLIGHT" SECTIONS

These sections add background information for key topics.

214 CHAPTER 5 MODELING AND DESIGN

Typical Features

Many CAD systems have aids to creating features that are a part of many engineering designs (see Figure 5.63) that can make creating your model even easier.



5.63 Commonly Manufactured Features

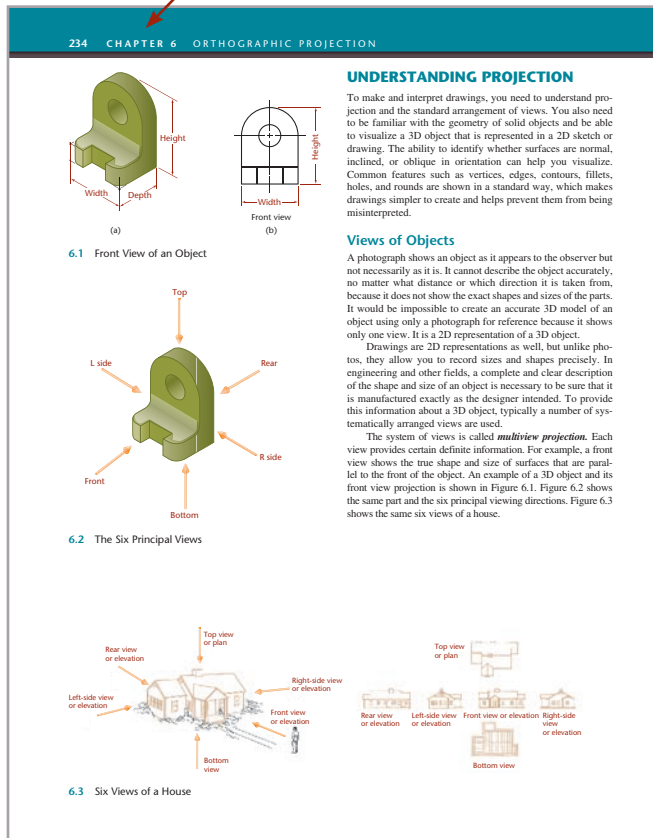
SPOTLIGHT

Feature	Example
Fillet: A rounded interior blend between surfaces; used, for example, to strengthen adjoining surfaces or to allow a part to be removed from a mold	
Round: A rounded exterior blend between surfaces; used to make edges and corners easier to handle, improve strength of castings, and allow for removal from a mold	
Counterbore: A cylindrical recess around a hole, usually to receive a bolt head or nut	
Countersink: A cone-shaped recess around a hole, often used to receive a tapered screw head	
Spotface: A shallow recess like a counterbore, used to provide a good bearing surface for a fastener	
Boss: A short raised protrusion above the surface of a part, often used to provide a strong flat bearing surface	
Lug: A flat or rounded tab protruding from a surface, usually to provide a method for attachment	
Flange: A flattened collar or rim around a cylindrical part to allow for attachment	
Chamfer: An angled surface, used on a cylinder to make it easier to start into a hole, or a plate to make it easier to handle	

“FOUNDATIONS” SECTION

This introductory section covers the chapter topic's usage and importance, visualization tips, and theory related to the drawing and modeling techniques.

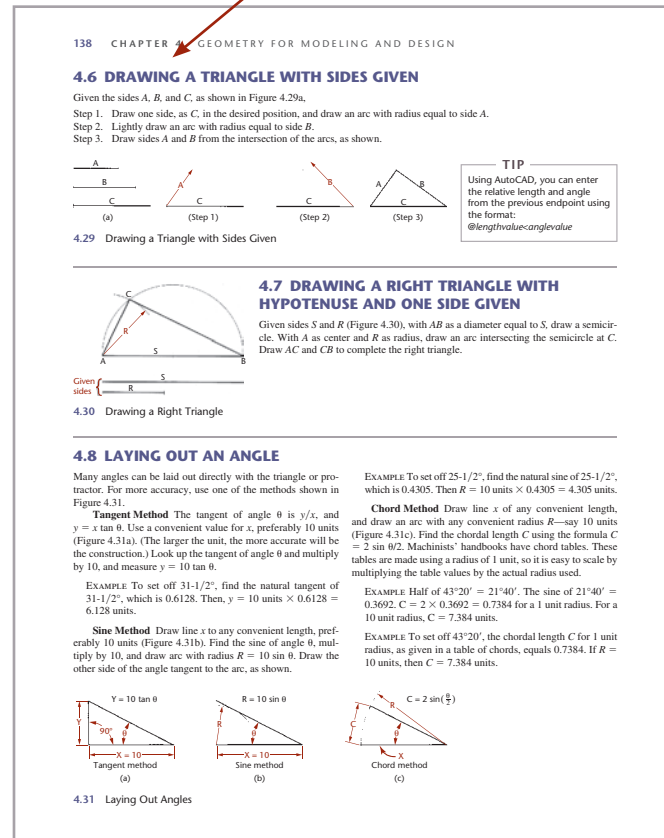
Color at the top of the page makes it easy to flip to the “Foundations” section.



“DETAIL” SECTION

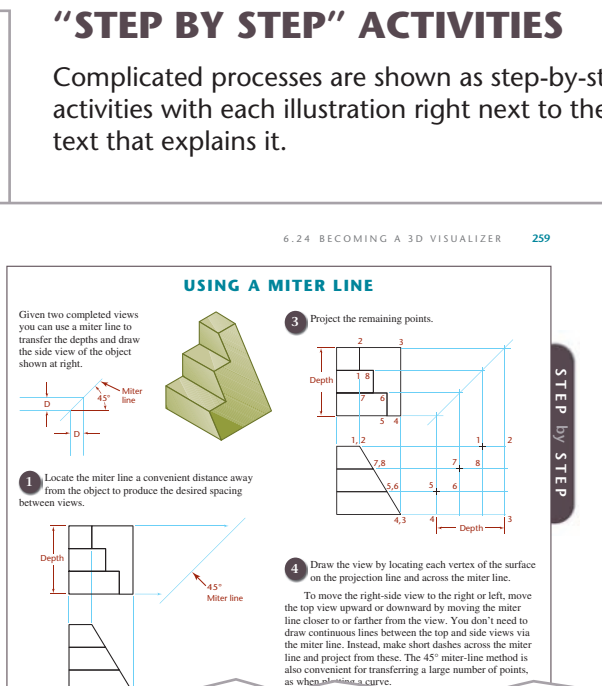
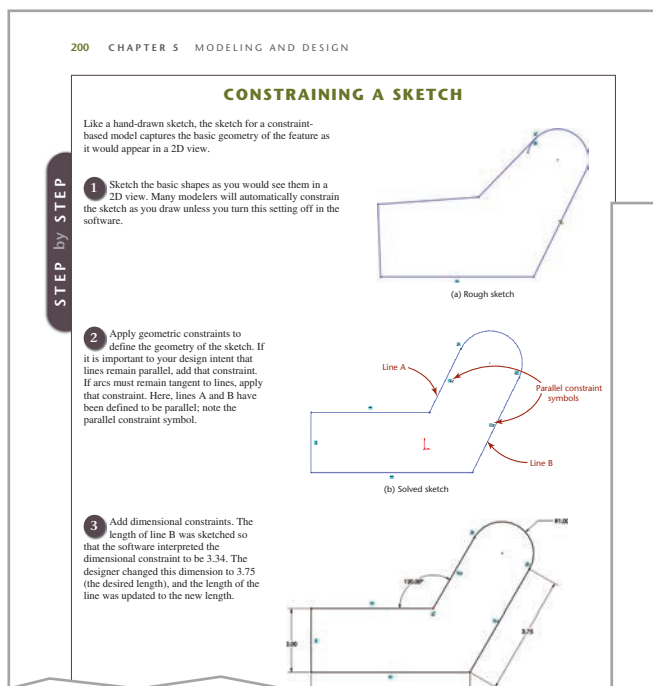
This is the “brass tacks” of the book, where detailed techniques, variations, and examples are organized into quick-read sections, numbered for easy reference.

Content is broken into individual, numbered sections.



“STEP BY STEP” ACTIVITIES

Complicated processes are shown as step-by-step activities with each illustration right next to the text that explains it.



"CAD AT WORK"

CAD at Work sections break out examples related to using the 2D or 3D CAD model to generate drawings.

"INDUSTRY CASE"

Several industry practitioners share their approaches to modeling and documenting design.

INDUSTRY CASE

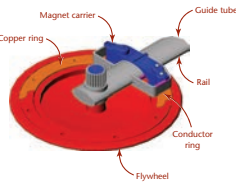
THE GEOMETRY OF 3D MODELING: USE THE SYMMETRY

Stratix ID used magnets to create a clean, quiet, zero maintenance brake for the exercise bike it designed for Park City Entertainment. When copper rings on the bike's iron flywheel spin past four rare-earth magnets, they create current in circular flow (an eddy current) that sets up a magnetic field.

This opposing magnetic field dissipates power and slows the wheel. Moving the magnets onto and off the copper rings varies the amount of resistance delivered. When Marty Albini, Senior Mechanical Engineer, modeled the plastic magnet carrier for the brake, he started with the magnets and their behavior as the carrier moved them onto and off the copper rings (see Figure 4.84). "There is no one way to think about modeling a part," Albini said. "The key is to design for the use of the part and the process that will be used to manufacture it." To make the magnet carrier symmetrical, Albini started by modeling half of it.

The magnet carrier was designed as a part in the larger flywheel assembly, parts of which were already completed.

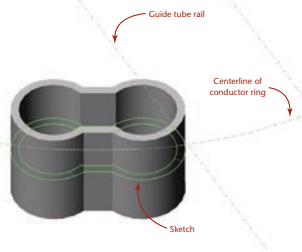
Each pair of magnets was attached to a backing bar that kept them a fixed distance apart. To begin, Albini started with the geometry he was sure of: the diameter of the magnets, the space between them, and the geometry of the conductor ring. He sketched an arc sized to form a pocket around one of the magnets so that its center point would be located on the centerline of the conductor ring (see Figure 4.85). He then sketched another similar arc but with its center point positioned to match the distance between the centers of the two magnets. He connected the two arcs with parallel lines to complete the sketch of the inside of the carrier. This outline was offset to the outside by the thickness of the wall of the holder. (Because this is an injection-molded plastic part, a uniform wall thickness was used throughout.) One final constraint was added to position



4.84 Flywheel Assembly. The magnet carrier for the brake was designed to move onto and off the conductor ring by sliding along an elliptical guide tube, pulled by a cable attached to the small tab in the middle of the carrier.

the carrier against the rail on the elliptical tube along which it would slide; the outside of the inner arc is tangent to this rail. With the sketch geometry fully defined, Albini extruded the sketch up to the top of the guide tube and down to the running clearance from the copper ring.

To add a lid to the holder, Albini used the SolidWorks **Offset** command to trace the outline of the holder. First, he clicked on the top of the holder to make its surface the active sketch plane. This is equivalent to changing the user coordinate system in other packages: it signals to SolidWorks that points picked from the screen lie on this plane. He then selected the



4.85 Extruding the Carrier. The magnet carrier was extruded up and down from the sketch, shown here as an outline in the middle of the extruded part. Notice that the sketch is tangent to the guide tube rail, and the centers of the arcs in the sketch are located on the centerline of the conductor ring.

"PORTFOLIO"

These pages offer examples of finished drawings showing real-world application of topics presented.

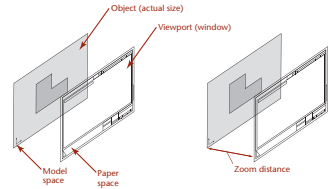
CAD at WORK

MODEL SPACE AND PAPER SPACE IN AUTOCAD

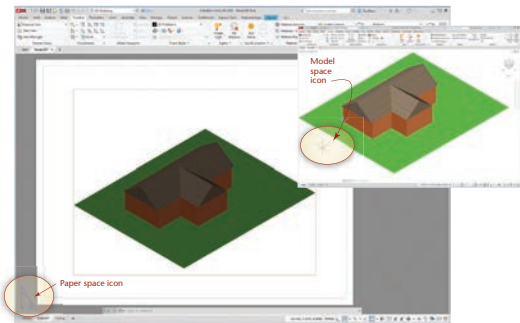
Using CAD, you can make an accurate model of the device or structure. To do this, you create the object at the actual size that it exists in the real world, using whatever system of measurement that you would use when constructing it.

On paper it is a different matter. You would have to have some really large sheets to print your building full size. AutoCAD software uses the concept of two "spaces," model space and paper space, to describe how to transform the full-size CAD model to proportionate views that fit your sheet of paper.

Understanding scale as it relates to paper drawings or as it relates to creating layouts from a CAD drawing is an important concept for technical drawing because the ultimate goal is for drawings to be interpreted and used in the real world. Therefore, they must be easy to print and read.

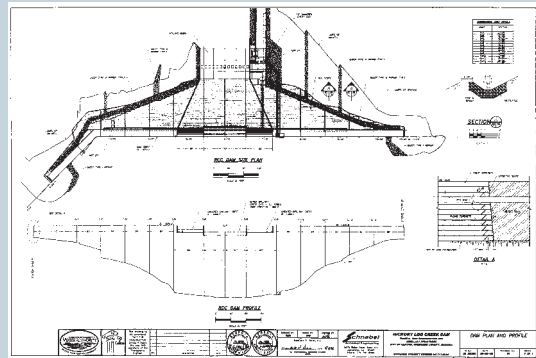


(A) In AutoCAD, paper space allows you to see how various views of the full-size model can be shown on a sheet of paper.

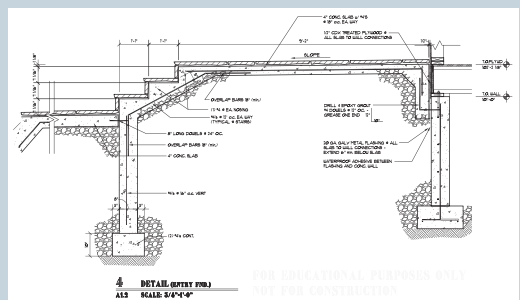


(B) The window at left shows a paper space representation of the full-size CAD model in the smaller window at right. Note that AutoCAD uses icons to help users differentiate the two "spaces." (Autodesk screen shots reprinted courtesy of Autodesk, Inc.)

542 CHAPTER 11 DIMENSIONING



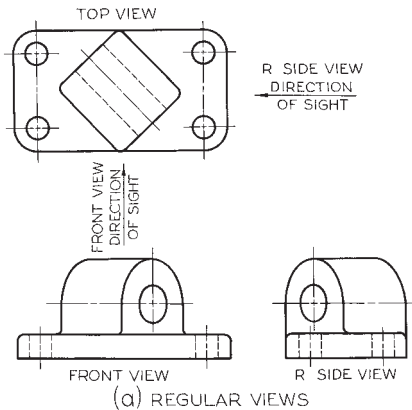
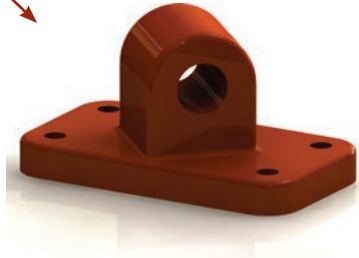
Plan and Profile for Dam Site (Courtesy of Schnabel Engineering.)



Portion of a Drawing Showing Dimensioned Architectural Details (Courtesy of Locati Architects.)

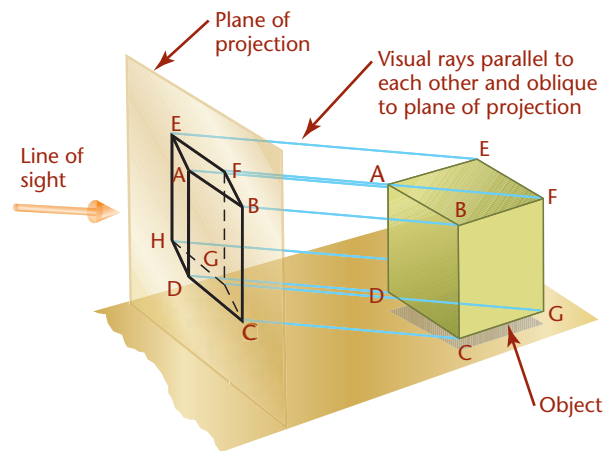
SOLID MODEL VISUALIZATION ART

Solid models bring views to life on the page to help you visualize the drawing.



ILLUSTRATIONS

Colored callouts differentiate explanatory text from annotations in technical drawings. Consistent use of color helps differentiate the meaning of projection lines, fold lines, and other drawing elements. A color key is provided for easy reference.



Color Key for Instructional Art

Item	In instructional art	In a technical drawing
Callout arrow		*
Dimension line		a thin (0.3mm) black line
Projection line		a lightly sketched line
Folding line		used in descriptive geometry
Picture plane on edge		*
Plane of projection		*
Cutting plane on edge		(see Chapter 6)
Cutting plane		*
Reference plane on edge		used in descriptive geometry
Reference plane		*
Viewing direction arrow		
Horizon + ground line		
Rotation arrow		30°

* Not a typical feature of technical drawings. (Shown in this book for instructional purposes.)

CHAPTER REVIEW

Each chapter ends with Key Words, a Chapter Summary, and Review Questions.

Review and exercises are tabbed to make them easy to find. The color stripe corresponds to the alternating chapter color.

CHAPTER EXERCISES

The Giesecke problem sets feature updated exercises including plastic and sheet metal parts, constraint-based modeling, sketching problems, and reverse engineering projects.

KEY WORDS

Depth
Edge
First-Angle Projection
Folding Lines
Frontal Plane
Glass Box
Height
Horizontal Plane
Inclined Edge
Inclined Surface
Multiview Projection
Necessary Views
Normal Edge
Normal Surface
Oblique Edge
Oblique Surface
Orthographic
Plane
Plane of Projection
Point
Principal Views
Profile Plane
Projection Symbols
Surfaces
Third-Angle Projection
Three Regular Views
Width

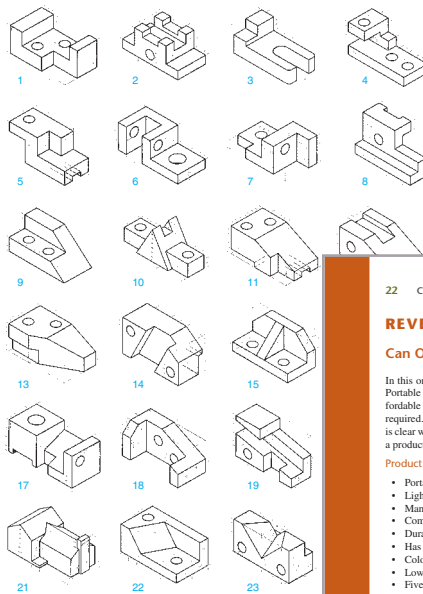
- Choice of scale is important for representing objects clearly on the drawing sheet.
- Hidden lines are used to show the intersections of surfaces, surfaces that appear on edge, and the limits of curved surfaces that are hidden from the viewing direction.
- Centerlines are used to show the axis of symmetry for features and paths of motion, and to indicate the arrangement for circular patterns.
- Creating CAD drawings involves applying the same concepts as in paper drawing. The main difference is that drawing geometry is stored more accurately using a computer than in any hand drawing. CAD drawing geometry can be reused in many ways and plotted to any scale as necessary.

REVIEW QUESTIONS

1. Sketch the symbol for third-angle projection.
2. List the six principal views of projection.
3. Sketch the top, front, and right-side views of an object of your design having normal, inclined, and oblique surfaces.
4. In a drawing that shows the top, front, and right-side view, which two views show depth? Which view shows depth vertically on the sheet? Which view shows depth horizontally on the drawing sheet?
5. What is the definition of a normal surface? An inclined surface? An oblique surface?
6. What are three similarities between using a CAD program to create 2D drawing geometry and sketching on a sheet of paper? What are three differences?
7. What dimensions are the same between the top and front view: width, height, or depth? Between the front and right-side view? Between the top and right-side view?
8. List two ways of transferring depth between the top and right-side views.
9. If surface A contains corners 1, 2, 3, 4, and surface B contains 3, 4, 5, 6, what is the name of the line where surfaces A and B intersect?

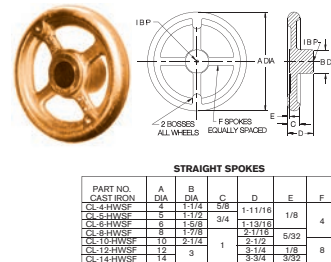
CHAPTER SUMMARY

- Orthographic drawings are the result of projecting the image of a 3D object onto one of six standard planes of projection. The six standard views are often thought of as an unfolded glass box. The arrangement of the views in relation to one another is important. Views must line up with adjacent views, so that any point in one view projects to line up with that same point in the adjacent view. The standard arrangement of views shows the top, front, and right side of the object.
- Visualization is an important skill. You can build your visual abilities through practice and through understanding terms describing objects. For example, surfaces can be normal, inclined, or oblique. Normal surfaces appear true size in one principal view and as an edge in the other two.



Exercise 6.5 Multiview Sketching Problems. Sketch necessary orthographic view either one or two problems per sheet as assigned by your instructor. These exercises are on metric A4 paper. The units shown may be either .500" and .250" or 10 mm and 5 mm.

Exercise 6.19 Create a constraint-based model of the four-spoke hand wheel shown such that it can be resized to match the dimensions in the table.



REVERSE ENGINEERING PROJECTS

Can Opener Project

In this ongoing project, you will reverse engineer an Amco Swing-A-Way 407WH Portable Can Opener. It is recommended you purchase a readily available and affordable product similar to this one, so you can make measurements directly when required. This effective and low-cost can opener seems simple in its familiarity, but it is clear when you begin to take one apart that considerable effort went into designing a product that is inexpensive, reliable, and easy to operate for most people.

Product Features

- Portable
- Lightweight
- Manually operated
- Comfortable to hold
- Durable construction
- Has a bottle opener
- Colored handles available
- Low maintenance
- Five-year warranty



SHOULDER
1/2" (12.7mm)
1/4" (6.35mm)
1/8" (3.18mm)
1/16" (1.59mm)

CLM-1000-SS
CLM-1000-SS
CLM-1000-SS
CLM-1000-SS

Exercises for Chapter 1

RE 1.1 This is far from the only can opener on the market. Use the Web to research manual can opener designs. Find at least three can opener models that are different from the Amco Swing-A-Way. Make a list of the features of each of the three.

RE 1.2 Create a diagram for the can opener. How many distinct parts are used in its manufacture? Which parts can be grouped together and preassembled as a unit?

Exercises for Chapter 2

RE 2.1 Make a table listing the dimensions of the can opener parts. Do not worry about measurements for now. Give names to the dimensions, such as lower handle length, lower handle height, and hole diameter.

RE 2.2 Which dimensions in the list you created are critical to the function of the can opener? Identify in your list the dimensions that must match dimensions on other parts for the can opener to function. Which dimensions will not be very important to the can opener's function?

RE 2.3 To accurately reverse engineer the can opener, you will need to make measurements for the part features. Metrology is the science of making measurements. The digital caliper is one commonly used measurement tool. The accuracy of a measurement is dependent on several factors, including the following:

- the skill of the operator
- the temperature at which the measurements are taken

Exercises for two reverse engineering projects are keyed to the chapter they best accompany.

PREFACE

For many decades, *Technical Drawing with Engineering Graphics* has been recognized as an authority on the theories and techniques of graphics communication. Generations of instructors and students have used and retained this book as a professional reference. The long-standing success of *Technical Drawing with Engineering Graphics* can be attributed to its clear and engaging explanation of principles, and to its drawings, which are unsurpassed in detail and accuracy.

Although not a departure from its original authoritative nature and hallmark features, the book is thoroughly revised and updated to the latest technologies and practices in the field. With the addition of topics related to the role of the 3D CAD database in design and documentation, this sixteenth edition of *Technical Drawing with Engineering Graphics* will prepare students to enter the marketplace of the twenty-first century and continue to serve as a lasting reference.

Shawna Lockhart, contributing author since the ninth edition, first used Giesecke's *Technical Drawing* when teaching engineering graphics at Montana State University. Throughout her 15 years as an award-winning professor, she selected this text because, in her words, "It was the most thorough and well-presented text with the best graphic references and exercises on the market."

The quality of the illustrations and drawing examples was established by the original author, Frederick E. Giesecke, who created the majority of the illustrations in the first edition of *Technical Drawing*, published in 1933.

Giesecke, founder of the first formal architectural education program in Texas at what is today Texas A&M University, has been described as "a wunderkind of the first magnitude." He joined the A&M faculty at the age of 17, after graduating in 1886 with a B.S. in Mechanical Engineering, and by the age of 19, was appointed head of A&M's Department of Mechanical Drawing. Having studied architectural drawing and design at Cornell University and the Massachusetts Institute of Technology, Giesecke also served as head of the Department of Architecture and the official college architect at Texas A&M, designing many campus buildings that are still standing today.

A long-time admirer of Giesecke's legacy, Lockhart was honored to carry on the commitment to clear, engaging, thorough, and well-organized presentation that began with the original author.

Lockhart is known as an early adopter and authority on CAD technologies. She is an instructor noted for outstanding dedication to students and for encouraging a broad

spectrum of individuals, particularly women and minorities, to follow careers in engineering-related fields. Lockhart now works full time to ensure that the Giesecke graphics series continually applies to an evolving variety of technical disciplines.

ONLINE RESOURCES

An Instructor's Manual (9780138065676) and Lecture Slides in PowerPoint format (9780138104405) are available on the companion site for this book at <https://www.pearson.com/en-us/subject-catalog/p/technical-drawing-with-engineering-graphics/P200000009880>.

Web chapters on axonometric projection and perspective drawing may be downloaded from www.peachpit.com. To access and download the bonus chapters:

1. Visit www.peachpit.com/techdrawing16e.
2. Log in with your Peachpit account, or if you don't have one, create an account.
3. Register using this book's ISBN, 9780138065720, then click the Access Bonus Content link next to this book on your account's Registered Products page.

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TECHNICAL DRAWING WITH ENGINEERING GRAPHICS

CHAPTER ONE

THE WORLDWIDE GRAPHIC LANGUAGE FOR DESIGN

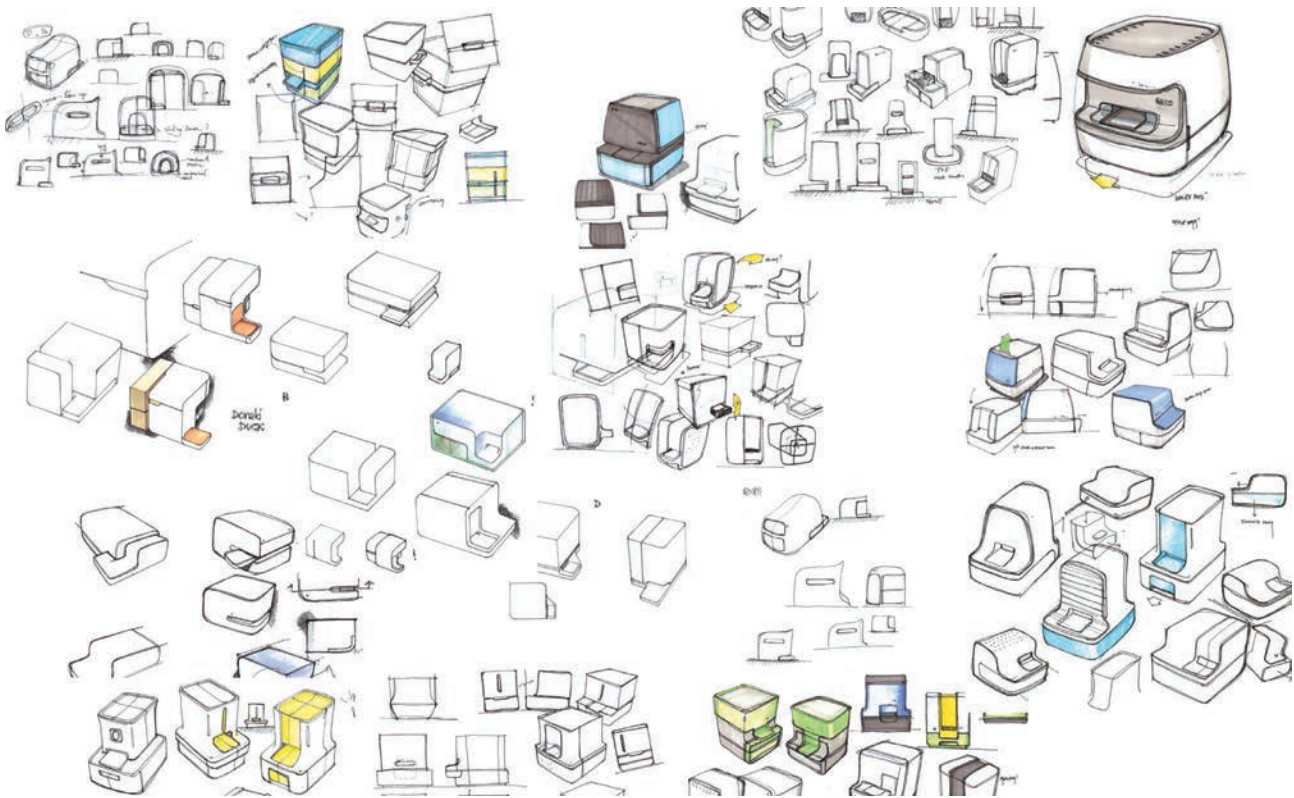
OBJECTIVES

After studying the material in this chapter, you should be able to:

1. Describe the role of graphics in the design process.
2. Identify stages in the design process.
3. Contrast concurrent versus traditional design processes.
4. List five professions that use technical drawings.
5. Describe four creativity techniques.
6. Explain why standards are important.
7. Identify three purposes for technical drawings.

Refer to the following standard:

- *ANSI/ASME Y14.100 Engineering Drawing Practices*



Conceptual Sketches. Exploring many design options through quick sketches is one method that Lunar, recently named one of the top 10 award-winning American product design firms by BusinessWeek magazine, uses to create beautiful products and successful brands. (Images courtesy of LUNAR.)

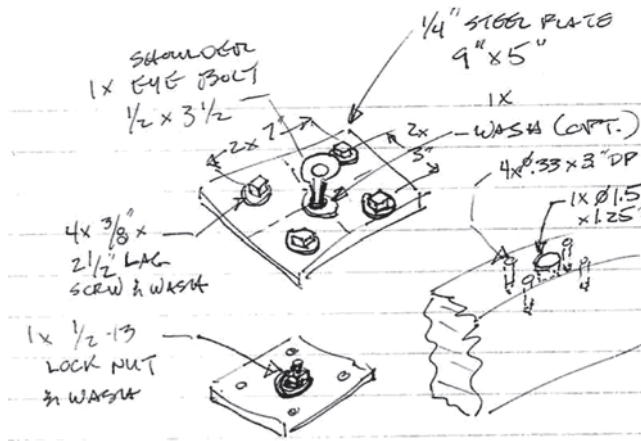
OVERVIEW

Regardless of the language they speak, people all over the world use technical drawings to communicate their ideas. Graphic representation is a basic, natural form of communication that isn't tied to a particular time or place. It is, in a sense, a universal language.

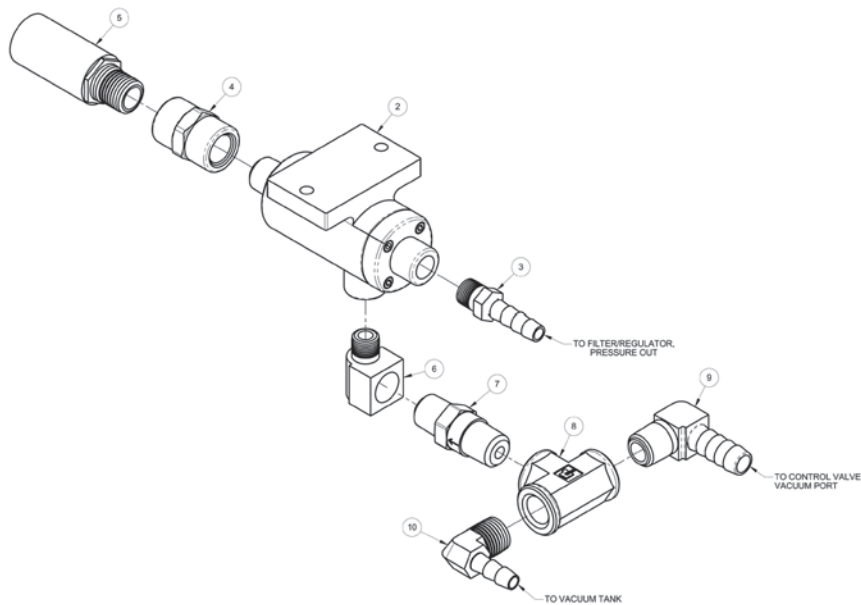
Accomplishing ideas, from the simplest to the most elaborate, requires teamwork. A new product, machine, structure, or system may exist in the mind of the engineer or designer, but before it can become a reality, the idea must be communicated to many different people. The ability to communicate design concepts quickly and accurately through technical

drawings is key to meeting project budgets and time constraints. Effective graphic communication is also an advantage in the global marketplace, where team members may not always share a common language.

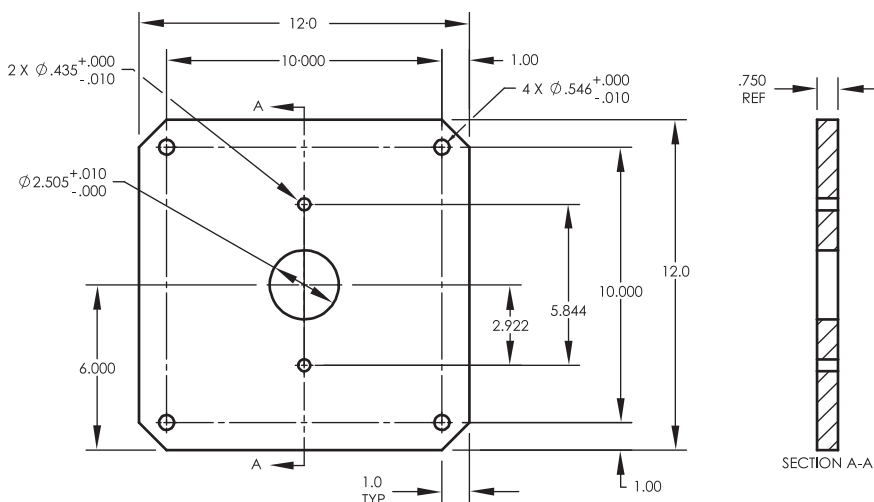
Like carpenters who learn to use the tools of their trade, engineers, architects, drafters, designers, manufacturers, and technicians learn the tools of technical drawing. They learn specific methods to represent ideas, designs, and specifications in a consistent way that others can understand. Being an effective graphic communicator ensures that the product, system, or structure that you envision is produced as you specified.



1.1 Computation Sketch Detail (Courtesy of Jeffrey J. Zerr.)



1.2 Excerpt from an Assembly Drawing (Courtesy of Woods Power-Grip Co., Inc.)



1.3 A Part Drawing (Courtesy of Dynojet Research, Inc.)

UNDERSTANDING THE ROLE OF TECHNICAL DRAWINGS

Drawings and specifications control the many details of product manufacture, assembly, and maintenance. Both ease in freehand sketching and the ability to use computers to produce CAD models and technical drawings are valued skills in the global marketplace. Conveying information graphically requires knowledge of the standards that allow drawings to concisely communicate designs around the world.

Technical drawings can take many forms: idea or concept sketches (such as the sketches on the previous page), computation sketches, design sketches, layout drawings, part drawings, working or construction drawings, electrical drawings, installation drawings, and assembly drawings are all examples. Sketches, 2D CAD drawings, and 3D CAD models are all forms of technical drawing. Some of these types of technical drawings are shown in Figures 1.1 through 1.3. Each of these types of drawings and others have a place in the process of designing and building a product, system, or structure. In general, technical drawings serve one of three purposes:

- **Visualization**
- **Communication**
- **Documentation**

A wide variety of professions use technical drawings to communicate and document designs. Some examples are civil engineering, mechanical engineering, electrical engineering, architecture, bio-resource engineering, landscape design, landscape architecture, industrial design, construction engineering, construction technology, pattern making, project management, fabrication, and manufacturing. There are many others.

Whether you are designing a bridge, installing underground power lines in a subdivision, or designing a plastic housing for a new toaster, understanding and using technical drawing is a key skill you will need.



1.4 These drills all perform similar functions, but a consumer's choice of one over another may depend on the product's styling and aesthetics as well as function. (©Shutterstock/Vereshchagin Dmitry.)



1.5 Project engineers in hard hats examine a blueprint. (Goodluz/Shutterstock.)

The Design Process

The organized and orderly approach to solving problems is known as the **design process**. The engineering design process addresses society's needs, desires, and problems by applying scientific principles, experience, and creativity.

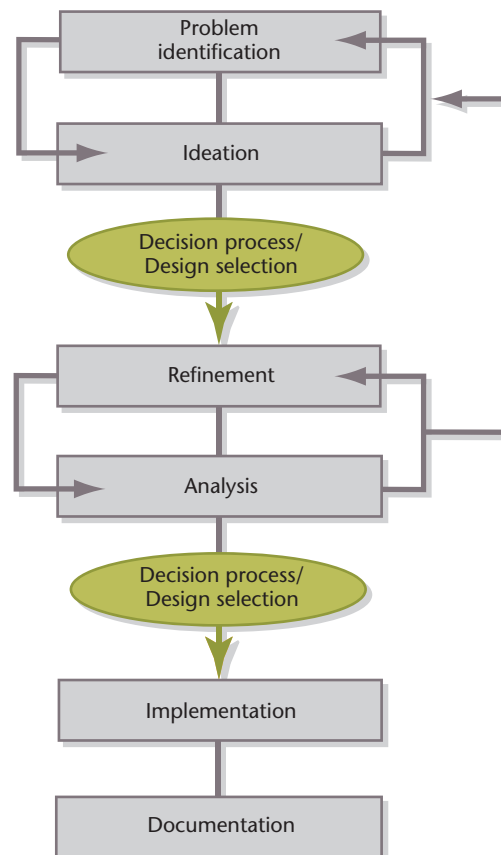
Different types of technical drawings have a specific function in the engineering design process. For example, freehand sketches capture and document the ideation process. Later in the process, CAD models and drawings capture the design and specify the details necessary for manufacture.

The design process for any product requires a clear understanding of the functions and the performance expected of that product. The construction project shown in Figure 1.5 is very different from the styling and functional requirements of the products shown in Figure 1.4, but the stages in the design process are similar for both.

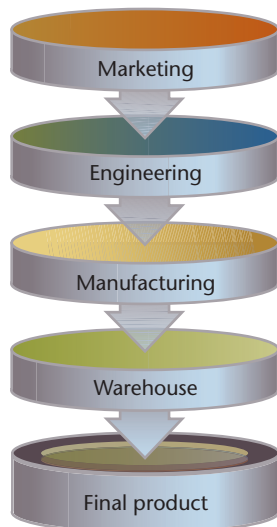
It has been estimated that 70% to 80% of the cost of product development and manufacture is determined during the initial design stages. Although many industrial groups may identify them in their own way, one procedure for designing a new or improved product follows the stages shown in Figure 1.6:

1. **Problem identification:** First, a clear statement of the need for and objectives for the design must be written.
2. **Ideation:** Technical sketches are often used to convey concepts to multidisciplinary teams.
3. **Refinement/analysis:** Designs may be rethought, based on engineering analysis. CAD models and sketches are useful during the analysis and compromise stage. Accurate 2D or 3D CAD models and drawings are created to refine the design.
4. **Implementation/documentation:** The production and/or working drawings that provide the details of manufacture and assembly are finalized and approved.

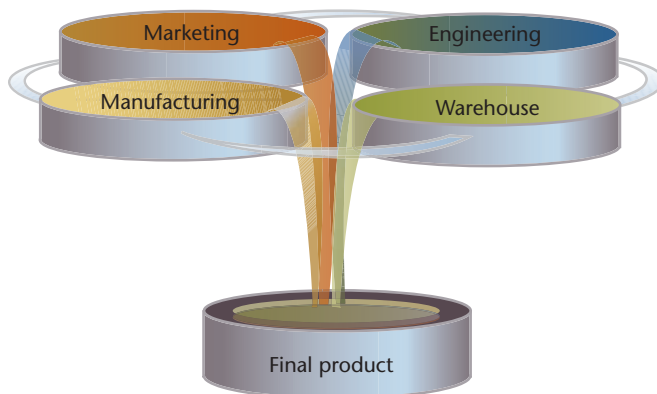
Ideally, the design moves through these stages, but as new information becomes available, it may be necessary to return to



1.6 The Stages of the Design Process

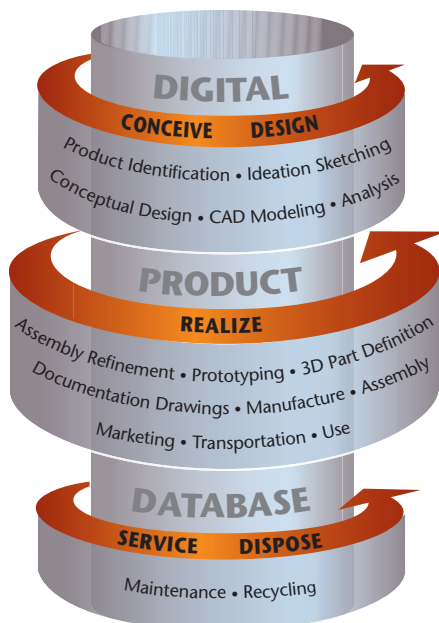


(a) Sequential Process



(b) Concurrent Process

1.7 A Model of the Concurrent Design Process



1.8 The Life Cycle of a Product

a previous stage and repeat the process. For example, based on engineering analysis, the familiar phrase “back to the drawing board” might come into play at the refinement/analysis stage.

Concurrent Engineering

Traditionally, design and manufacturing activities have taken place in sequential order rather than concurrently (simultaneously). Designers would spend considerable effort and time analyzing components and preparing detailed part drawings, and then forward them to other departments. For example, the materials department would use the drawing to identify the particular alloys and source vendors to use. The manufacturing department would then identify the processes needed for efficient production. This step-by-step approach seems logical, but in practice it has been found to be wasteful.

For example, if a manufacturing engineer wanted to taper the flange on a part to make it easier to cast in metal, or to choose a different alloy, the design analysis stage would have to be repeated to ensure that the product would still function effectively. These iterations, illustrated in Figure 1.7a, may be necessary, but they waste resources, and more importantly, time. Time is important because early product introduction makes possible a greater market percentage (and hence greater profits) and a longer life before the product becomes obsolete (clearly a concern with products such as consumer electronics).

Concurrent engineering is a systematic approach that integrates the design and manufacture of products with the goal of optimizing all elements involved in the life cycle of the product. Figure 1.7b illustrates the concurrent design process.

Life cycle design means that all aspects of a product (such as design, development, production, distribution, use, and its ultimate disposal and recycling) are considered simultaneously. Figure 1.8 depicts the life cycle of a product from conception to disposal.

The basic goals of concurrent engineering are to minimize product design and engineering changes and to reduce the time and cost involved in taking a product from concept through production and ultimately to the marketplace. In concurrent engineering, all disciplines are involved in the early design stages, so that natural iterations result in less wasted effort and lost time.

Communication between and within disciplines is especially important in a concurrent design process. Effective interaction among engineering, marketing, and service functions, as well as among engineering subdisciplines, is recognized as crucial to this type of process. Cross-disciplinary communication also helps provide a fertile environment for innovative approaches that can lead to savings in material and production costs.

Computer-Aided Design and Product Development

For both large and small companies, product design often involves preparing analytical and physical models of the product that can be used to study factors such as forces, stresses, deflections, and optimal part shape. The necessity for these



A Mechanical Engineer Working with CAD (Gorodenkoff/Shutterstock.)

types of models depends on how complex the product is. Today, the process of constructing and studying analytical models is simplified by using **computer-aided design** (CAD), **computer-aided engineering** (CAE), and **computer-aided manufacturing** (CAM) techniques. These systems allow rapid design analysis for simple objects as well as complex structures.

CAD allows for a range of activities, from modeling 2D and 3D geometry to creating drawings that document the design for manufacturing and legal considerations.

CAM provides computerized control for manufacturing processes. Examples include using a computer interface to control a lathe, or generating the path for milling machine tools directly from the CAD model. In more sophisticated systems, CAM can be used in materials handling, assembly, and inspection.

CAE allows users to simulate and analyze structures that will be subject to various temperatures, static loads, or fluctuating loads. Kinematic analysis studies moving parts. Some of these functions are integrated with CAD software, and other packages import data from a CAD system. Using these tools, engineers can simulate, analyze, and test designs efficiently, accurately, and quickly.

Designing Quality into Products

Companies use a number of systematic ways to try to design “quality” into their products as well as to measure performance and make decisions based on data. Designers may feel that their creative approach to problem solving is stifled when these systems are poorly implemented in the workplace; but when these systems are well implemented, organizations can show quality improvement.

DFSS Design for Six Sigma is an approach that uses engineering and statistical tools to design products in a way that predicts and minimizes customer and manufacturing problems.

Six Sigma is a process originated at Motorola to improve quality by reducing or eliminating defects.

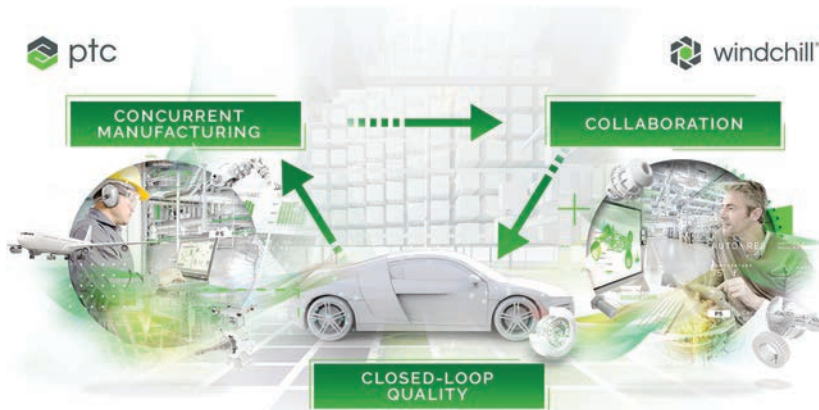
DMAIC Define, Measure, Analyze, Improve, and Control are steps defined in a continuous improvement process that attempts to define and ensure critical to function (CTF) characteristics.

QFD Quality Function Deployment is a tool for decision making that helps companies focus on a customer-driven approach and set of product characteristics.

The Digital Database

Computer use continues to change the way products are produced. All the information to manage, design, analyze, simulate, package, market, and manufacture a product can be shared with a diverse (and perhaps geographically distant) group of users through a single complex digital database.

Product data management (PDM) systems or **enterprise data management** (EDM) systems electronically store the various types of data associated with designing and manufacturing a product. An effective PDM system allows all the product data to be quickly stored, retrieved, displayed, printed, managed, and transferred to anywhere in the organization. This allows for designs to be optimized or directly modified at any time. Costs, product revisions, and **engineering change orders** (ECOs) can be analyzed, tracked, and implemented quickly. Managing enterprise data requires commitment and planning, but companies who implement PDM effectively can capture product data once and utilize it many ways to achieve a competitive advantage (Figure 1.9).



1.9 Product Data Management. PTC's Windchill is an enterprise-wide product data management system designed to be extended to serve data over the internal network (or intranet). Data from the database are provided to external users via a secure, password-protected site (to share nonpublic information with suppliers and other partners in remote locations). (Image courtesy of PTC, Inc. All rights reserved.)

1.1 GRAPHICS TOOLS IN ACTION

Designers and engineers use graphics for visualization, communication, and documentation throughout the design process. A case study from Santa Cruz Bicycles will illustrate the tools used in the redesign of its V10 model, a downhill racing mountain bike, shown in Figure 1.10.

Design Phase: Problem Identification

The impetus for the V10, as for most racing bikes, was higher performance. In 2000, Santa Cruz Bicycles (SCB) purchased three patents for suspension systems that had never been developed. Most bikes have suspension on the front wheel; these patents showed how rear suspension could be manipulated to improve performance and control.

The bicycle's suspension has two roles: absorbing bumps and translating the pulsing motion coming from the human pedaler into forward movement. The balance between how far the wheel can move up and down and maximal forward motion determines bike performance.

The first V10 was ridden in 2002. Its rear suspension allowed the wheel to travel vertically 10 inches. At that time, 8–8.5 inches of travel was common among competitors.

When SCB set out to update the V10 design, better performance was a key criterion. The design team included the product manager (who interfaced with marketing), two design engineers, and racers who rode the V10. The team members had plenty of information for defining the problem: they were familiar



1.10 The V10 is a generation of downhill racing bikes first sold in 2002 and continually improved based on race team results and rider feedback. The 3D model of the new design is shown here. (Courtesy of Santa Cruz Bicycles.)

with the design drawings, performance specifications, and manufacturing costs of the V10 already being produced. Racing teams provided feedback about how the current design handled and where they envisioned improvement. Racers and engineers viewed race videos to see how the bike performed on courses with more or less rugged terrain. Marketing provided information about competitors' bikes and how the V10 measured up against them.

At the problem identification stage, the team worked to enlarge the criteria for judging the solution. The team identified three primary goals for the redesign:

- Get maximum forward movement from a suspension that also absorbs bumps.
- Improve the handling to give the rider better control.
- Minimize the weight to make a lighter bike.

SPOTLIGHT

Santa Cruz Bicycles

The team at Santa Cruz Bicycles believes that to build a good bike, you have to first really love riding bikes. They all ride daily—from the owner to the engineers to the salespeople. This creates a very strong relationship between the design team and their customers—each model is constantly being refined and improved based on feedback from riders who want to—and do—win races with a Santa Cruz bike. Their engineers are proficient 3D modelers who use

the many features of Pro/ENGINEER to visualize, analyze, and test designs before they are built. More important, the 3D model supports the continual improvement that is built into the Santa Cruz culture. They can quickly modify parts, see how the new part affects the movement of the bike, assess the stresses on the new part, and send it to be prototyped for the ultimate test: riding on unpredictable terrain.

1.11 The V10's rear wheel attaches to a triangular swingarm that connects to the front part of the frame with a pivot. This allows the rear wheel to travel up and down. (Courtesy of Santa Cruz Bicycles.)



Design Phase: Ideation

The design team focused first on the geometry of how the parts of the bike are arranged. Unlike the frame of a conventional bicycle, the V10's frame is in two parts. The back wheel connects to a triangular “swingarm” that attaches to the front part of the frame with a pivot (see Figure 1.11). The length of the parts, the angles between them, and their movement determine how the bike handles. In the ideation phase, the team captured and evaluated different combinations of handlebar placements, seat, crank, pivots to the wheels and shock absorbers, and more. How much the shock absorbers allow the wheel to travel vertically, the pivot locations, and link sizes all affect how the bike will perform.

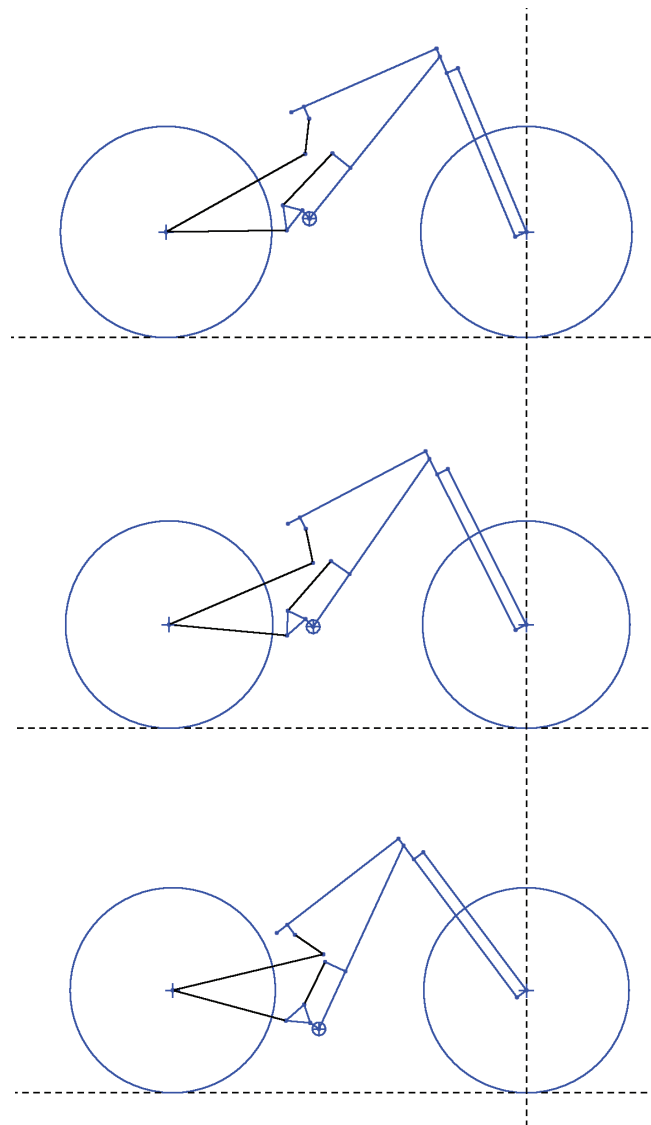
At the ideation stage, being able to draw a readable free-hand sketch is vital to “selling” your idea to your design team colleagues. This stage is called *universal possibilities* because the group seeks to consider every possible solution and not limit the design by preconceived notions of what will be best.

For each subsystem of the bike, the team generated and evaluated many concepts. For example, they brainstormed all the ways a pivot assembly might be made. They also employed a kind of “contest” to spur creativity and generate options. One engineer would be given 1.5 days to come up with a combination of parts and materials to make a subsystem, such as a link mechanism. Then, the design tasks would be shuffled and a different engineer would have 1.5 days to define an alternative approach to the link mechanism. Using a whiteboard, the team would then list the pros and cons of each subsystem design.

Design Phase: Decision Process/Design Selection

The design selection process varies by company and by the design challenge. It may be informal and conducted by a single individual, or it may be a highly formal process in which different teams compete for development dollars for new products. In all cases, design selection narrows the field of options for the final design.

SCB's team needed to evaluate how the parts would *move*. They created 3D skeleton models in CAD. A skeleton model is a simplified representation of the centerlines and other shapes that capture the design geometry. These skeleton models are useful for kinematic analysis (Figure 1.12).



1.12 Skeleton models were used for kinematic analysis of different combinations of the swingarm geometry. The model shows how much the swingarm can pivot. (Courtesy of Santa Cruz Bicycles.)



1.13 This graph compares the measure of rear-wheel travel produced from the skeleton model during ideation. (Courtesy of Santa Cruz Bicycles.)

Data generated from each model showed the behavior of the suspension and how the bike handled. Data from the kinematic analysis were captured in a spreadsheet and used to compare the various arrangements of lengths and angles in the designs (Figure 1.13).

Even at this early stage, drawings and models can be used to elicit customer feedback. If the user interface is important to a product, a 3D model or a computer simulation provides a way to get early user feedback. These models may be shown to customers informally, or more formally through focus groups.

At SCB, promising combinations were translated into simple prototypes that could be used to verify the concept. Called “mules,” these proto-bicycles were fabricated so that a human rider could evaluate the handling (see Figure 1.14).

For a new product, the design team generally prepares a product proposal to show the management group approving new products. Drawings or sketches are used to communicate key features of the device. Rendered models or **prototypes** convey options for the final look. The proposal may include a product plan, sales estimates, and cost figures for developing the product. It estimates the profit to be generated by the product and describes how it will meet the criteria for a new product, such as helping to achieve company goals or promising the best return on investment. The ability of the design team to communicate effectively through drawings, spreadsheets, written communication, and oral presentation all play a role in the product’s acceptance for development.

The preliminary design selection group for the V10 consisted of design engineers, manufacturing engineers, and the product manager. (At some companies, the team might also include members from finance or executive management.)

Function was the primary criterion for this review of the V10’s redesign. Spreadsheets with data from the skeleton models, graphs comparing different options, and measurements from the prototypes were used to make the selection. Feedback from the riders who rode the “mules” was an important part of the process as well.



1.14 This prototype “mule” for the V10 can be ridden and evaluated by a human rider. (Courtesy of Santa Cruz Bicycles.)

Design Phase: Refinement

When a product is approved for further development, the team moves into the refinement and analysis phases of the project. Because these two phases form another “loop” in the design process, they are often combined into a single stage called the **development phase**.

During the refinement stage, the product concept is solidified into an accurate plan for making the product. Different options are evaluated in terms of how much they will cost, how long they will take to make, how well they satisfy the customer’s requirements, how durable they will be, how difficult they are to assemble and service, how environmentally sustainable the design is, and so forth. At this stage members may join the design team to provide expertise in these different areas.

At Santa Cruz Bicycles, the team had chosen a functional design and was now ready to begin defining the part features, styling, and materials for the bike. During refinement, the design is updated as more information is determined about the designed components and their manufacturing processes (see Figure 1.15).

At this point, time and money are invested in creating a model to represent the product accurately. At Santa Cruz, two or three combinations of features were modeled in 3D and used to analyze and test the design further.

The following were criteria at this point:

Weight: How much will a component or choice of material add to the weight of the bike?

Strength: Can the material used to make the part stand up to the stresses generated during a bike race?

Cost: Is a lower-cost option available? Or is the cost justified because of what it adds to performance?

Optimization: Does the choice result in fewer parts to make or better clearance between parts?

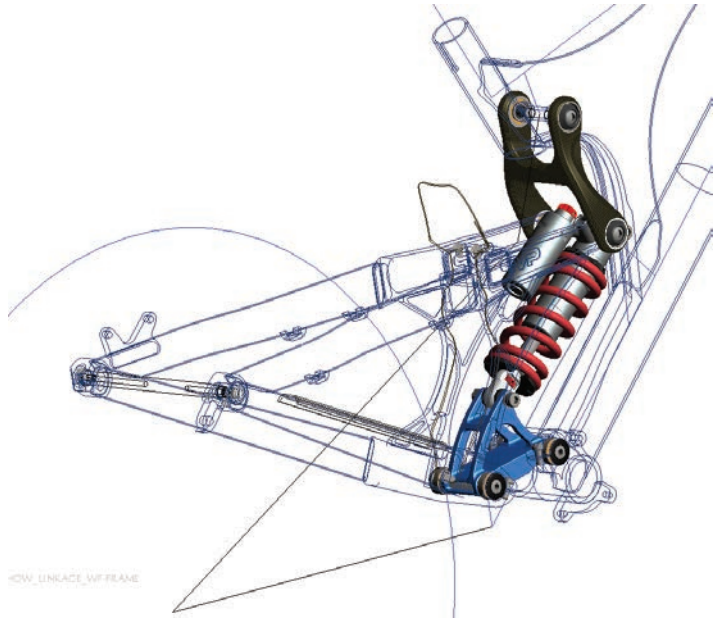
The team used an informal concept/criteria matrix to evaluate the options against multiple criteria. For example, manufacturing the frame using carbon fiber instead of aluminum would affect the cost but would decrease the weight of the bike by 15%–17%. The matrix can tally the pluses and minuses of various options to aid in comparing them.

At Santa Cruz, bicycles are modeled in Pro/ENGINEER, a parametric solid modeling package. Parametric or **constraint-based modeling** captures *relationships* between part features and the sizes of the features. These relationships capture the **design intent** for the part. They define features in terms of characteristics that must be preserved when the part changes. For example, the key design criteria for a pair of holes for mounting a bracket might be that each be the same distance from the center of the part. Defining the location of the holes to be half the width of the bracket from its midline ensures that when the bracket gets wider, the holes will move accordingly. Similarly, when a part is defined in terms of its relationship to another part, changes in that part will cause any related parts to be updated automatically (see Figure 1.16). This flexibility makes it easy to test many different versions of a design—and is ideal for continually refining and improving racing bikes like the V10.

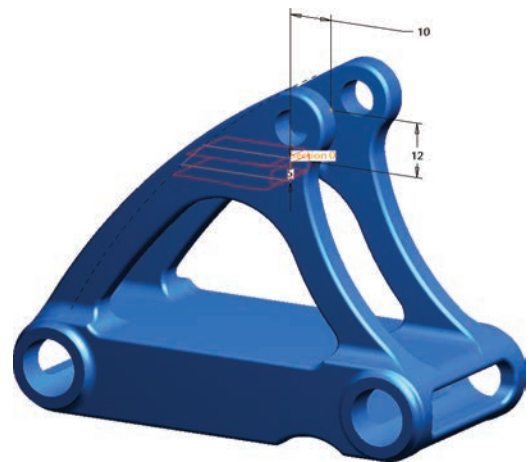
Design Phase: Analysis

The analysis phase tests the design and feeds back information to refine the design further, forming the second key iteration or “loop” in the design process. Crucial here are the performance criteria that the design must meet.

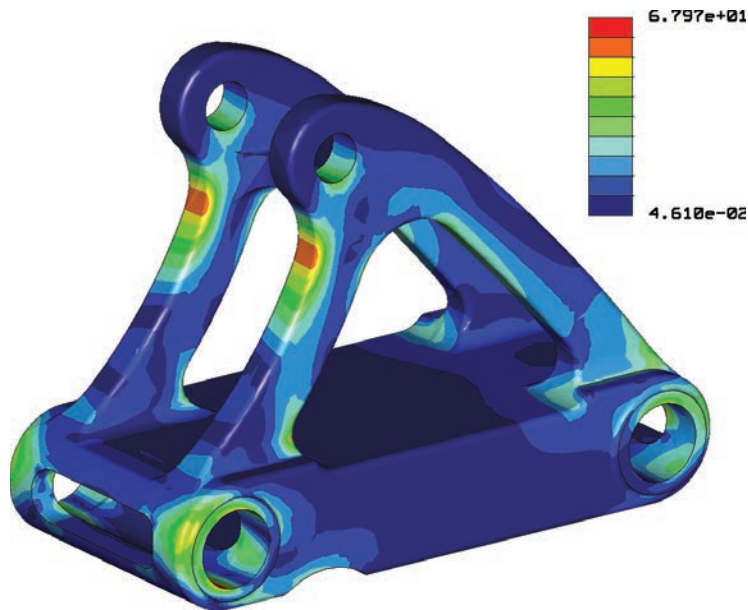
For the V10, the assembled solid model could be tested like a real bike. By specifying the material for each part, the team assessed the total weight of the bike as designed. Stress analysis (finite element analysis) predicted where parts or the frame might be likely to break (see Figure 1.17). The model was also used to evaluate the bike’s fit for riders of different sizes. Human factors software provides a database of different body types and sizes for this purpose. The team can simulate what an Asian female in the 50th percentile for height and weight sees while riding, what she can reach, and the location of her center of mass. The same can be done for a white male in the 50th percentile (see Figure 1.18).



1.15 This subassembly is a combination of features that were evaluated during refinement. (Courtesy of Santa Cruz Bicycles.)



1.16 The rib shown here in red is constrained to be 10 mm back of and 12 mm down from the pivot point. If the pivot location is changed, the rib will be updated to preserve the relationship defined by the constraints. (Courtesy of Santa Cruz Bicycles.)



1.17 Finite element analysis can be performed within the solid modeling software, or the model can be imported into a separate analysis package. (Courtesy of Santa Cruz Bicycles.)



1.18 “Mannequins” of various shapes and sizes can be added to the solid model to assess ergonomic factors. (Courtesy of Santa Cruz Bicycles.)

At this point, the model can also be used for manufacturing analysis. At Santa Cruz, manufacturing engineers identify aspects of the design that will be hard to manufacture: a part shape that will require special tooling or be difficult to weld, or parts that must be made to a high degree of precision (tolerance).

Tools that work within the solid modeling packages are available for much of the analysis. Analysis for moving parts, such as the rear wheel suspension, can be done in Pro/ENGINEER (and other modelers) and tested under a wide range of physical conditions—even zero gravity! Human factors and stress testing are two more examples of analysis that can be done in the 3D modeler. For more advanced or specialized testing, a wide range of software can import the 3D model data.

Whether the analysis software recommends a change or simply reports results, it is up to the design engineer to interpret the tests and take steps to refine and test the design further until the desired results are achieved.

Design Phase: Decision Process/Design Selection

Once a design is ready to manufacture, another design review may be required. At each stage of the design process, more is invested in the design, and design changes become more costly. At the ideation stage, many ideas are investigated at relatively low cost. In the refinement stage, more time and money are devoted to building and testing models, and the cost of making major design changes increases. After the second design review, companies commission molds and tooling, purchase materials, and commit other resources to manufacturing the product. After this point, major design changes become very costly and can threaten a product’s success. The second selection process, often called *critical design review*, is a filter that allows only the products with the greatest chance of success to proceed.

The selection team and review process may be very similar to the one that precedes refinement, or it may involve a completely different set of criteria. Understanding the review criteria helps you make an effective presentation and also helps you make better design decisions in the context of your company’s goals.

For the V10, more stakeholders evaluated the design at this point. Representatives from marketing, sales, and graphic design were added to the team, and the aesthetics of the design were considered (see Figure 1.19). How will the logo fit onto



1.19 Multiple options for the look of the frame triangle were considered in the final design review. (Courtesy of Santa Cruz Bicycles.)

the bike as designed? Which look is more marketable? Should a straight or a curved tube be used for the swingarm? What are the implications of the manufacturing cost for the bike's sales potential?

Design Phase: Implementation

During the implementation phase, the design is communicated to those responsible for manufacturing, assembling, and distributing the product. Up to this point, drawings and models served a variety of purposes. In the implementation phase, they become contracts with suppliers. The model and the accompanying drawings (if any) must clearly specify what constitutes an acceptable part.

As in refinement, the 3D solid model serves as the information database. Dimensioned multiview drawings may be generated directly from the solid model. Increasingly often, manufacturing processes are directed from the model itself, with no need for an intermediate drawing. In these cases, tolerances may be noted in the model, or critical dimensions and tolerances marked on a drawing may be sufficient.

Tooling for a bicycle's design can cost as much as \$120,000, so care is taken to be sure that the tools are correct. SCB uses a detailed checklist to ensure that they are ready to have the tools made for manufacturing the bicycle. Because the 3D model is used to communicate the design to the toolmakers, it is sent to them for review and analysis before the final design is documented.

Sample parts are requested so that the physical parts can be measured against the model and tested to see how they fit together. Mechanical testing, ride testing, and destructive testing are done on the sample parts while there is still time to make a change to alleviate problems (see Figure 1.20). Once engineering is satisfied with parts as manufactured, the models for the design are approved.

Although the design has been finalized, there still may be changes during the implementation phase (as well as later when the product is in production). With rapid prototyping and design simulations using the solid model, it is easier and more cost effective to anticipate these issues in the refinement stage, but it is not uncommon for fine-tuning to occur during implementation. Unavailable parts or materials can force a substitution, a supplier may recommend a cheaper alternative, standard parts may be substituted for newly designed ones, or unanticipated problems may crop up during assembly. Sustainability issues can affect a design throughout the lifetime of the product.

Assembly drawings may be generated to document the parts included in an assembly and to illustrate how they fit together. The parts list, or bill of materials, is an important facet of inventory planning and assembly. Parts from the solid model may be catalogued in a company-wide parts database. Parts in the database can be reused or shared by other products, and standard parts inventories can be tracked and monitored.

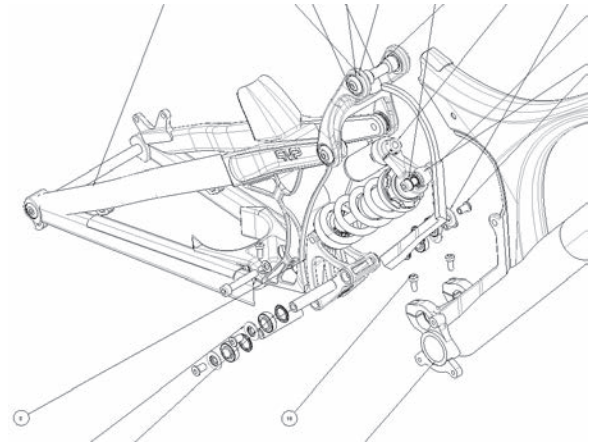


1.20 This sample part for the swingarm failed during testing. (Courtesy of Santa Cruz Bicycles.)

Design Phase: Documentation

electronic format can be altered so that they no longer represent the actual product design, companies must control access to the information in the solid model—or capture it in another medium—to create a permanent record. Once a model has been archived, only authorized personnel may make changes to it.

The solid model can also be used to generate a wide range of graphics used to document a product or process. Views of the model are imported into step-by-step illustrated guides for assembly line personnel (Figure 1.22), and the same guides are sent to service personnel who need to disassemble and reassemble the products after sale.



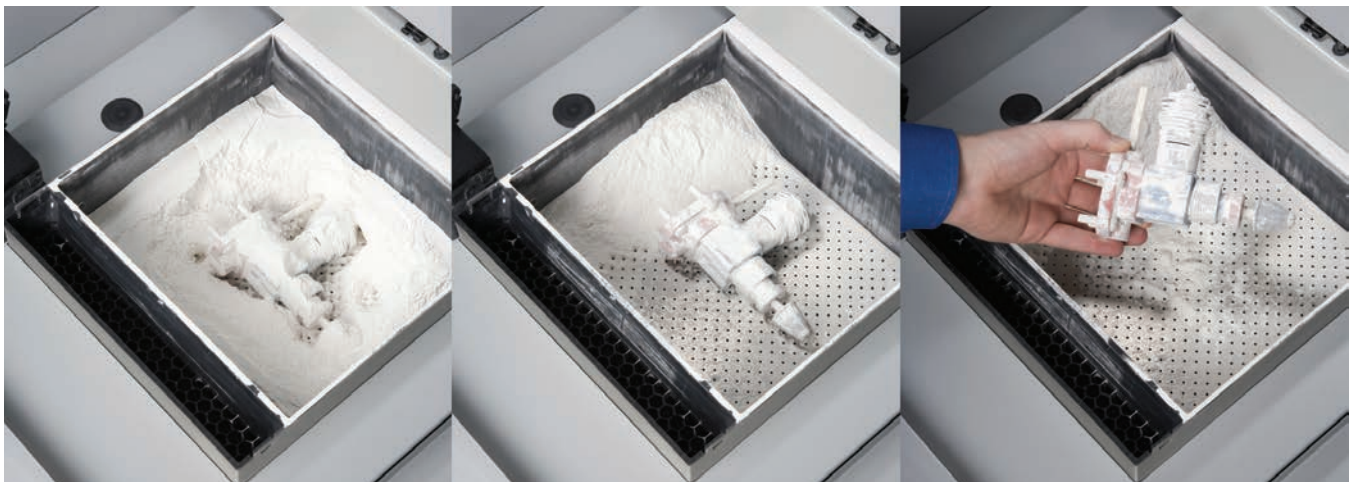
1.22 Exploded views such as this one can be used to illustrate assembly or service manuals for the V10. (Courtesy of Santa Cruz Bicycles.)



1.2 RAPID PROTOTYPING

While refining design ideas, the design team often works concurrently with manufacturing to determine the best ways to make and assemble the necessary parts. After several cycles of refining, analyzing, and synthesizing the best ideas, **rapid prototyping** systems allow designers to generate parts quickly, directly from 3D models, for mockup and testing. For example, they can check usability for humans or explore customer

reactions in consumer focus groups by showing them facsimile products created using rapid prototyping. 3D design data can be formatted for use by rapid prototyping equipment. Figure 1.23 shows the prototyped part emerging from the unfused material of the rapid prototyper, which can “print” a color part like the one shown in about four hours. You will learn more about rapid prototyping in Chapter 10.



1.23 Rapid Prototyping. The ZPrinter 450 “printed” the colored part shown in about four hours. (Courtesy of 3D Systems.)

1.3 DRAFTING STANDARDS

There are *standards* that support a uniform, effective graphic language for use in industry, manufacturing, engineering, and science. Technical drawing texts like this book can help you learn them. In the United States, providing these standards has been the work of the American National Standards Institute (ANSI) with the American Society for Engineering Education (ASEE), the Society of Automotive Engineers (SAE), and the American Society of Mechanical Engineers (ASME).

Participants in these organizations help develop the *American National Standard Drafting Manual—Y14*, which is composed of a number of separate sections that are published as approved standards as they are completed (see Appendix 1).

These standards are frequently updated to communicate information so that it meets the needs of modern industry and engineering practice. They are considered the most authoritative guide to uniform drafting practices in the United States today.

International standards, often defined by the International Organization for Standardization (ISO), and the ASME or ANSI standards for drawing practices are similar in many respects. The greatest differences are in the preferred method of projection: first-angle versus third-angle projection, and in the units of measurement in dimensioning. Third-angle projection is the projection method that you will be learning in this text. Where practical, information for both U.S. and international drawings will be presented.

1.4 CREATIVITY TECHNIQUES

Some people are naturally gifted at design and drawing, but all persons can improve their design ability if they learn to use the proper tools and techniques. It is similar to learning to play a musical instrument; it comes more naturally to some people than to others, but all can learn to play if they practice.

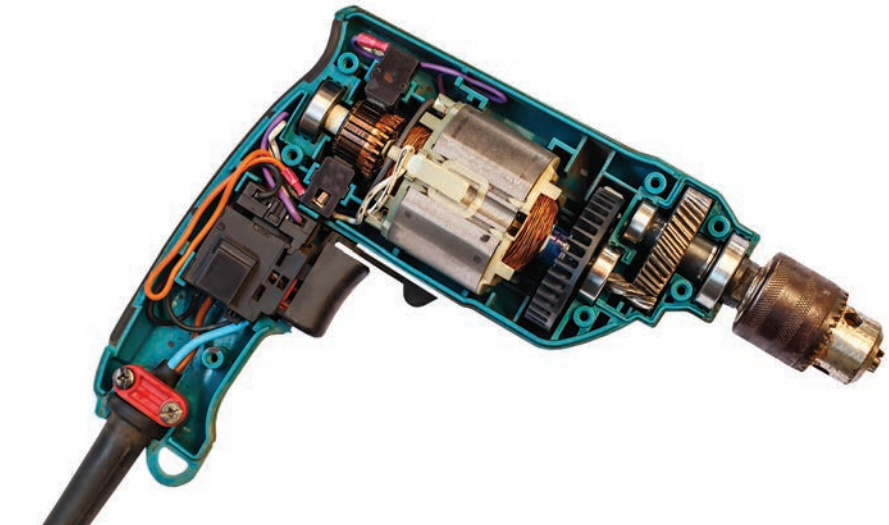
How do you develop new ideas? Here are some proven techniques you can practice.

Examine Manufactured Products

It is common for design engineers to dismantle manufactured products, evaluate them, and study how their parts are designed to work together. As you use and examine manufactured products, ask yourself, How could they be improved? What would you do differently? How could you expand/change the design to guarantee better performance? What could you do to expand the life of the product? How could you make it more efficient, more cost-effective, and so forth?

Reverse engineering is a term that refers to designing products based on existing designs, usually through measurement and deconstruction of an existing product. A **coordinate measuring machine** (CMM) aids in speeding the time to reverse engineer some products. A CMM measures the object using a probe or laser and stores the pertinent geometric information into a database where it can be manipulated using CAD.

Functional decomposition is a term for determining the subfunctions involved



A Drill Housing Cut in Half to Expose Interior Part Functions (*bit mechanic/Shutterstock.*)

in a design and then using those functions to reconstruct a similar product.

Study the Natural World

Noting how other creatures and organisms function and interact with their surroundings can provide a wealth of information and creativity. Such things as beehives and spiderwebs are masterpieces of structural design that have inspired human designers for centuries. A hummingbird's wings are aerodynamic wonders. There is much to be learned by studying natural forms and expanding on their designs, as Leonardo da Vinci did with his flying machine (Figure 1.24).

Watch the Web

Excellent resources for engineering and design are available on the World Wide Web. Reading technology blogs and news feeds and visiting engineering sites is a way to stay current and keep in touch with peers. The following websites are useful for engineering design:

- www.uspto.gov (U.S. Patent Office online search site)
- www.practicalmachinist.com
- www.engineeringtoolbox.com
- www.engnetglobal.com
- www.efunda.com
- www.thomasnet.com
- www.mcmaster.com
- www.howstuffworks.com



1.24 3D Model of the Ornithopter, a Flapping-Winged Flying Machine Conceived by Leonardo da Vinci (Leo Blanchette/Shutterstock.)

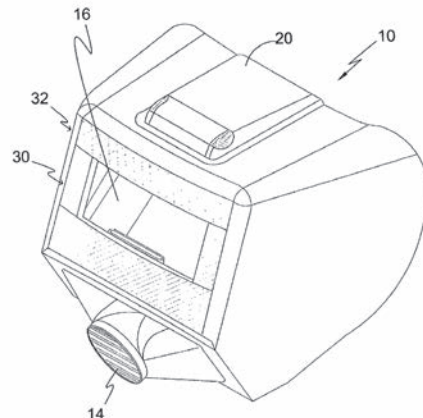
Research Patent Drawings

Patents can be a great source of ideas. A patent is issued by the U.S. government and grants the holder the “right to exclude others from making, using, or selling” a specific product for a specific time period. The patent process was first developed as a way to promote the disclosure of technical advances. You can research the current state of design for your product idea at the U.S. Patent and Trademark Office’s searchable website (www.uspto.gov). The U.S. Patent and Trademark Office has strict regulations as to how designs are presented in *patent drawings* to ensure that the drawings are easy to reproduce (Figure 1.25).

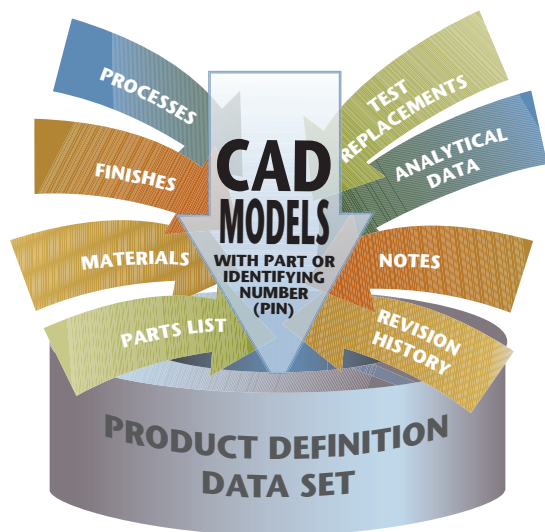
Design Groups

To bring complex products and systems to market, most individuals end up working in a team environment. Teams combine the expertise of individuals familiar with materials, production processes, marketing, finance, and so on, so most projects become a team effort long before a product is produced and marketed. In addition to supplying the varied expertise that it takes to bring an idea from concept to manufacture, interaction between people of varied talents plays an important role in the creative design process. The ability to create and understand technical drawings helps group members contribute to and benefit from team communication.

(12)	United States Patent Buckman	(10) Patent No.: US 7,178,932 B1
		(45) Date of Patent: Feb. 20, 2007
(54)	WELDING HELMET	5,031,237 A * 7/1991 Honrad 2/8.6
(76)	Inventor: Michael Allen Buckman, P.O. Box 792, Pablo, MT (US) 59855	5,123,114 A * 6/1992 Desanti 2/8.6
		5,561,855 A * 10/1996 McFall 2/8.6
		5,896,579 A * 4/1999 Johnson et al. 2/8.6
		6,340,234 B1 * 1/2002 Brown, Jr. 362/105
(*) Notice:	Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.	2005/0077278 A1 * 4/2005 Steinemann 219/147
FOREIGN PATENT DOCUMENTS		
(21) Appl. No.: 11/324,913	CA 2184929 * 3/1998	
(22) Filed: Jan. 4, 2006	GB 1511303 * 5/1978	
	WO 81/02514 * 9/1981	
(51) Int. Cl. F21V 21/084 (2006.01)	* cited by examiner	
(52) U.S. Cl. 362/105; 362/106; 362/373; 219/147; 2/8.6	Primary Examiner —Stephen F. Husar	
(58) Field of Classification Search 362/105; 362/106; 373; 219/147; 2/8.2, 8.6	(74) Attorney, Agent, or Firm —Michael I. Kroll	
See application file for complete search history.		
(56) References Cited	(57) ABSTRACT	
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4,309,774 A * 1/1982 Guzowski 2/8.6		
5,029,342 A * 7/1991 Stein et al. 2/8.6		
11 Claims, 9 Drawing Sheets		



1.25 Patent Drawing for a Healthier Helmet



1.26 Possible Contents of a 3D Data Set

1.5 PRODUCT DEFINITION

As stated before, technical graphics play three important roles: visualization, communication, and documentation. Drawings and CAD models are an important part of product definition. **Product definition** refers to the collection of digital or hard copy documents that specify the physical and functional requirements for a product. These can range from a 3D CAD model that specifies manufacturing requirements within the CAD file to a dimensioned paper sketch.

Many companies prefer storing information in a digital format because the storage cost is comparatively low and the data can be easily searched, retrieved, and utilized by many individuals. Figure 1.26 represents the range of contents in a product definition data set. A thorough understanding of the graphic language, from hand sketches to complex 3D models, is necessary to be a player in the competitive world marketplace.

1.6 SHOWING THE DESIGN PROCESS IN A PORTFOLIO

A portfolio is a representative sample of work that helps communicate your skills and talents, usually to a prospective employer or client. Many companies and individuals post portfolios on the Web to promote their capabilities and accomplishments. Larger companies tend to show more models and completed projects because working drawings may be proprietary information. Individuals, particularly recent students, may focus more on showing correct drawing practices and creative problem-solving skills.

Whether you are a consulting engineer, an architect, an industrial designer, or a draftsman, showing samples of your work offers an advantage over just talking about it. Drawings are proof that you not only have good ideas but also have the skill to communicate clearly and accurately throughout the stages of a project, from concept to completion.

If you are seeking an entry-level job, don't worry if your portfolio looks like an entry-level portfolio. Clean, correct student work is a good indication of your potential. Don't try to cover a lack of well-drawn examples with a more expensive portfolio case, or decorative presentation. Instead, spend that energy completing a project that shows your ability.

A portfolio is an opportunity to tell concise success stories. For example, you may want to show four to six projects, including the design brief (or assigned problem), sketches, development drawings, and final presentation (including working drawings, renderings, and physical or 3D models) for each. Your portfolio samples make great talking points in an interview, so take advantage of the chance to show how you tackled challenges such as cutting costs, improving function, or solving problems that arose along the way.

Some more tips for creating a successful portfolio are:

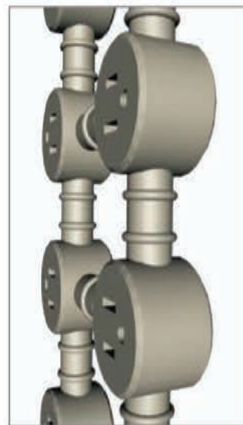
1. Less is more. Others will assume that your portfolio is your best work, so a few good examples are better than a million mediocre examples. Four to six projects are usually plenty.

2. Show correct drawing practices. Especially if you are seeking a drafting position, but even if your strength is creativity, be sure that lineweights, dimensioning, notes, and other parts of the drawing conform to technical drawing standards.
3. Keep several versions of your current portfolio, including hard copy, digital, and Web presence. Use widely accessible formats (such as PDF or PowerPoint) to make your digital portfolio easy to access. Including your Web address in correspondence allows people to view your work or refer to your samples at their convenience.
4. Never rely entirely on electronic equipment. Even if you have a great digital presentation, bring a portfolio of printed samples. Drawings should be clean and legible. Depending on your discipline, you may want a larger or smaller portfolio case, keeping in mind how convenient it is to transport and view. You may fold large drawings to fit, as long as they are easy to take out and look at.
5. Save good examples as you create them. It is easy to forget what you did last year, or four years ago. Keep copies (both digital and paper) of your best work in a file for portfolio samples and remember to save some development sketches from successful projects, too.
6. Depending on your career goals, let your portfolio reflect the area you want to explore. If you are seeking a creative position and you have related skills (such as artistic drawing or photography) you may want to include appropriate examples. If you are mainly interested in a drafting position, you may want to include more working drawings and models.

Throughout this book, the Portfolio sections give you some exposure to real-world drawings that apply to chapter topics. However, the Portfolio section in this chapter shows how an industrial design student successfully communicated the design process through a well-conceived portfolio page.

PROJECT: REDESIGN OF THE SURGE PROTECTOR

:: SURGE PROTECTOR



Bottom



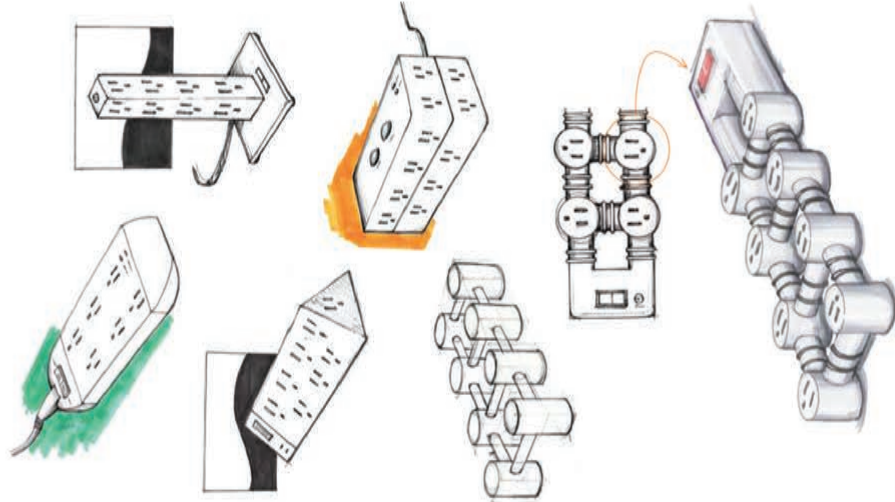
Power & Reset Buttons

**POWER STRIP**

The main focus for designing a surge protector was to take an ordinary and aesthetically boring product and give it a sense of "coolness". Giving it a unique look helps catch the eye of the consumer when displayed with other surge protectors.



Functionally was also considered by providing eight plug-ins spaced far enough apart to accommodate AC adapters. Also, a large power button and a flat head plug for tight areas.



Ideation

This single page from the portfolio of John Mountz begins with the design brief and rationale for the design. It includes initial concepts, more refined sketches, a solid model, and details. Mountz showed the final presentation drawing on an additional portfolio page. At the time, he was a student of industrial designer Doug Wintin, then Chair of the School of Design at ITT Technical Institute. (Courtesy of John Mountz.)

KEY WORDS

Communication
 Concurrent Engineering
 Computer-Aided Design
 Computer-Aided Engineering
 Computer-Aided Manufacturing
 Constraint-Based Modeling
 Coordinate Measuring Machine
 Define, Measure, Analyze, Improve, Control (DMAIC)
 Design for Six Sigma (DFSS)
 Design Intent
 Design Process
 Development Phase
 Documentation
 Engineering Change Orders
 Enterprise Data Management
 Functional Decomposition
 Ideation
 Implementation/Documentation Phase
 Life Cycle Design
 Parametric Modeling
 Patent Drawings
 Product Data Management
 Product Definition
 Prototype
 Quality Function Deployment (QFD)
 Rapid Prototyping
 Refinement/Analysis Phase
 Reverse Engineering
 Six Sigma
 Standards
 Universal Possibilities
 Visualization

CHAPTER SUMMARY

- The graphic language is the universal language used to design, develop, and construct products and systems throughout the world.
- The design process combines ideas, scientific principles, resources, and existing products into a solution for a problem. It consists of five specific stages.
- Every technical drawing is based on standards that prescribe what each symbol, line, and arc means.
- The members of an engineering design project team must be able to communicate among themselves and with the rest of the project team to contribute to the team's success.
- The basic principles for communicating information using technical drawings are the same whether drawings are created by hand or using CAD.
- Successful companies hire skilled people who can add value to their team. A thorough understanding of the graphic language is an essential skill that employers value.

REVIEW QUESTIONS

1. When is sketching used as graphic communication?
2. Why are standards so important for members of the engineering design team?
3. What is the design process?
4. What are the five phases of the design process?
5. Describe the difference between concurrent and traditional design process models.
6. What does PDM or EDM stand for? What are some advantages of PDM?
7. When are rapid prototypes useful?
8. List three ways a CAD database can be used.
9. List five techniques you can use to enhance creativity.

CHAPTER EXERCISES

If necessary, refer to the appropriate section of the chapter to check your answers.

Exercise 1.1 Define the following terms as they apply to the design process: needs statement, concept, compromise solutions, prototype, layout sketch.

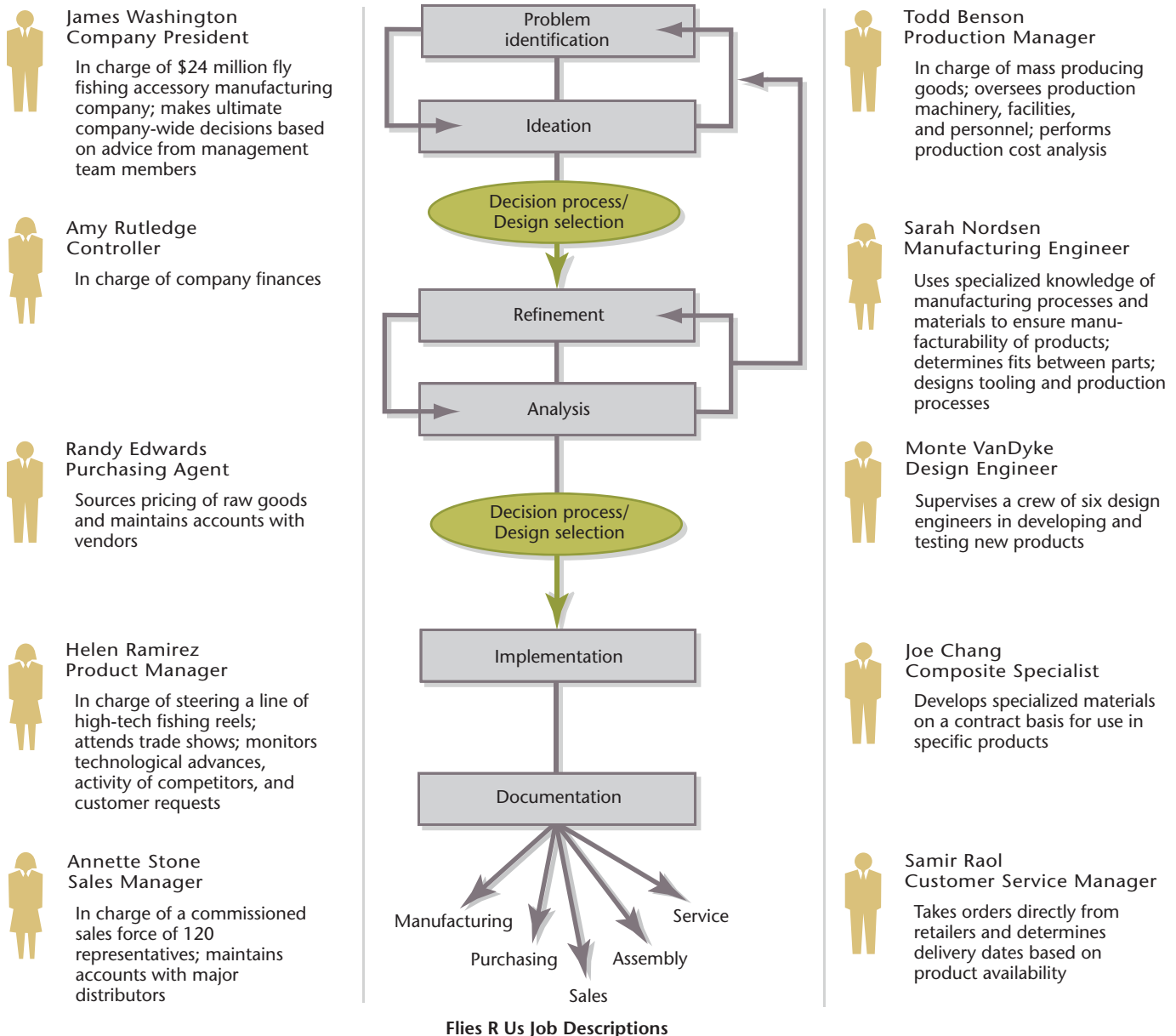
Exercise 1.2 Sketch a flowchart that depicts the design process as you understand it.

Exercise 1.3 The hypothetical “Flies R Us” fishing tackle company has contracted your services to restructure their design process and increase profitability. Read the job descriptions accompanying the “Flies R Us” design flowchart and decide at what stages of the design process each team member should be involved. Sketch out a flowchart

that illustrates the new reporting relationships for these individuals or groups.

Exercise 1.4 Using the “Flies R Us” flowchart as a guide, answer the following:

- What sort of information does Randy Edwards need before he can source raw goods for the manufacture of a new product?
- At what point should Todd Benson be involved in the design process, and why?
- What might happen if Samir Raol was out of the communication loop?
- Would Annette Stone be likely to see a prototype?
- Which team members do you think Monte VanDyke has on the speed-dial function of his office phone?



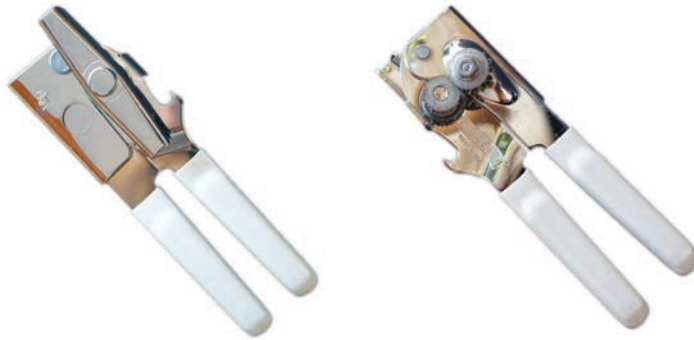
REVERSE ENGINEERING PROJECTS

Can Opener Project

In this ongoing project, you will reverse engineer an Amco Swing-A-Way 407WH Portable Can Opener. It is recommended you purchase a readily available and affordable product similar to this one, so you can make measurements directly when required. This effective and low-cost can opener seems simple in its familiarity, but it is clear when you begin to take one apart that considerable effort went into designing a product that is inexpensive, reliable, and easy to operate for most people.

Product Features

- Portable
- Lightweight
- Manually operated
- Comfortable to hold
- Durable construction
- Has a bottle opener
- Colored handles available
- Low maintenance
- Five-year warranty



Exercises for Chapter 1

RE 1.1 This is far from the only can opener on the market. Use the Web to research manual can opener designs. Find at least three can opener models that are different from the Amco Swing-A-Way. Make a list of the features of each of the three.

RE 1.2 Create a diagram for the can opener. How many distinct parts are used in its manufacture? Which parts can be grouped together and preassembled as a unit?

Exercises for Chapter 2

RE 2.1 Make a table listing the dimensions of the can opener parts. Do not worry about measurements for now. Give names to the dimensions, such as lower handle length, lower handle height, and hole diameter.

RE 2.2 Which dimensions in the list you created are critical to the function of the can opener? Identify in your list the dimensions that must match dimensions on other parts for the can opener to function. Which dimensions will not be very important to the can opener's function?

RE 2.3 To accurately reverse engineer the can opener, you will need to make measurements for the part features. Metrology is the science of making measurements. The digital caliper is one commonly used measurement tool. The accuracy of a measurement is dependent on several factors, including the following:

- the skill of the operator
- the temperature at which the measurements are taken
- how stationary the part is while being measured
- the accuracy of the measurement device

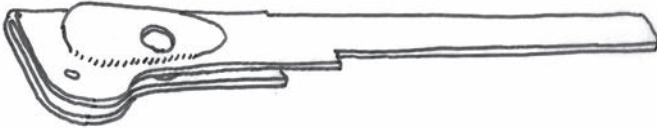
Measure the critical dimensions of the lower handle part. Make each measurement five times. Calculate the mean and standard deviation for each set of measurements. Determine what value you will use when modeling that dimension. Label the values on the sketch you drew for the lower handle.

RE 2.4 What factors influence the accuracy of the value you chose for the dimension?

Exercises for Chapter 3

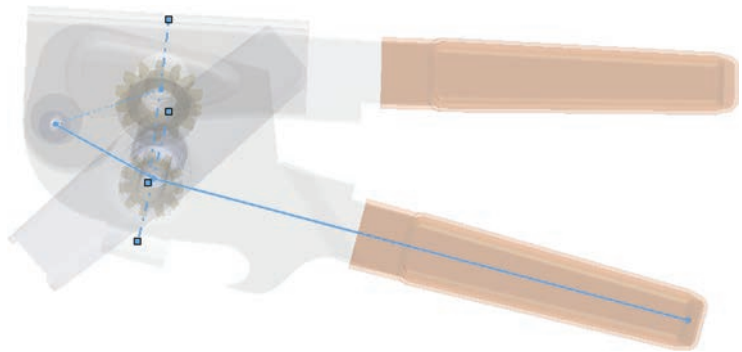
RE 3.1 Sketch a concept for an innovative can opener design.

RE 3.2 Sketch a pictorial drawing of the can opener part assigned by your instructor, such as the one shown here for the upper handle.



Exercises for Chapter 4

RE 4.1 Sketch and model a 3D wireframe drawing showing the main centerlines of the can opener parts. Use your 3D modeling software to create a single 3D sketch that shows the centerlines of each main part of the can opener, similar to that shown. Draw the 3D sketch lines only. You will model the can opener parts later.



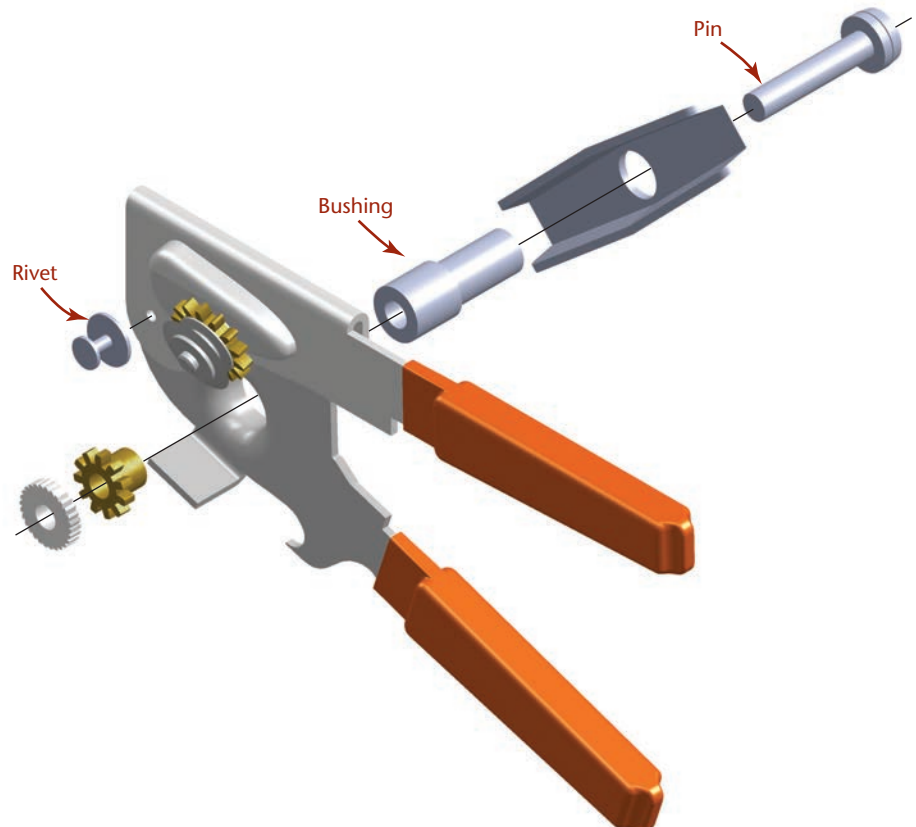
Draw the 3D sketch lines only.
You will model the can opener parts later.

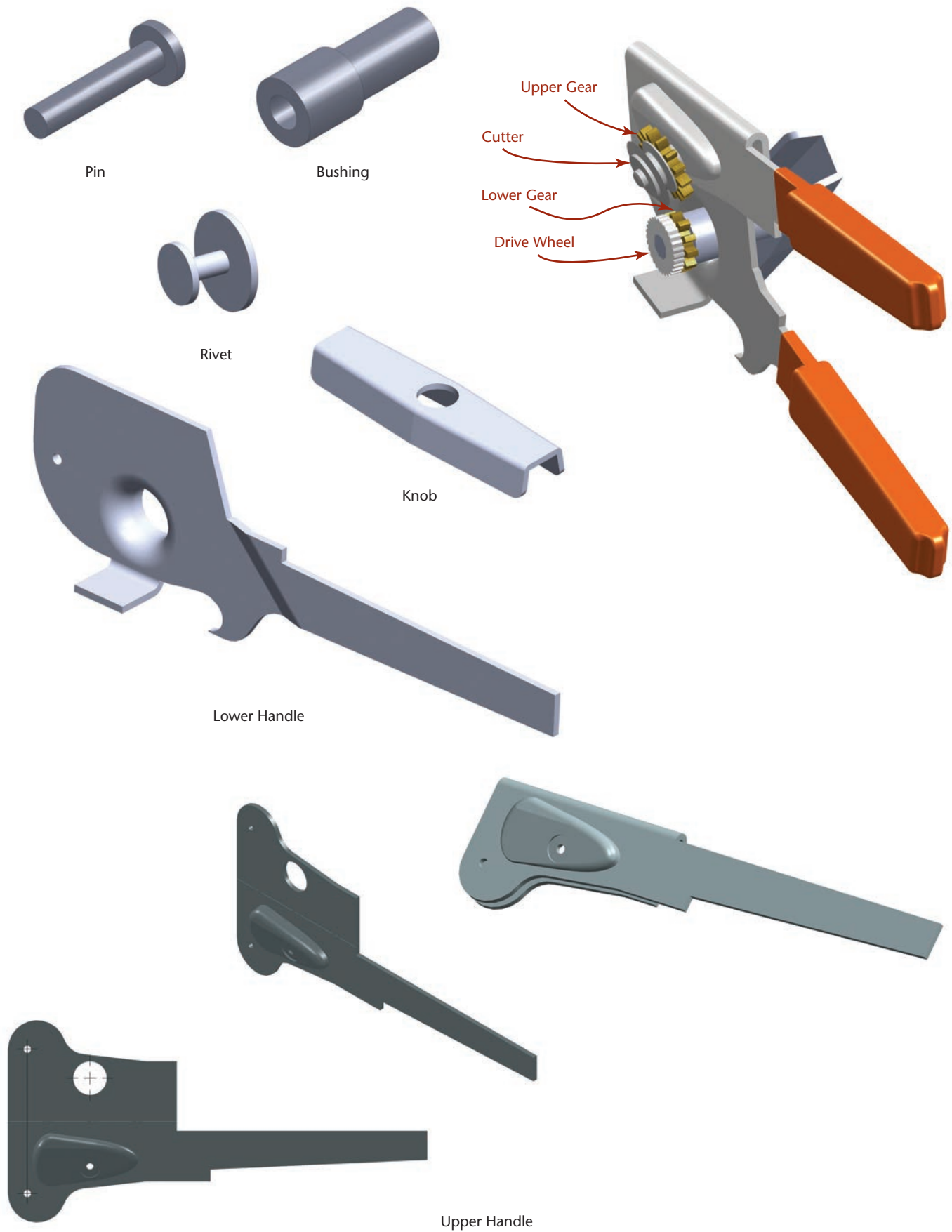
Exercises for Chapter 5

Begin modeling the can opener's parts, with emphasis on the dimensional and geometric constraints that are appropriate.

RE 5.1 Model the pin, rivet, and bushing for the can opener using your 3D modeling software.

RE 5.2 Create constraint-based part models of the top handle, bottom handle, cutter, knob, and gears (see next page). Consider part fit, and make your model so that the major dimensions can flexibly change. Review the dimension names you assigned for the lower handle, and revise them if needed. Assign dimension names to these parts wherever it makes sense for the can opener assembly.

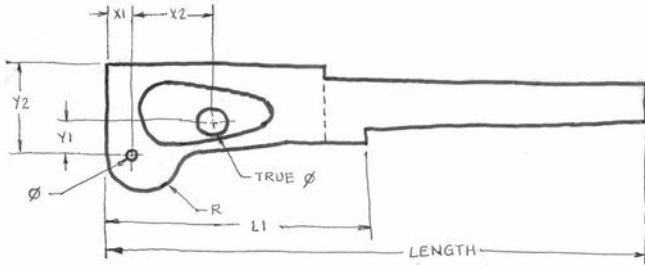




Exercises for Chapter 6

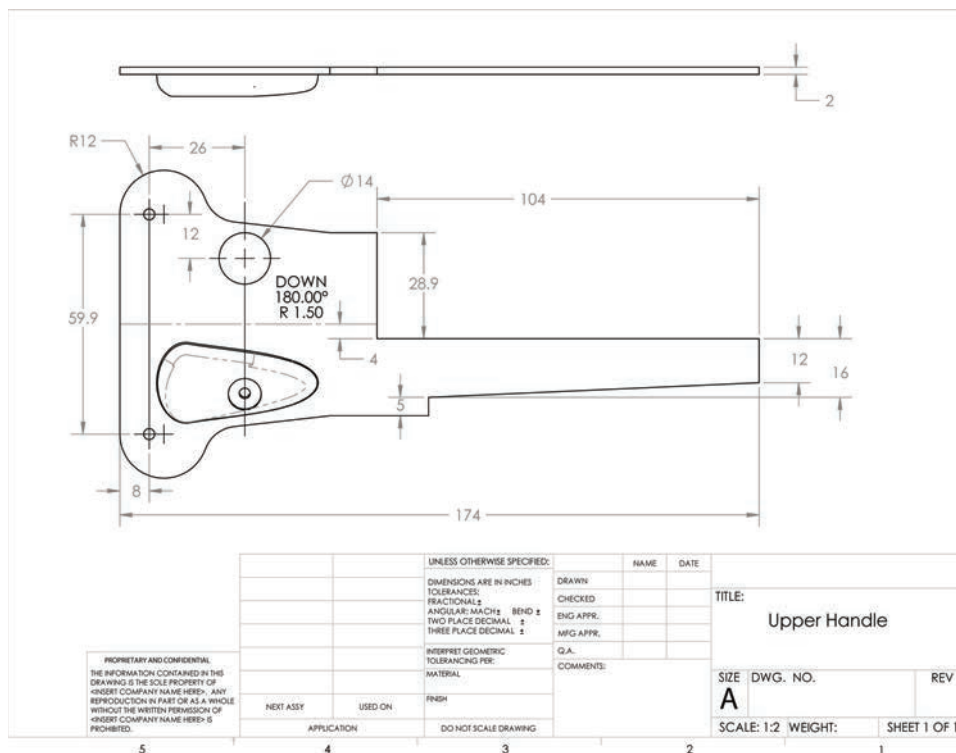
RE 6.1 Create an orthographic view sketch of the lower handle for the can opener similar to that shown here for the upper handle.

RE 6.2 Name the dimensions. Add dimensions to your orthographic sketch of the handle.



Exercises for Chapter 7

RE 7.1 Create a multiview part drawing for the upper handle. The upper handle is a sheet metal part that is bent to form its shape. Consider this manufacturing process as you create the dimensioned drawing. An example drawing is shown below with dimensions added. Complete Chapter 11 before adding the dimensions to your drawing.



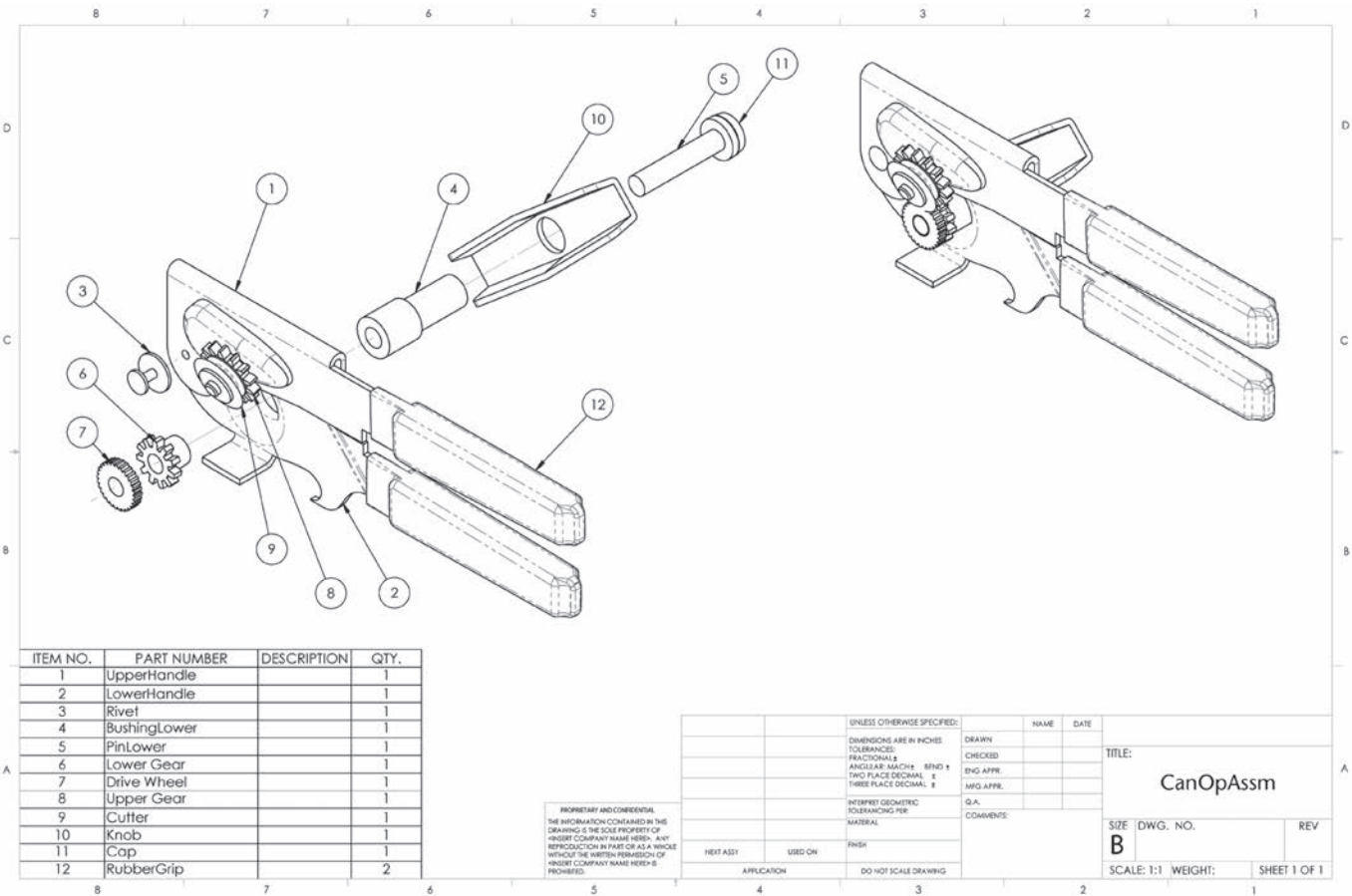
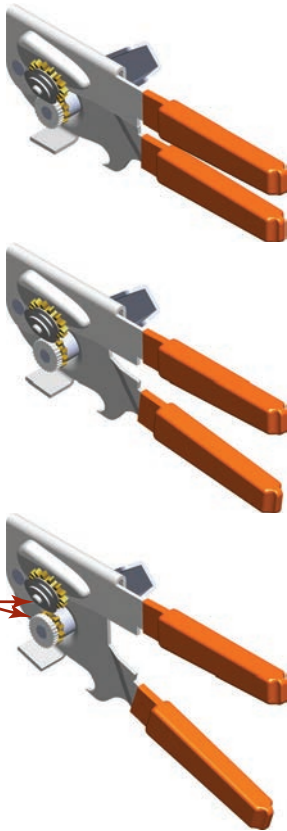
Exercises for Chapter 8

RE 8.1 Create an orthographic drawing for the bushing part. Show the front view in section.



Exercises for Chapter 10

- RE 10.1** Use the parts you created to create an assembly of the can opener.
- RE 10.2** Use the capabilities of your software to animate the motion of the can opener. What is the minimum distance between the cutting wheel and the lower gear?
- RE 10.3** Assign appropriate materials to each of the parts and determine the weight of the can opener from your assembly model. Weigh the actual can opener and compare the two. Are they close?
- RE 10.4** Next, determine the center of gravity (COG) for the can opener assembly. Do you think the can opener will feel balanced when you are operating it based on the location of its center of gravity?
- RE 10.5** Change the handle parts to make a new pistol-grip design. Analyze the part again to determine the mass properties and COG.
- RE 10.6** Experiment with different options by making changes to your can opener parts and updating the assembly. List two options for changing the design if you are required to reduce the overall weight by 20%.
- RE 10.7** Create an assembly drawing for your can opener, similar to the one shown. Use an exploded view and a fully assembled view. Add balloon numbers and a parts list.



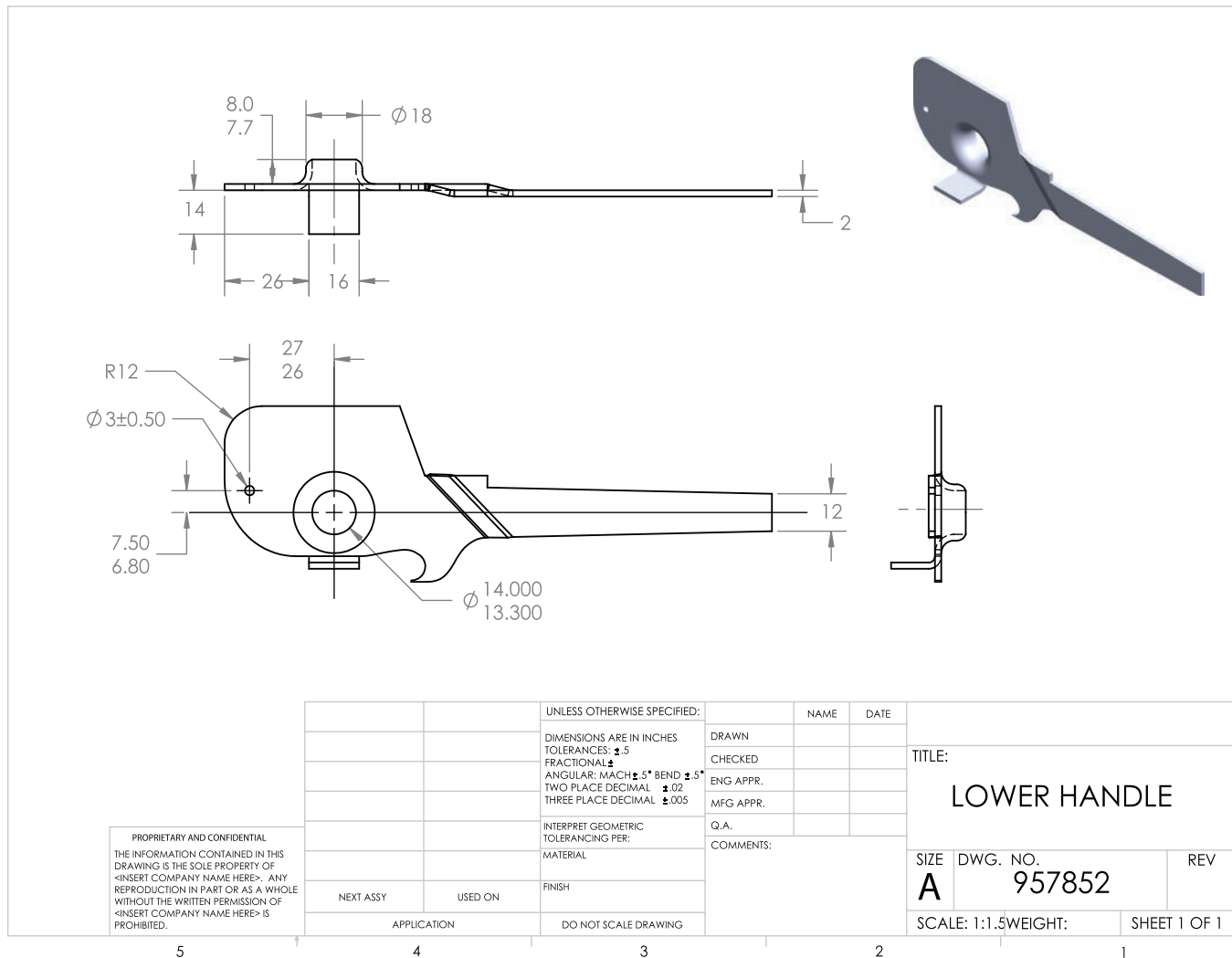
Exercises for Chapter 11

RE 11.1 Add dimensions to the multiview orthographic part drawings you created earlier.

Exercises for Chapter 12

RE 12.1 Assign tolerances to the part model of the lower handle of your can opener. Then, add tolerances to your drawing of the lower handle part. Your drawing should look similar to the example below.

RE 12.2 What parts of the can opener have fits that are critical to the can opener function? This can opener sells for around \$5.00. Consider and describe some of the effects on the price of the can opener if highly-precise fits are required.



Locking Pliers Project

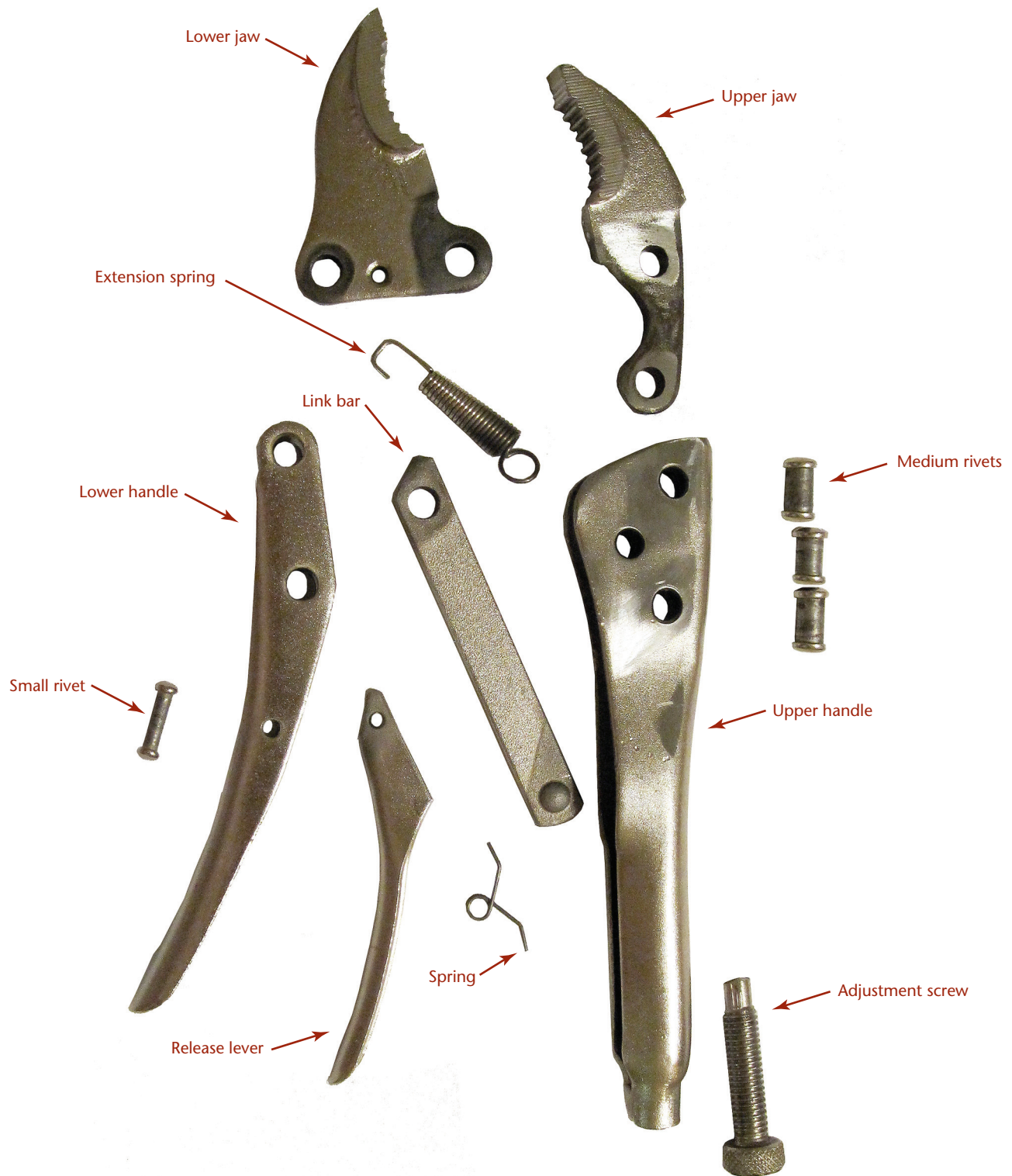
RE Locking Pliers 1. Use stages similar to those for the can opener to create sketches, models, and drawings for the locking pliers shown below.

RE Locking Pliers 2. Create solid models for the locking pliers parts (shown on the next page). Use dial calipers, a micrometer, or machinist scales to make measurements from your pliers. Consider the mating features and alignment that make the locking plier mechanism function. If assigned, make an assembly skeleton as represented below and assemble your completed parts onto it.

RE Locking Pliers 3. Ensure your parts will fit and function in the assembly. Perform a fit study on the completed assembly to identify any interference issues in your model.

RE Locking Pliers 4. Make part drawings as assigned by your instructor.





CHAPTER TWO

LAYOUTS AND LETTERING

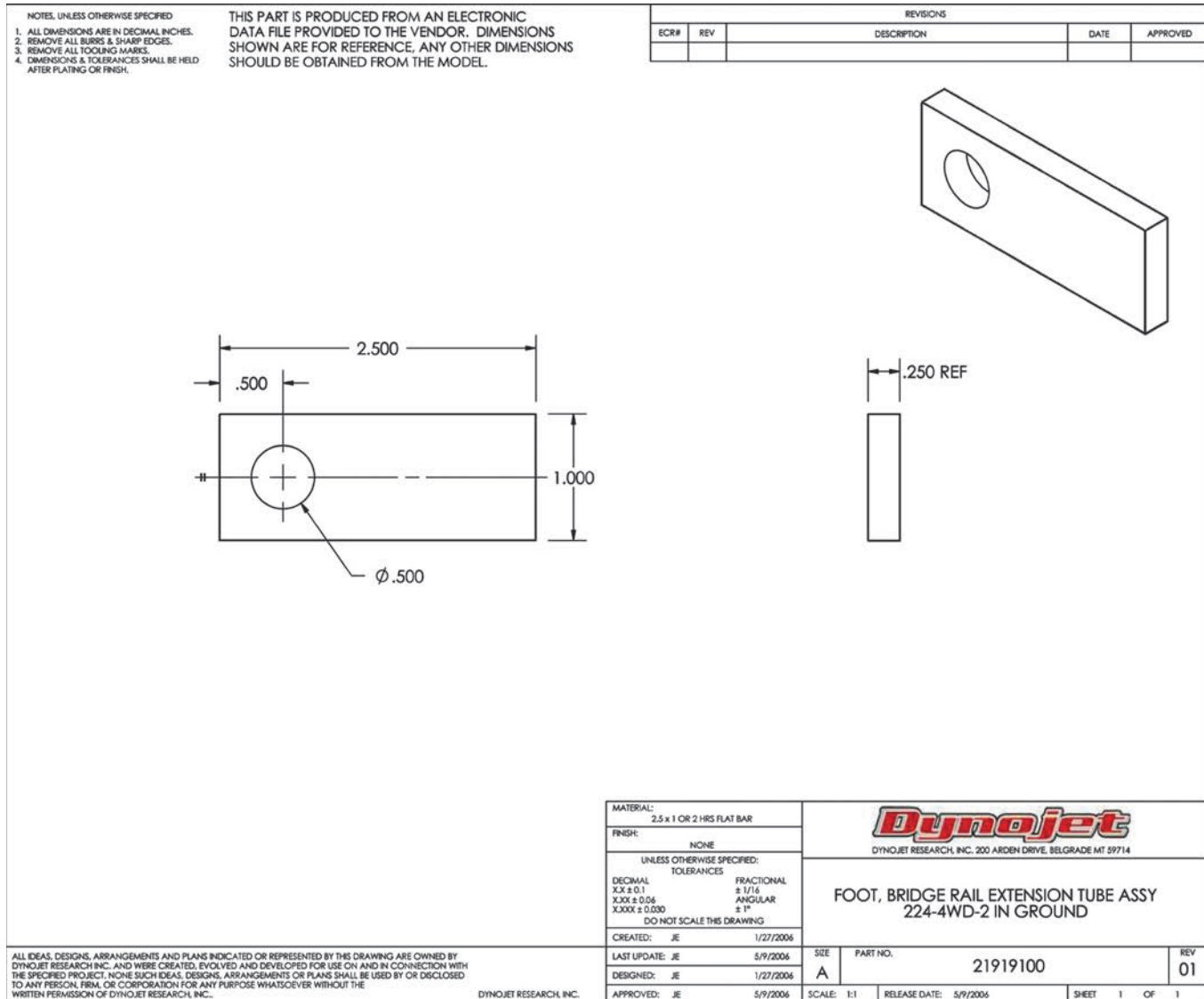
OBJECTIVES

After studying the material in this chapter, you should be able to:

1. Identify six types of technical drawings based on the projection system they use.
2. Identify the line patterns used in technical drawings and describe how they are used.
3. Identify standard drawing media and sheet sizes.
4. Label drawing scale information.
5. Add lettering to a sketch.
6. Fill in a standard title block with the appropriate information.
7. Lay out a drawing sheet.

Refer to the following standards:

- *ANSI/ASME Y14.100 Engineering Drawing Practices*
- *ANSI/ASME Y14.2 Line Conventions and Lettering*
- *ANSI/ASME Y14.1 Drawing Sheet Size and Format*

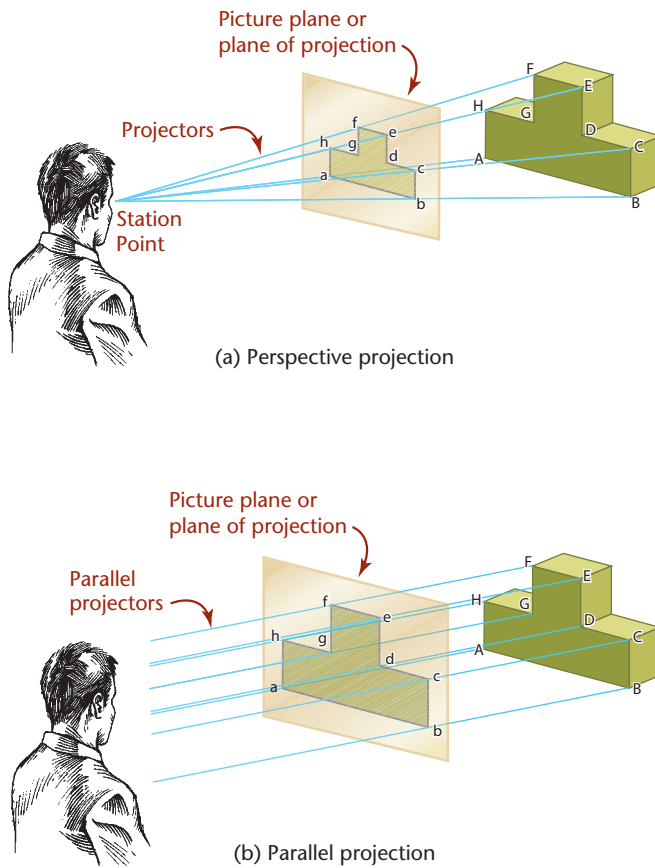


Layout with Title Block of a Small Part at Scale 1:1 (Courtesy Dynojet Research, Inc.)

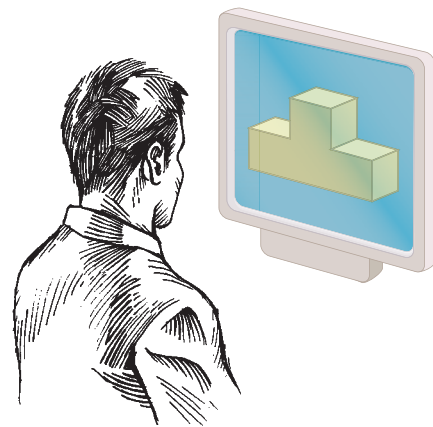
OVERVIEW

Two-dimensional technical drawings, whether they are sketched by hand, drawn using instruments, drawn using a CAD program, or generated from 3D solid models, follow certain rules so that they can be correctly interpreted. Unlike artistic drawings, which communicate self-expression and emotional content, technical drawings communicate size, shape, and feature information about a product, system, or device. To clearly describe this information, technical drawings adhere closely to formal standards.

These formal standards include systems of projection for developing and understanding drawing views. They also include an “alphabet of lines,” in which each line of the drawing represents certain information. Lettering is also standardized, to make drawings quick to create and easy to read and reproduce. Standard sheet sizes for drawings include a title block that provides important information such as the drawing name, company information, scale, revision numbers, and approvals for release of the drawing.



2.1 The Concept of Projection



2.2 A View of a 3D Object "Projected" onto a Computer Monitor

UNDERSTANDING PROJECTION

Behind every 2D drawing of an object is a space relationship involving the object and three "imagined" things:

1. The observer's eye, or **station point**
2. The **plane of projection**
3. The **projectors** (also called *visual rays* or *lines of sight*)

Figure 2.1 shows two drawings of a shape projected onto a plane as viewed by an observer, whose eye represents the station point. The lines projecting from the corners (or vertices) of the object are the imagined lines, called projectors.

To understand projection, imagine that the drawing is produced by points, called **piercing points**, where the projectors would pierce the projection plane. The drawing may be a two-dimensional representation on a sheet of paper, or it may be a two-dimensional representation shown on your computer screen, as shown in Figure 2.2, but the basic principles are the same. One reason 2D projection skills remain relevant, even with the advent of 3D modeling, is that computer monitors still display a 2D view on their flat screens.

Types of Projection

There are two main types of projection: perspective and parallel. These are broken down into subtypes, as shown in Figure 2.3.

In **perspective projections**, the projectors come together at the station point to form a cone, as in Figure 2.1a. Perspective drawings represent objects as we see them or as they would appear in a photograph.

In **parallel projections**, the projectors are parallel, as shown in Figure 2.1b.

Orthographic projections are one type of parallel projection. In orthographic (meaning right-angle) projections, the parallel projectors are perpendicular to the plane of projection. Because orthographic projections show objects in a way that their features can be represented at true size or scaled at a proportion of true size, they are especially useful in specifying the dimensions needed in technical applications.

If the projectors are parallel to each other, but are at an angle *other than 90°* to the plane of projection, the result is called an **oblique projection**.

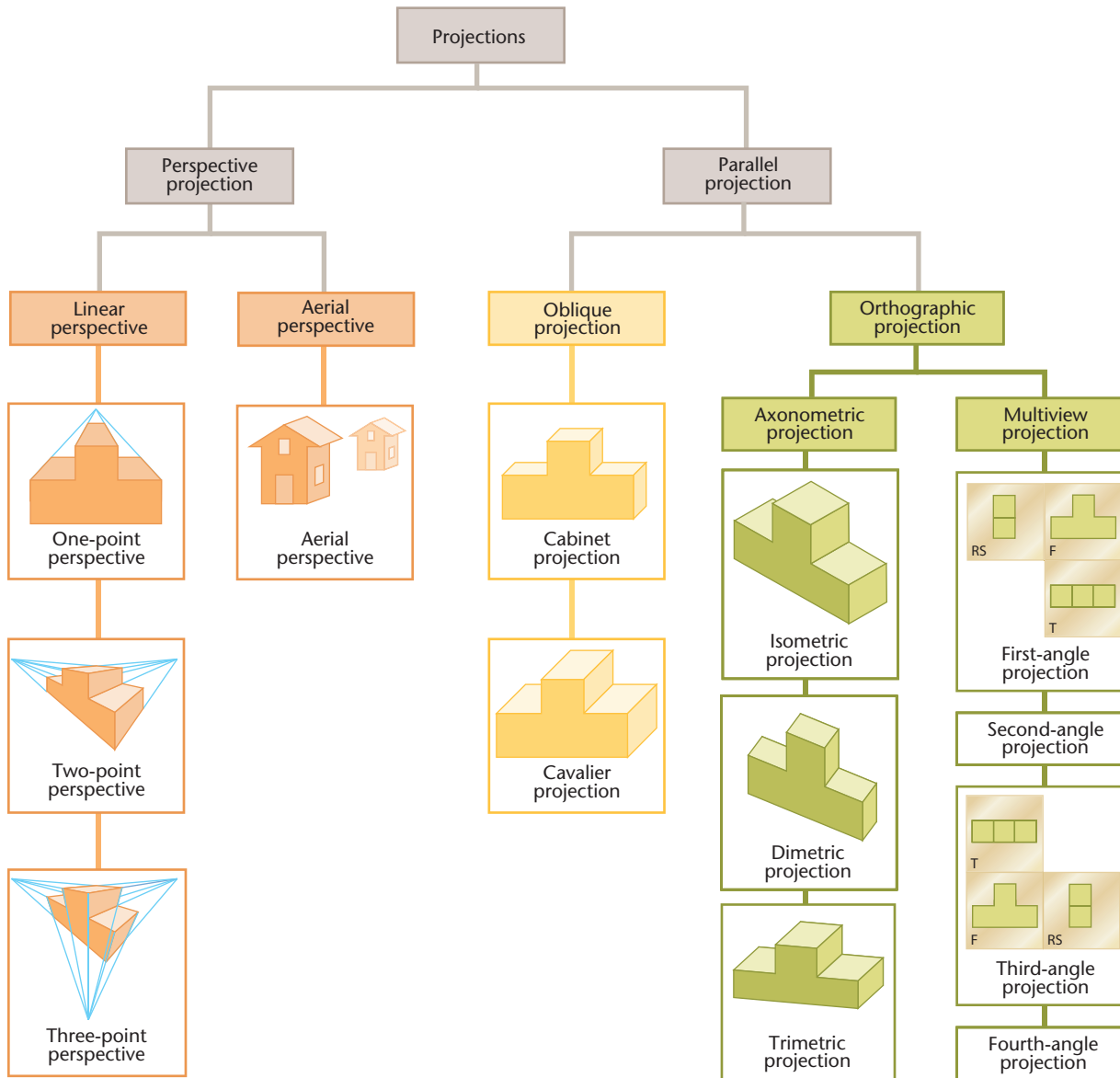
Technical drawings of 3D objects usually use one of four standard types of projection, shown in Figure 2.3:

- Multiview
- Axonometric (includes isometric)
- Oblique
- Perspective

Multiview projection shows one or more necessary views. Either of two systems is used to arrange the views in a multiview drawing: third-angle or first-angle. You will learn about multiview projection in Chapter 6.

Axonometric, oblique, and perspective sketches are methods of showing the object pictorially in a single view. They will be discussed in Chapter 3.

The main types of projection are listed in Table 2.1.



2.3 Classification of Projections

Table 2.1 Classification by Projectors

Class of Projection	Distance from Observer to Plane of Projection	Direction of Projectors
Perspective	Finite	Radiating from station point
Parallel	Infinite	Parallel to each other
Oblique	Infinite	Parallel to each other and oblique to plane of projection
Orthographic	Infinite	Perpendicular to plane of projection
Axonometric	Infinite	Perpendicular to plane of projection
Multiview	Infinite	Perpendicular to plane of projection

Drawing Vocabulary

Drawing lines, lettering, measurement systems, scale, sheet sizes, and title blocks are presented in this chapter.

Drawing Lines Projected drawing views use specific line patterns to represent object features. For example, when showing a 3D object, some lines represent the edges of surfaces that are hidden from that viewing direction. These hidden lines have a dashed line pattern to help the reader understand the drawing. Another type of line indicates the location of the center of a symmetrical feature, such as a hole. Familiarity with the types of lines used in technical drawings helps you read drawings and create drawings that others can easily understand.

Lettering The shapes of letters that are easy to read and write are described as part of drawing standards. Often, free-hand sketching is used early in the design process to present ideas. Showing notes and information legibly helps present

your ideas to others clearly. Good lettering often makes or breaks a sketch.

Measurement Systems Two measurement systems are used for technical drawings: the metric system and U.S. customary units. It is important to be familiar with both measurement systems to create and read drawings that are used worldwide.

Scale Obviously, a large item (a house or bridge for example) cannot be shown full size on a paper sheet. To clearly convey important information about particularly large or small objects, you need to select an appropriate sheet size and show drawings to scale (proportionately smaller or larger than the actual size). Standard lettering sizes for drawings depend on the sheet size.

Title Blocks Company information, the drawing scale, sheet size, and other information is included in a standard title block located in the lower right corner of the drawing to make it easy to locate these important details on every drawing layout.

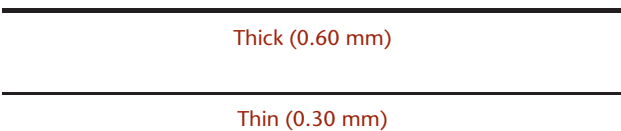
2.1 ALPHABET OF LINES

The meaning of each line on a technical drawing is indicated by its width (thick or thin) and its particular line style. The person who reads the drawing will depend on these line styles to know if a line is visible or hidden, if it represents a center axis, or if it conveys dimension information.

To make your drawings easy to read, make the contrast between thick and thin lines distinct. **Thick lines** (0.6 mm) should be twice the width of **thin lines** (0.3 mm), as shown in Figure 2.4. The line gage in Figure 2.5 shows various widths.

Figure 2.6 shows freehand line technique. You may find it helpful to use 1/8" graph paper at first to get a feel for the length of dashes used in hidden lines and centerlines. Soon you will be able to estimate the lengths by eye.

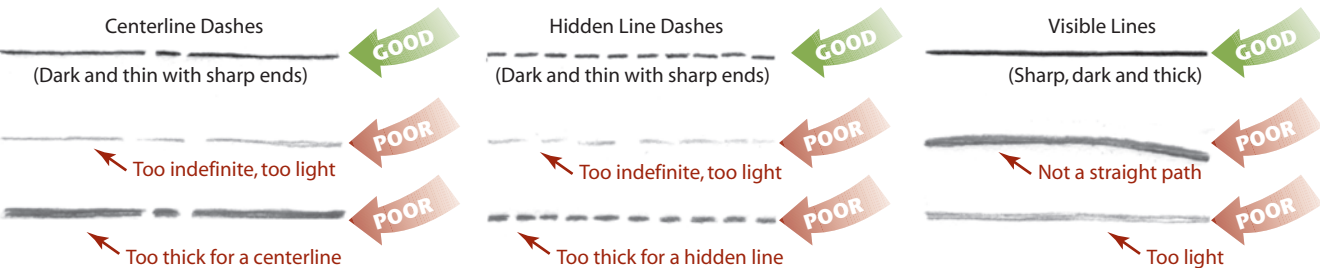
Figure 2.7 illustrates line styles for technical drawings. All lines (except construction lines) must be sharp and dark. For visible, cutting-plane, and short-break lines use thick lines. Thin drawing lines should be just as sharp and black, but only half the thickness of thick lines. Construction lines and lettering guidelines should be thin and light so that they can barely be seen at arm's length and need not be erased. All lines should be uniform in width and darkness. Ideal lengths of the dashes used to form the line patterns are also shown in Figure 2.7.



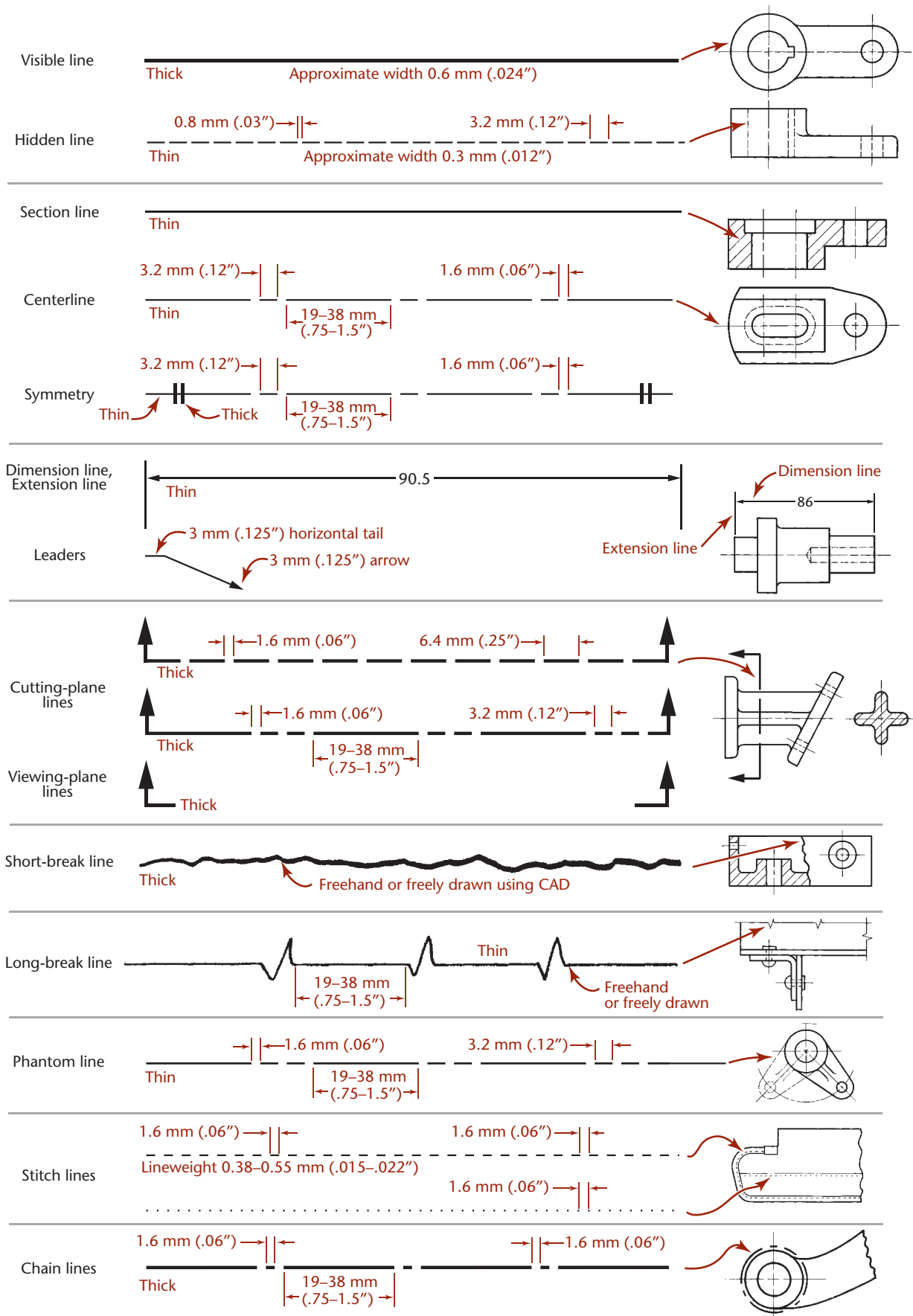
2.4 Thick and Thin Drawing Lines

.007"	.010"	.012"	.014"	.020"	.024"
.18 mm	.25 mm	.30 mm	.35 mm	.50 mm	.60 mm
.028"	.031"	.039"	.047"	.055"	.079"
.70 mm	.80 mm	1.00 mm	1.20 mm	1.40 mm	2.00 mm

2.5 Line Gage



2.6 Good and Poor Freehand Line Technique



2.7 Alphabet of Lines (Full Size)

2.2 FREEHAND LINES

The main difference between an instrument or CAD drawing and a freehand sketch is in the appearance of the lines. A good *freehand line* is not expected to be precisely straight or exactly

uniform, as is a CAD or instrument-drawn line. Freehand lines show freedom and variety. Freehand *construction lines* are very light, rough lines. All other lines should be dark and clean.

2.3 MEASUREMENT SYSTEMS

When you create a technical drawing, the item you show will be manufactured or constructed using a particular system of measurement, which you indicate on the drawing. The metric system is the world standard used for measuring lengths.

U.S. Customary Units

U.S. customary units based on inch-foot and yard measurements (where there are 3 feet to the yard, and 12 inches to the foot; a yard equals exactly 0.9144 meter) continue to be used in the United States. Drawings may use either measurement system and still follow ANSI/ASME drawing standards as long as the system of measurement is stated clearly on the drawing. Figures 2.8 and 2.9 show the same part dimensioned with the two different measurement systems.

The Metric System

Today's metric system is the International System of Units, commonly referred to as SI (from the French name, le Système International d'Unités). It was established in 1960 by international agreement and is now the international standard of measurement, with all countries in the world adopting it, although some continue using traditional U.S. units to a greater or lesser degree.

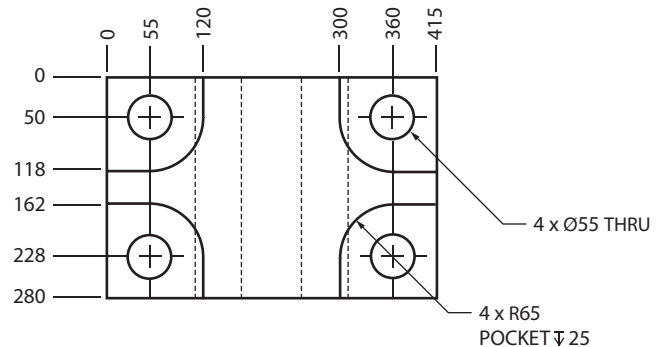
The meter was established by the French in 1791 as the length of one ten-millionth of the distance from the Earth's equator to the pole along the meridian that, coincidentally, passes through Paris. Since then the definition has been updated to be the distance that light travels in a vacuum in $1/299,792,458$ of a second. A meter equals 39.37 inches or approximately 1.1 yards.

The metric system for linear measurement is a decimal system similar to the U.S. system of counting money. For example,

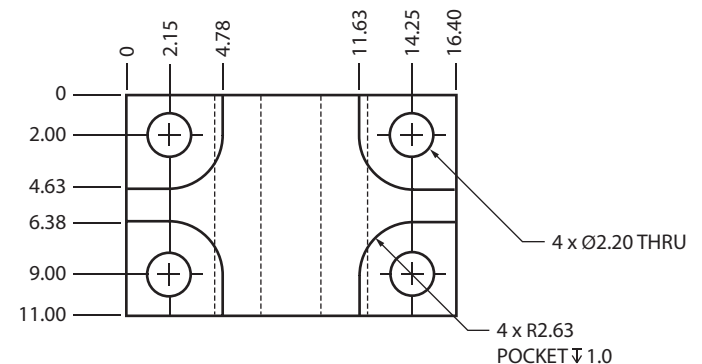
- 1 mm = 1 millimeter (1/1000 of a meter)
- 1 cm = 1 centimeter (1/100 of a meter)
= 10 mm
- 1 dm = 1 decimeter (1/10 of a meter)
= 10 cm = 100 mm
- 1 m = 1 meter
= 100 cm = 1000 mm
- 1 km = 1 kilometer = 1000 m
= 100,000 cm = 1,000,000 mm

The primary unit of measurement for engineering drawings and design in the mechanical industries is the millimeter (mm). Secondary units of measure are the meter (m) and the kilometer (km). The centimeter (cm) and the decimeter (dm) are rarely used on drawings.

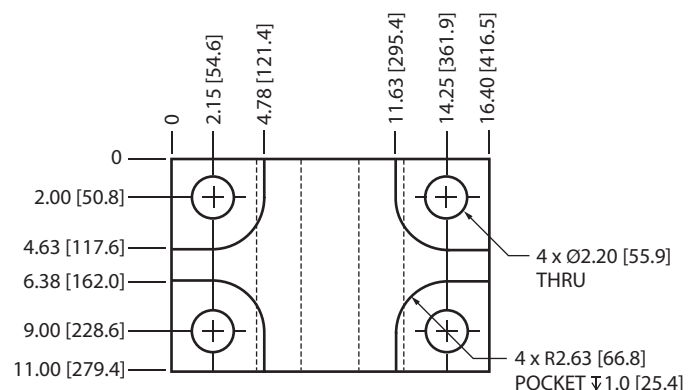
Some industries have used a dual dimensioning system of millimeters and inches on drawings. However, this practice can



2.8 A Drawing Dimensioned Using Metric Units



2.9 A Drawing Dimensioned Using U.S. Customary Units



2.10 A Dual-Dimensioned Drawing Using U.S. Customary Units as the Primary Units

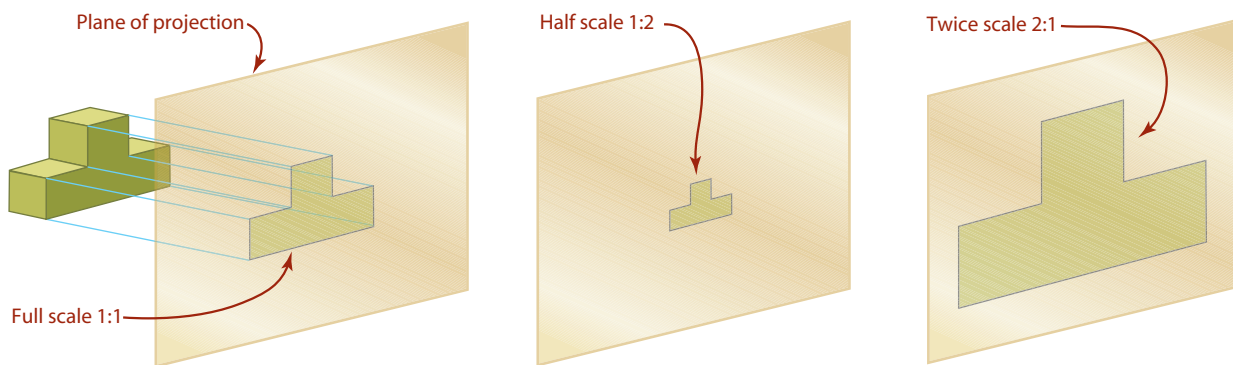
be confusing because the sizes displayed in the two systems may contain rounding errors. If two systems are shown, the primary units are used for all manufacturing measurements, and the secondary system units (shown in parentheses or brackets) are for general information purposes only. Figure 2.10 shows a drawing using dual dimensioning. Most large manufacturers use all metric dimensions on the drawing for ease and consistency.

Dimensions that are given in U.S. customary units (inches and feet, either decimal or fractional) can be converted easily to metric values. In standard practice, the ratio 1 in. = 25.4 mm is used. Decimal-equivalents tables can be found inside the back cover, and conversion tables are also given in Appendix 30. Many handy unit conversion sites are also available on the Web at sites such as www.onlineconversion.com.

2.4 DRAWING SCALE

Unlike a computer drawing (where an object is drawn actual size so that the information stored in the computer file is accurate), a printed or paper drawing may represent the object at its actual size (full size) or may be larger or smaller than the object, depending on the size of sheet used. **Drawing scale** is the reduction or enlargement of the drawn object relative to the real object (Figure 2.11).

Scale is stated as a ratio of the number of drawing units to the number of actual units. For example, a machine part may be shown on a sheet at half its actual size, a scale of 1:2; a building may be drawn 1/48 of its size, a scale of 1:48 (or in U.S. customary units, $1/4'' = 1'$); a map may be drawn 1/1200 actual size, a scale of $1'' = 100'$ or 1:1200; or a printed circuit board may be drawn four times its size, a scale of 4:1.



2.11 Reduced and Enlarged Scale. Many drawings must be shown at reduced scale for the object to fit on the paper.

2.5 SPECIFYING THE SCALE ON A DRAWING

There are several acceptable methods of noting scale on the drawing, but all of them show the relationship of the size of the object as drawn to the size of the actual object. For a part that is shown on the paper at half its actual size, the scale is listed in one of these three ways:

SCALE: 1:2
SCALE: 1/2
SCALE: .5

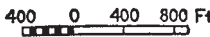
For machine drawings, the scale indicates the ratio of the size of the drawn object to its actual size, regardless of the unit of measurement used. Expansion or enlargement scales are given as 2:1, 4:1; 5:1, 10:1, and so on. Figure 2.11 illustrates how the actual object relates to a drawing at half size and how that might be noted in the title block of the drawing. Figure 2.12 shows the scale for a 1 to 24 reduction noted in a title block.

Architectural drawings in the United States typically list the scale based on the number of fractions of an inch on the drawing that represent one foot on the actual object. For example, SCALE: $1/8'' = 1'$.

The various scale calibrations available on the metric scale and the engineers' scale provide almost unlimited scale

ratios. Preferred metric scale ratios are 1:1; 1:2; 1:5, 1:10, 1:20, 1:50, 1:100, and 1:200.

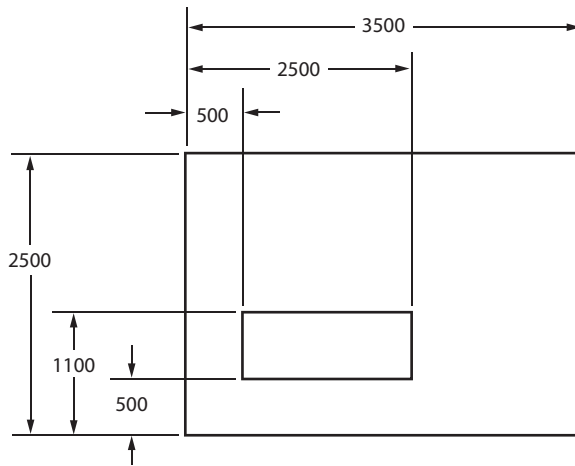
Map scales are indicated in terms of proportions such as Scale 1:62500, fractions such as Scale 1/62500, or graphically, such as:



Dynojet DYNJET RESEARCH, INC. 200 ARDEN DRIVE, BELGRADE MT 59714			
224/224-4WD IN GROUND DYNO Standard Wheel Base 88 X 130			
DATE: 7/19/2006	SIZE: B	PART NO.: 81224103	REV: 04
2/23/2007	SCALE: 1:24	RELEASE DATE: 2/23/2007	SHEET 1 OF 2

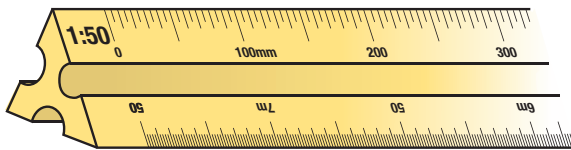
2.12 List the predominant drawing scale in the title block. (Courtesy of Dynojet Research, Inc.)

USING A MEASURING SCALE TO LAY OUT A ONE-FIFTIETH SIZE METRIC DRAWING

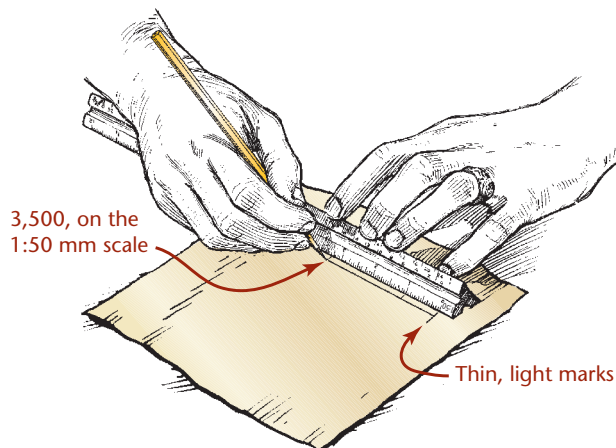


1 Determine the full-size measurements of the object you will draw. This example will lay out a 3500 × 2500-mm flat plate with a rectangular slot in it. A picture of the part to be drawn with dimensions representing its actual size is shown above.

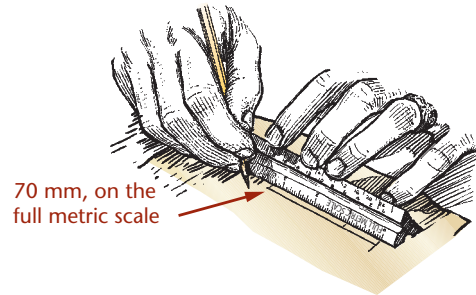
2 Many measuring scales are available that use standard drawing scales, like this 1:50 ratio metric one.



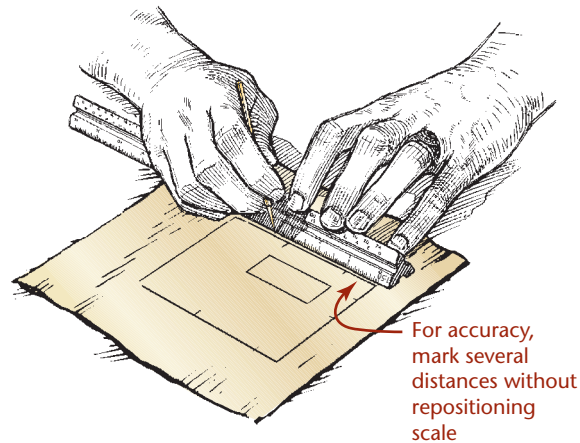
3 Starting from the 0 end of the 1:50 scale, use a sharp pencil to make a thin, light, short line to mark off the length of the 3500-mm line. To make accurate measurements, be sure to place the scale edge parallel to the line you are measuring on the drawing, and make your dashes at right angles to the scale, at the correct graduation mark, as shown.



4 Check the length of your scaled line by calculating how many millimeters the length should be, then measuring the line you have drawn with a full-scale metric scale. In this case the 3500-mm length should be 70 mm when shown at 1:50 scale.



5 Continue to lay out the remaining lengths. Even slight errors in measurements when using a scale may accumulate to produce a significant error, so work carefully.

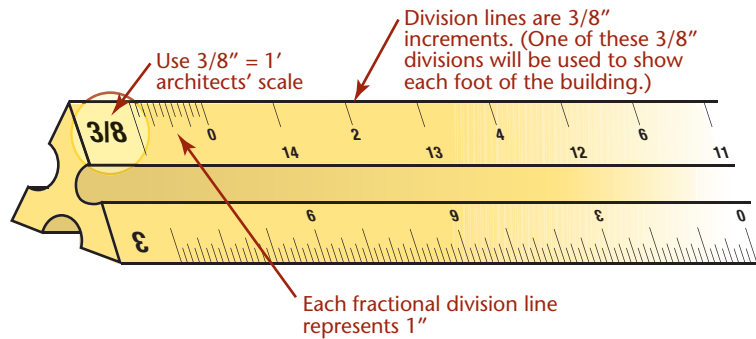
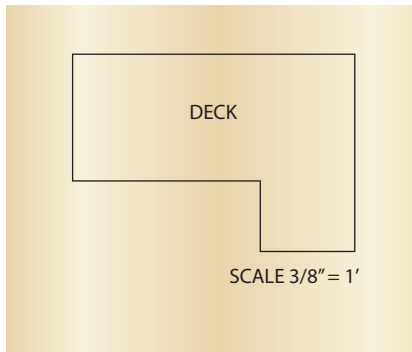


To avoid cumulative errors, instead of setting off distances individually by moving the scale to a new position each time, position the scale once and mark all of the distances by adding each successive measurement to the preceding one.

This is useful in dimensioning drawings, too. Keep in mind that providing dimensions from one end to each successive location (say, in the case of building a wall) makes it easier for the worker to lay it out quickly and accurately.

For more about the use of measuring scales, see Appendix 51.

MEASURING WITH WITH AN ARCHITECTS' SCALE



1 To make measurements with an architects' scale, first determine which scale to use by reading the scale noted in the title block or noted below the view. In the example above, $\frac{3}{8}$ inch = 1 foot.

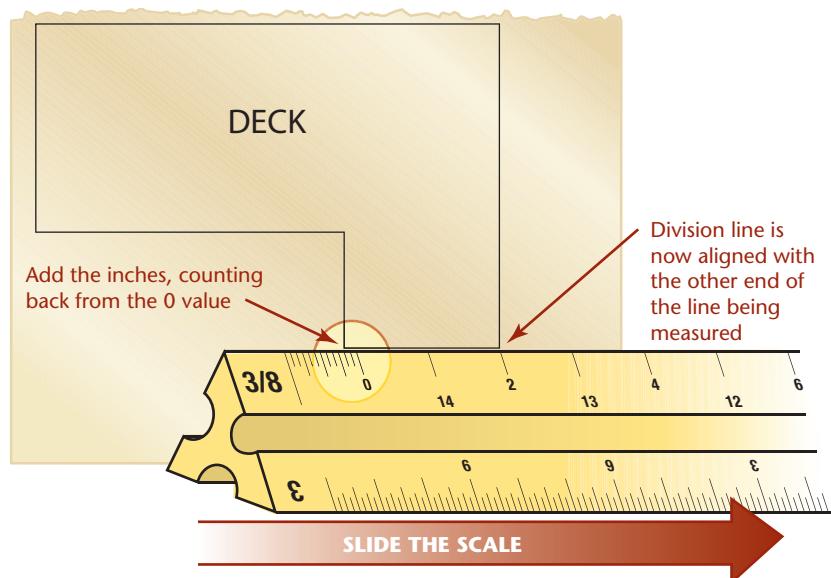
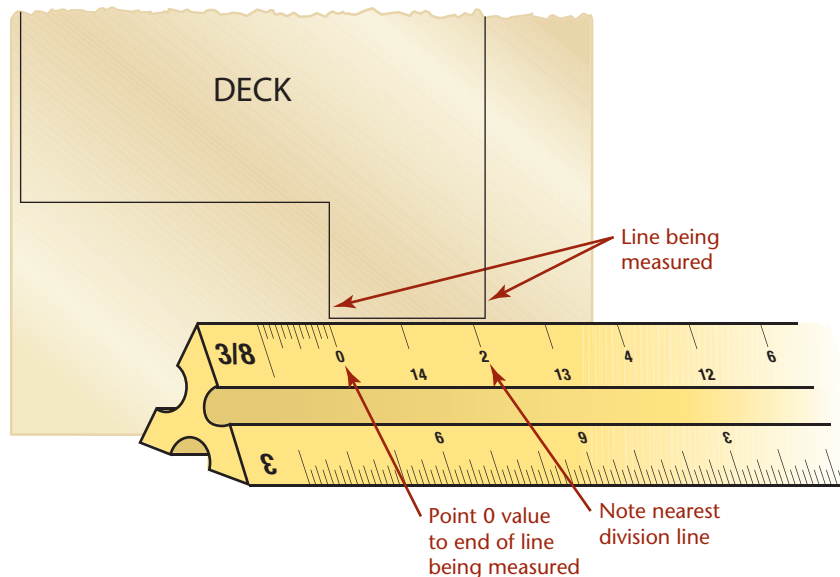
2 Position the scale so that the 0 value is aligned with the left end of the line being measured and note the division mark nearest to the line's right end (in this case, 2).

3 Slide the scale to the right so that the closest whole division you noted in Step 2 lines up with the right end of the line you are measuring.

A fractional portion of the line you are measuring now extends on the left, past the scale's 0 mark.

Counting toward the left, note how many fractional division marks are between zero and the left end of the line. (In this example, there are two.)

Add the fractional value to the whole value that you noted in Step 2. In this example, you noted two whole division lines, plus two fractional division lines, so the length of the line is 2'-2" at actual size.



**On architects' scales, there are 12 fractional divisions because there are 12 inches per foot.*

A B C D E F G H
a b c d e f g h

Sans serif lettering has no serifs,
or spurs, at the ends of the strokes

A B C D E F G H
a b c d e f g h

Serif letters have serifs and are accented
by thick and thin lineweights

A B C D E F G H
a b c d e f g h

Roman refers to the upright form of letters,
but it was used to mean letters with serifs

A B C D E F G H
a b c d e f g h

Italic letters are slanted,
whether serif or sans serif

2.13 Distinctions Among Letter Forms

AUTOCAD TXT FONT
ROMAN SIMPLEX
ROMAN DUPLEX
BASKERVILLE
TIMES NEW ROMAN
PLAYBILL
ARIAL
LETTER GOTHIC

2.14 Examples of Lettering Using CAD

2.6 LETTERING

Lettered text is often necessary to completely describe an object or to provide detailed specifications. Lettering should be legible, be easy to create, and use styles acceptable for traditional drawing and CAD drawing.

Engineering drawings use single-stroke sans serif letters because they are highly legible and quick to draw. (*Sans serif* means without serifs, or spurs.) The sans serif letters used for drawings are also referred to as Gothic. (Serif letters are sometimes called Roman, but today that term is commonly used for the upright form of the letters.) A *font* is the name for a set of letters with the same style. Figure 2.13 shows the distinctions among *Roman*, *italic*, *serif*, and *sans serif* fonts.

Lettering is a standard feature available in computer graphics programs. With CAD software, you can add titles, notes, and dimensioning information to a drawing. Several fonts and a variety of sizes may be selected. When modifications are required, it is easy to make lettering changes on the drawing by editing existing text.

Freehand lettering ability has little relationship to writing ability. You can learn to letter neatly even if you have terrible handwriting. There are three necessary aspects of learning to letter:

- Knowing the proportions and forms of the letters (to make good letters, you need to have a clear mental image of their correct shape)
- Spacing of letters and words for legibility
- Practice

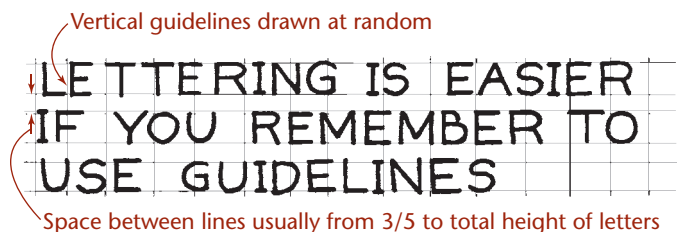
2.7 LETTERING STANDARDS

Most hand-drawn notes use lettering about 3 mm (1/8") high. Light horizontal *guidelines* are useful for producing consistent letter heights. CAD notes are set using the keyboard and sized to be in the range of 3 mm (1/8") tall according to the plotted size of the drawing. Lettering heights vary with the size of the sheet and the intended use of the drawing.

CAD drawings typically use a Gothic (sans serif) lettering style but often use a Roman (serif) style for titles. When adding lettering to a CAD drawing, a good rule of thumb is not to use more than two fonts within the same drawing. See Figure 2.14 for a sample of the fonts available using CAD. You may want to use one font for the titles and a different font for notes and other text. Keep in mind that if you open a drawing created with software such as AutoCAD, you must have the fonts available that were used in the drawing, otherwise the software will have to substitute different fonts. This can be a problem because the horizontal spacing may be different and the text will no longer fit correctly. It may be tempting to use many different fonts in a drawing because of the wide variety available, but this tends to look distracting on the drawing. Drawings that use too many lettering styles and sizes have been jokingly referred to as having a “ransom note” lettering style.

2.8 USING GUIDELINES FOR HAND LETTERING

Use extremely light horizontal *guidelines* to keep letter height uniform, as shown in Figure 2.15. Capital letters are commonly made 3 mm (1/8") high, with the space between rows of lettering being from three-fifths to full height of the letters. Do not use vertical guidelines to space the distance from one letter to the next within a word or sentence. This should be done by eye while lettering. If necessary, use a vertical guideline at the beginning of a column of hand-lettered text to help you line up the left edges of the following rows. Beginners can also use randomly spaced vertical guidelines to practice maintaining the correct slant.



2.15 Using Guidelines

2.9 VERTICAL AND INCLINED LETTERS AND NUMERALS

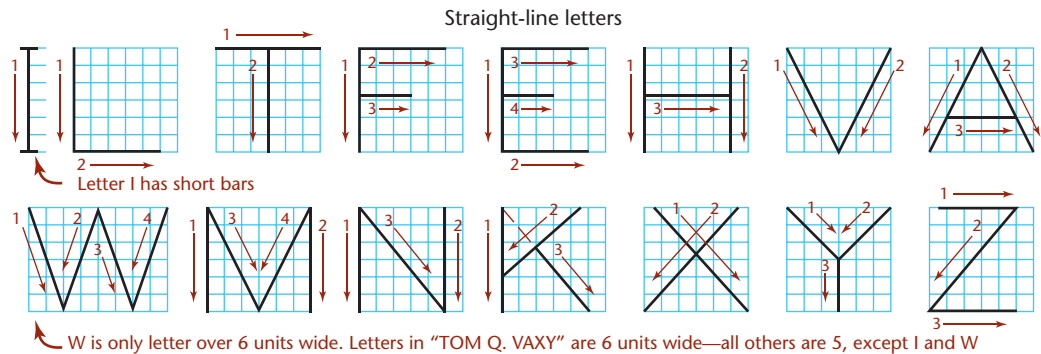
The proportions of *vertical* capital letters and numbers are shown in Figure 2.16. The letter shapes are probably a little wider than your usual writing. Hand lettering and text added to engineering drawings is typically upper case. Lowercase letters are rarely used except for large volumes of notes or when there is some other particular need for it. Lowercase letters are shown in Figure 2.17. The lower part of the letter (or descender) is usually two thirds the height of the capital letter.

Inclined (italic) capital letters and numerals are shown in Figure 2.18. They are similar to vertical lettering, except

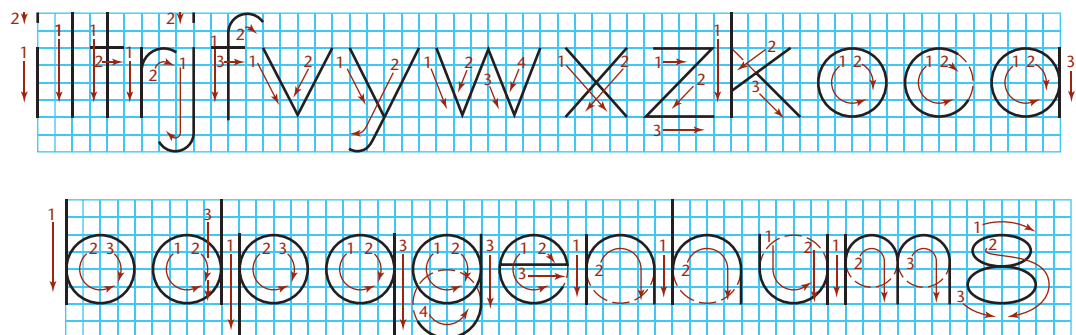
the slope is about 68° from the horizontal. Although you may practice hand lettering slanted at approximately this angle, it is important in CAD drawings to always set the amount of incline for the letters at the same value within a drawing so that the lettering is consistent. Inclined lowercase letters, shown in Figure 2.19, are rarely used.

Keep in mind that only one style of lettering or font, either vertical or inclined, should be used throughout a drawing.

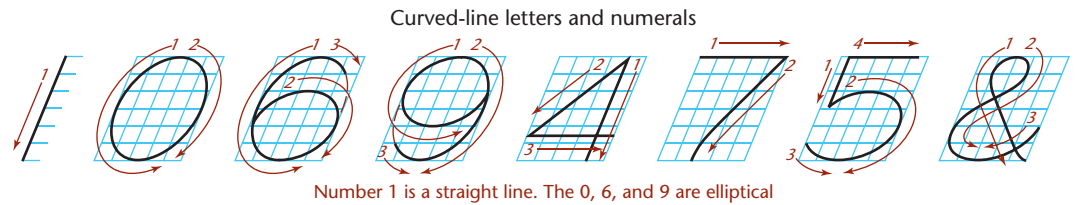
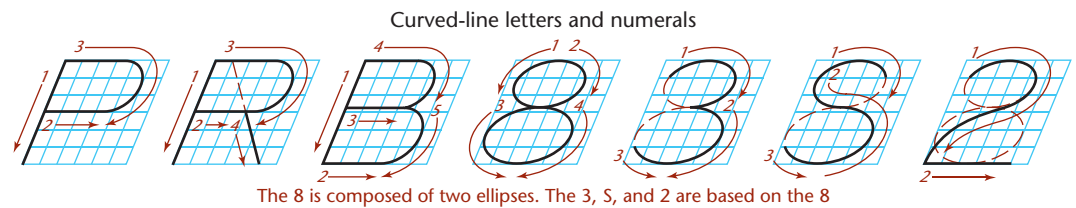
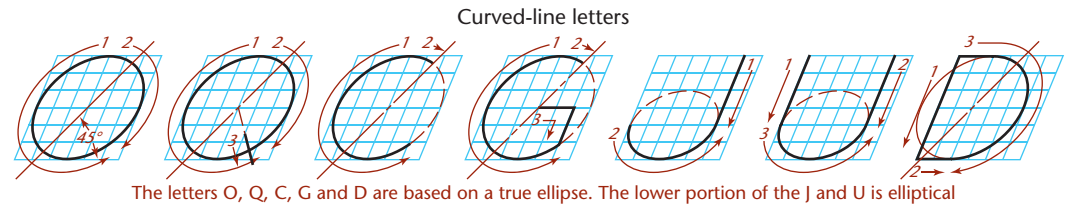
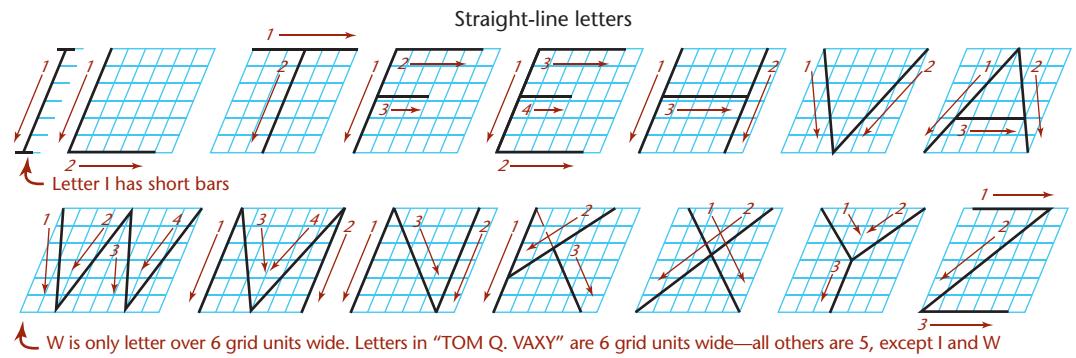
2.16 Vertical Capital Letters and Numerals



2.17 Vertical Lowercase Letters



2.18 Inclined Capital Letters and Numerals



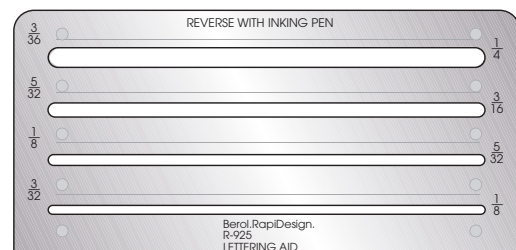
2.19 Inclined Lowercase Letters



TIP

For Even Freehand Letters

- Use 1/8" gridded paper for drawing to make lettering easy.
- Use a scale and set off a series of spaces, making both the letters and the spaces between lines of letters 1/8" high.
- Use a guideline template like the Berol Rapidesign 925 shown in Figure 2.20.
- For whole numbers and fractions, draw five equally spaced guidelines.



2.20 The Berol Rapidesign 925 template is used to create guidelines quickly for lettering.

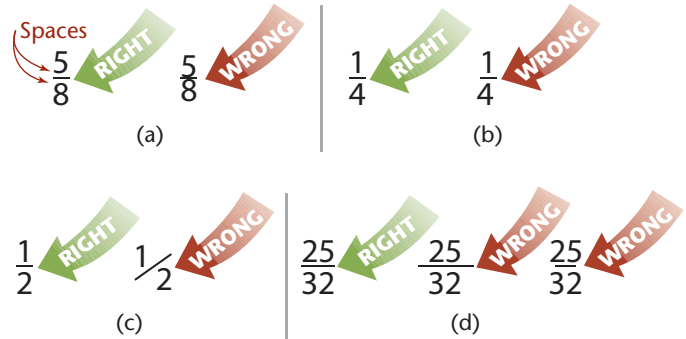
2.10 FRACTIONS

Fractions are shown twice the height of the corresponding whole numbers. Make the numerator and the denominator each about three-fourths as high as the whole number to allow enough space between them and the fraction bar. For dimensioning, the most commonly used height for whole numbers is 3 mm ($1/8''$), and for fractions 6 mm ($1/4''$), as shown in Figure 2.21.

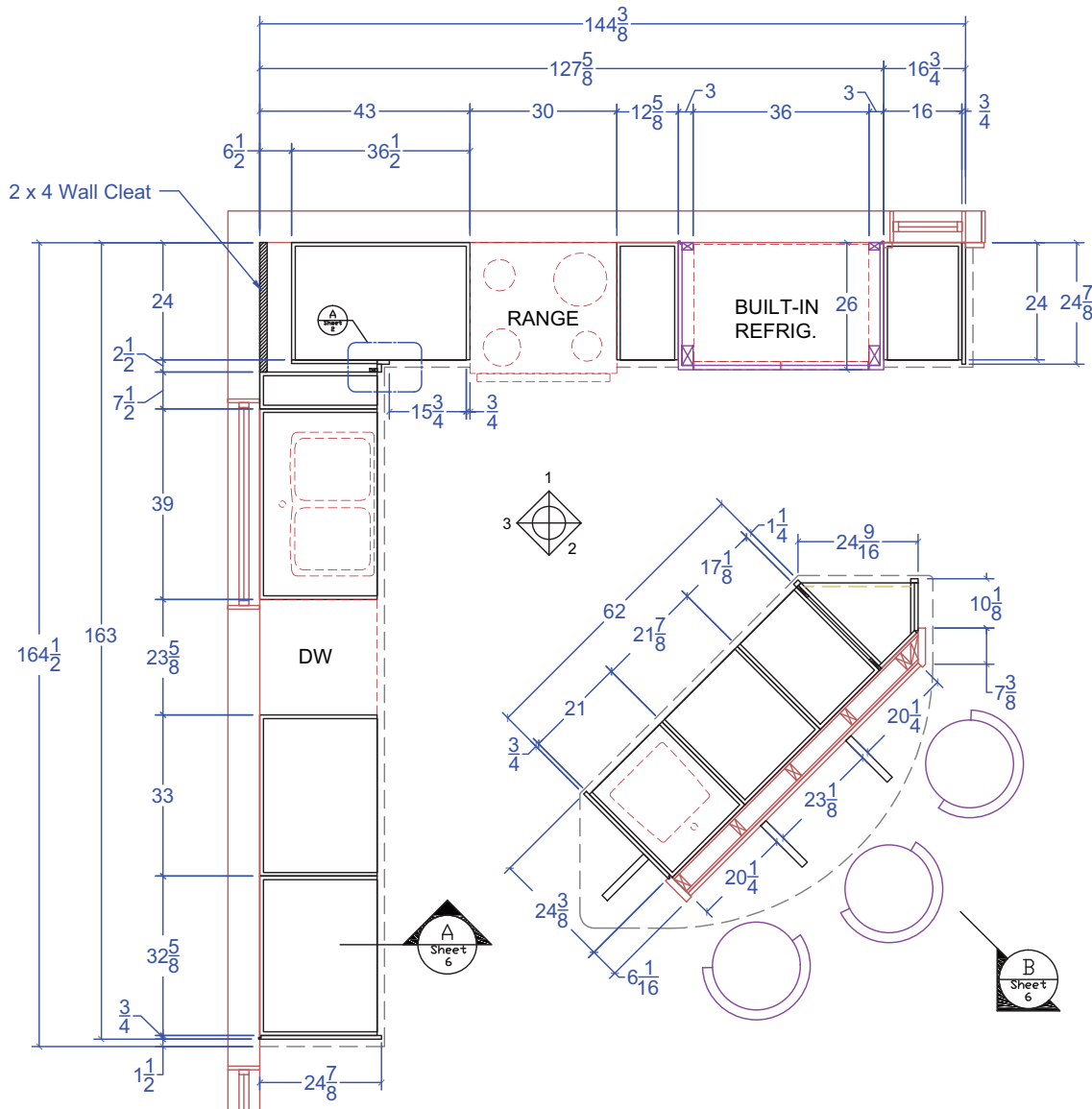
- Never let numerals touch the fraction bar.
- Center the denominator under the numerator.
- Avoid using an inclined fraction bar, except when lettering in a narrow space, as in a parts list.
- Make the fraction bar slightly longer than the widest part of the fraction.

Dimensioning in fractions is still used in the US, but even there it is less and less common. Calculating with fractions often takes manufacturing workers extra time and errors are common. Much of the numerically-controlled manufacturing

for cabinets and countertops uses millimeters as the default units, even in the United States.



2.21 Common Errors in Lettering Fractions



2.22 Custom Cabinetry Dimensioned in Fractional Inches (Courtesy of Centermark Industries, Inc.)

2.11 SPACING OF LETTERS AND WORDS

Spacing between Letters

Uniform spacing between letters is done by eye. Contrary to what might seem logical, putting equal distances from letter to letter causes them to appear unequally spaced. The background areas between letters, not the distances between them, should be approximately equal to get results that appear balanced. Figure 2.23 illustrates how using equal spacing from one letter to the next does not actually appear equal. Space your lettering so that background areas appear equal, like the example shown in the bottom half of the figure.

Some combinations, such as LT and VA, may have to be slightly closer than other letters to look correctly spaced. In some cases, the width of a letter may be decreased slightly. For example, the lower stroke of the L may be shortened

when followed by A. In typesetting, pairs of letters that need to be spaced more closely to appear correctly are called *kerned pairs*.

Spacing between Words

Space letters closely within words to make each word a compact unit, but space words well enough apart to clearly separate them from adjacent words. For both uppercase and lowercase lettering, make spaces between words approximately equal to a capital O.

Spacing between Rows

Be sure to leave space between rows of lettering, usually equal to the letter height. Rows spaced too closely are hard to read. Rows spaced too far apart do not appear to be related.

LATHING

literate

If equal spacing is used from one letter to another, spacing does not actually appear equal, as in this example

LATHING

literate

Space your lettering so that background areas appear equal, as in this example

2.23 Visually Balancing Letter Spacing

TIP

Creating Letters that Appear Stable

Certain letters and numerals appear top-heavy when they are drawn with equal upper and lower portions as in the example below.

To correct this, reduce the size of the upper portion to give a balanced appearance, as in the example below.

If you put the central horizontal strokes of the letters B, E, F, and H at midheight, they will appear to be below center.

To overcome this optical illusion, draw the strokes for B, E, F, and H slightly above the center as you letter, keeping letters uniform, as in the second example below.

C G B E K S X Z

C G B E K S X Z

The same practice applies to numerals. In the illustrations below, the example at left looks top-heavy. Note how the example at right looks more balanced.

3 8 5 2 3 8 5 2

A good example of uniform lettering

RELATIVELY

These examples show what not to do

Nonuniform style

Relatively

Nonuniform letter height

RELATIVELY
RELATIVELY

Nonuniform angle

RELATIVELY
RELATIVELY

Nonuniform stroke thickness

RELATIVELY
RELATIVELY

Nonuniform letter spacing

RELATIVELY

Nonuniform word spacing

NOW IS THE TIME FOR EVERY
GOOD PERSON TO COME TO THE
AID OF HIS OR HER COUNTRY

2.12 LETTERING FOR TITLES

In most cases, the title and related information are lettered in title boxes or title strips as shown in Figure 2.24. The main drawing title is usually centered in a rectangular space, which is easy to do using CAD.

In any kind of title, give the most important words prominence by making the lettering larger, heavier, or both. Other data, such as scale and date, can be smaller. One technique

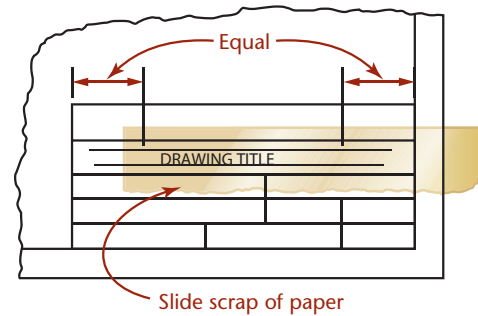
TOOL GRINDING MACHINE
TOOL REST SLIDE
SCALE : FULL SIZE
AMERICAN MACHINE COMPANY
NEW YORK CITY

DRAWN BY _____ CHECKED BY _____

2.24 Balanced Machine-Drawing Title

for centering words in the title block when hand lettering is to roughly letter the words on a scrap of paper, slide it in the available space, and make a light mark to indicate where to begin lettering the title (Figure 2.25).

Figure 2.26 shows examples of freehand lettering at actual size.



2.25 Centering Title in Title Box

THE IMPORTANCE OF GOOD LETTERING CANNOT BE
OVEREMPHASIZED. THE LETTERING CAN MAKE OR
BREAK AN OTHERWISE GOOD DRAWING.

PENCIL LETTERING SHOULD BE DONE WITH A FAIRLY
SOFT SHARP PENCIL AND SHOULD BE CLEAN-CUT
AND DARK. ACCENT THE ENDS OF THE STROKES.

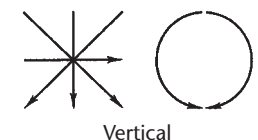
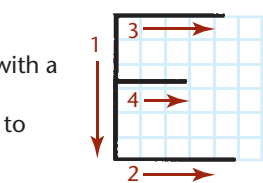
2.26 Pencil Lettering (Full Size)

TIP

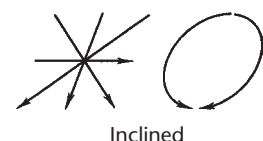
Lettering with a Pencil

- Because practically all pencil lettering will be reproduced, the letters should be dense black, not gray or blurred. Use a sharp, soft pencil, such as an F, H, or HB to make lettering dark and sharp.
- If you like using wooden pencils, sharpen them to a needle point, then dull the point very slightly.
- Don't worry about making the exact letter strokes unless you find it difficult to make the letters look right, but do use them as a reference if you are having trouble drawing uniform, symmetrical letters.
- Use extremely light, 1/8" (3 mm) horizontal guidelines to regulate the height of letters. A few light vertical or inclined lines randomly placed help you visually keep the letters uniformly vertical or inclined.

- Draw vertical strokes downward with a finger movement.
- Draw horizontal strokes from left to right with a wrist movement and without turning the paper.
- Draw curved strokes and inclined strokes with a downward motion.



Vertical



Inclined

Left-handers: Traditional lettering strokes were designed for right-handed people. Experiment with each letter to develop a system of strokes that works best for you.

2.13 DRAWING PENCILS

High-quality drawing pencils help produce good-quality technical sketches and drawings. Use light lines for construction lines, lettering guidelines, and precise layout work. Use dark, dense black lines for the final lines, lettering, and arrowheads. Drawings are often reproduced, and the lines need to be dark for the copies to turn out well.

Drawing pencils are made of graphite with a polymer binder or clay binder. They are divided into 18 grades from 9H (the hardest) to 7B (the softest) as shown in Figure 2.27. Specially formulated leads of carbon black particles in a polymer binder are also available in several grades for use on polyester film (Mylar).

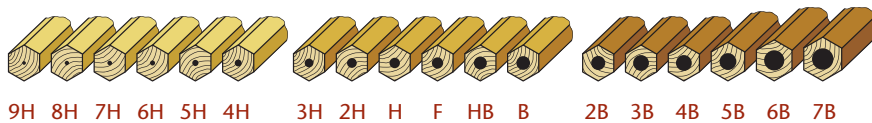
Hard leads are used where accuracy is required, such as on graphical computations and charts and diagrams. For other uses, their lines are apt to be too light.

Medium leads are used for general-purpose technical drawing, such as sketching, lettering, arrowheads, and other freehand work on mechanical drawings.

Soft leads are not useful in technical drawing. They make smudged, rough lines that are hard to erase, and the lead dulls quickly. These grades are generally used for artistic drawing.

Which grade of pencil works best for you depends on your hand pressure, the humidity, and the type of paper you are using, among other things. For light lines, use a hard lead in the range of 4H to 6H. For dark lines, use a softer lead in the range of 2H to B.

Mechanical pencils are available with 0.3-, 0.5-, 0.7-, or 0.9-mm-diameter drafting leads in several grades (Figure 2.28). Their thin leads produce uniform-width lines without sharpening. The .5-mm lead is a good general size, or you can use a .7-mm lead for thick lines and .3 mm for thin lines.



Hard
The hard leads in this group (left) are used where extreme accuracy is required, as on graphical computations and charts and diagrams. The softer leads in this group (right) are sometimes used for line work on engineering drawings, but their use is limited because the lines are apt to be too light.

Medium
These grades are for general-purpose work in technical drawing. The softer grades (right) are used for technical sketching, lettering, arrowheads, and other freehand work on mechanical drawings. The harder leads (left) are used for line work on machine drawings and architectural drawings. The H and 2H leads are widely used on pencil tracings for reproduction.

Soft
These leads are too soft to be useful in mechanical drafting. They tend to produce smudged, rough lines that are hard to erase, and the lead must be sharpened continually. These grades are used for artwork of various kinds, and for full-size details in architectural drawing.

TIP

You might be surprised how much your drawings benefit from finding a style of pencil that suits your use. Soft pencils, such as HB or F, are mainly used in freehand sketching. Choose a pencil that:

- Is soft enough to produce clear black lines, but hard enough not to smudge too easily.
- Is not so soft that the point breaks easily.
- Feels comfortable in your hand.
- Grips the lead without slipping.

Be aware that some lead holders require special sharpeners.

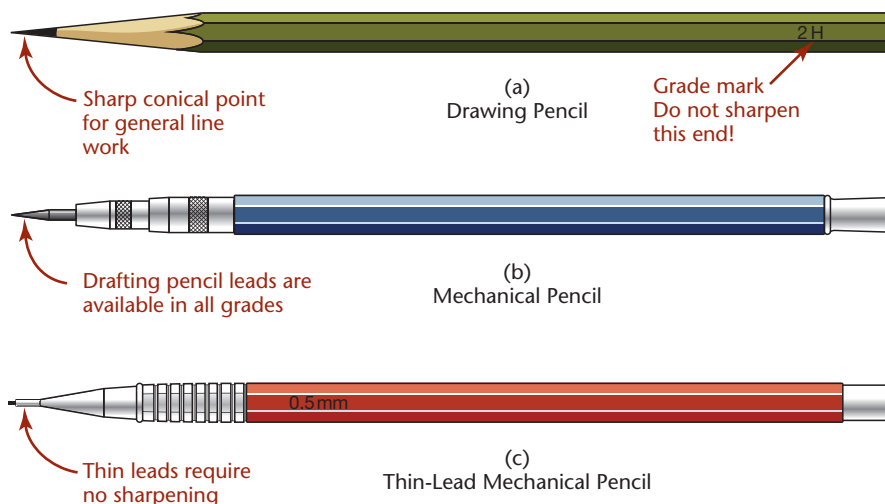
You can sometimes tell the difference in hardness of a mechanical pencil lead just by looking at it. Smaller-diameter leads are used for the harder grades, and larger-diameter leads are used to give more strength to the softer grades.

Plain wooden pencils work great. They are inexpensive, and it is easy to produce thick or thin lines by varying the amount that you sharpen them. An old trick to keep the lead sharp longer is to turn the pencil frequently as you work to wear it down evenly.

Gum erasers and nylon erasers work well to pick up smudges without leaving much eraser dust.

Nylon eraser strips that come in refillable holders like mechanical pencils can be convenient for areas that require some precision. A trick for erasing fine details is to sharpen the end of the eraser strip in a small handheld pencil sharpener.

2.27 Lead Grade Chart



2.28 Drawing Pencils

2.14 TEMPLATES

Templates are available for a great variety of specialized needs (Figure 2.29). Templates may be found for drawing almost any ordinary drafting symbol or repetitive feature.

2.15 CAD TOOLS

Most people who create technical drawings use CAD. Advantages include accuracy, speed, and the ability to present spatial and visual information in a variety of ways.

However, these advantages do not eliminate the need for drawings to be easily and accurately interpreted. CAD drawings use the same general concepts and follow the same drafting standards as drawings created by hand.

Most CAD drawings are plotted on standard sheet sizes and to similar scales as hand drawings. Both CAD and hand drawings should contrast thick lines for objects with thin lines for hidden, center, and dimension lines to make the printed drawing easy to read.



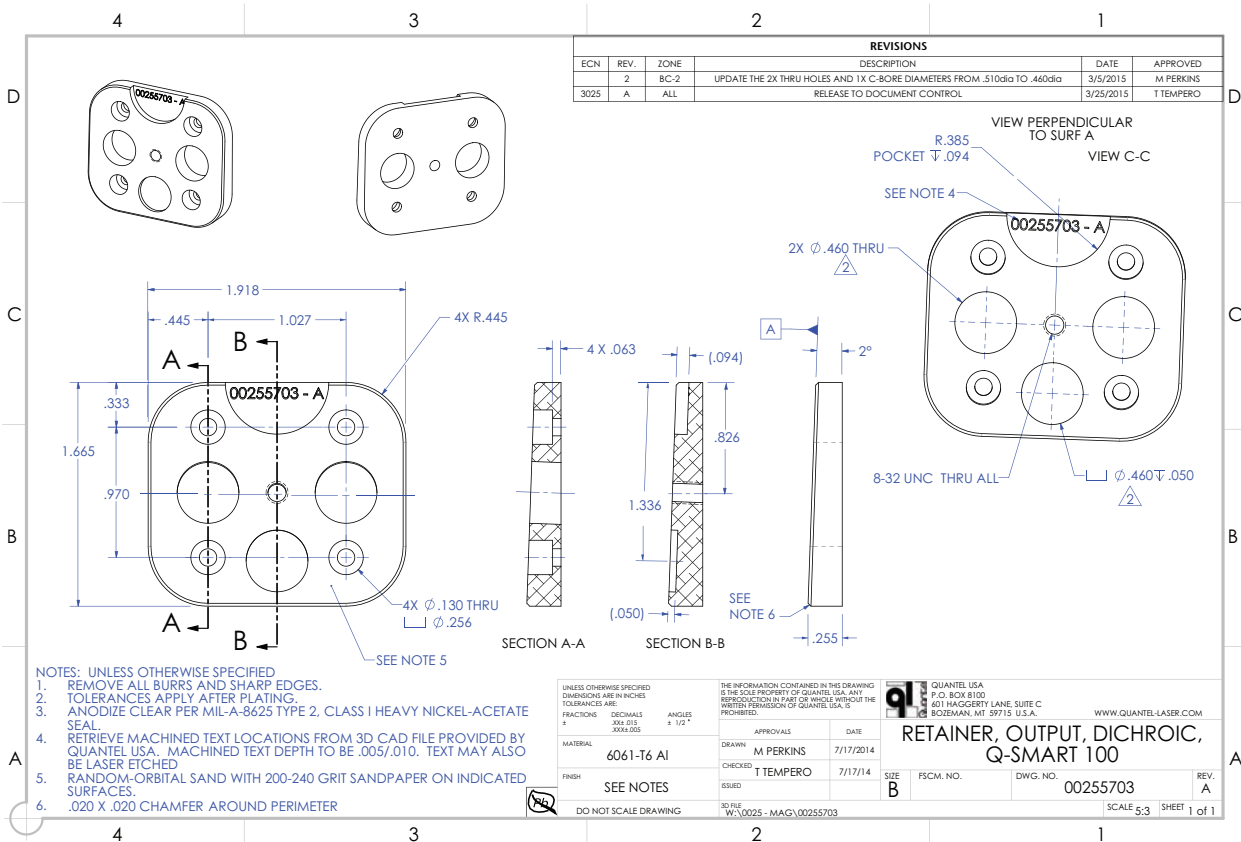
2.29 Drawing Templates

CAD drawings should use correct line patterns. Likewise, lettering on CAD drawings should follow these same general guidelines as for hand drawings.

One benefit of CAD is the ability to draw perfectly straight uniform lines and other geometric elements. Another is the ability to quickly represent the various styles of lines (Figure 2.30). Making changes to a CAD drawing takes about a tenth the time that it takes

to edit a drawing by hand. Using CAD, you can quickly plot drawings to different scales.

Keeping CAD drawing files organized, backing up data regularly, and following conventions for naming files so that you can find them again are important considerations. Even skilled CAD users use freehand sketching, to quickly get ideas down on paper and to show their ideas on a whiteboard.



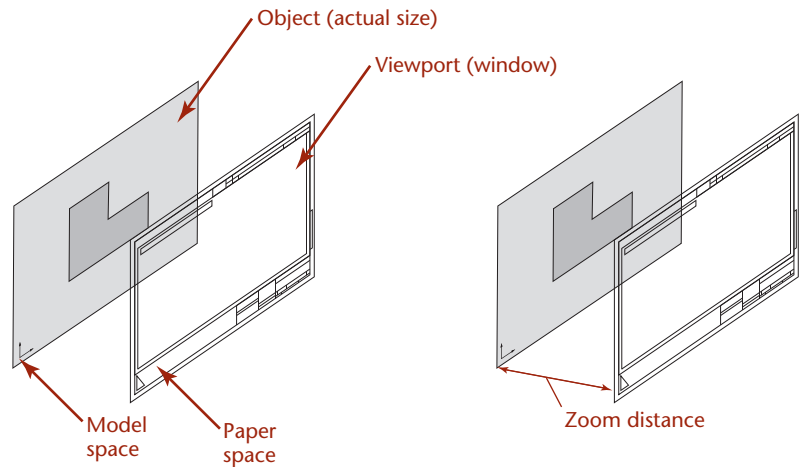
2.30 A Drawing Created Using CAD (Courtesy of Mark Perkins.)

MODEL SPACE AND PAPER SPACE IN AUTOCAD

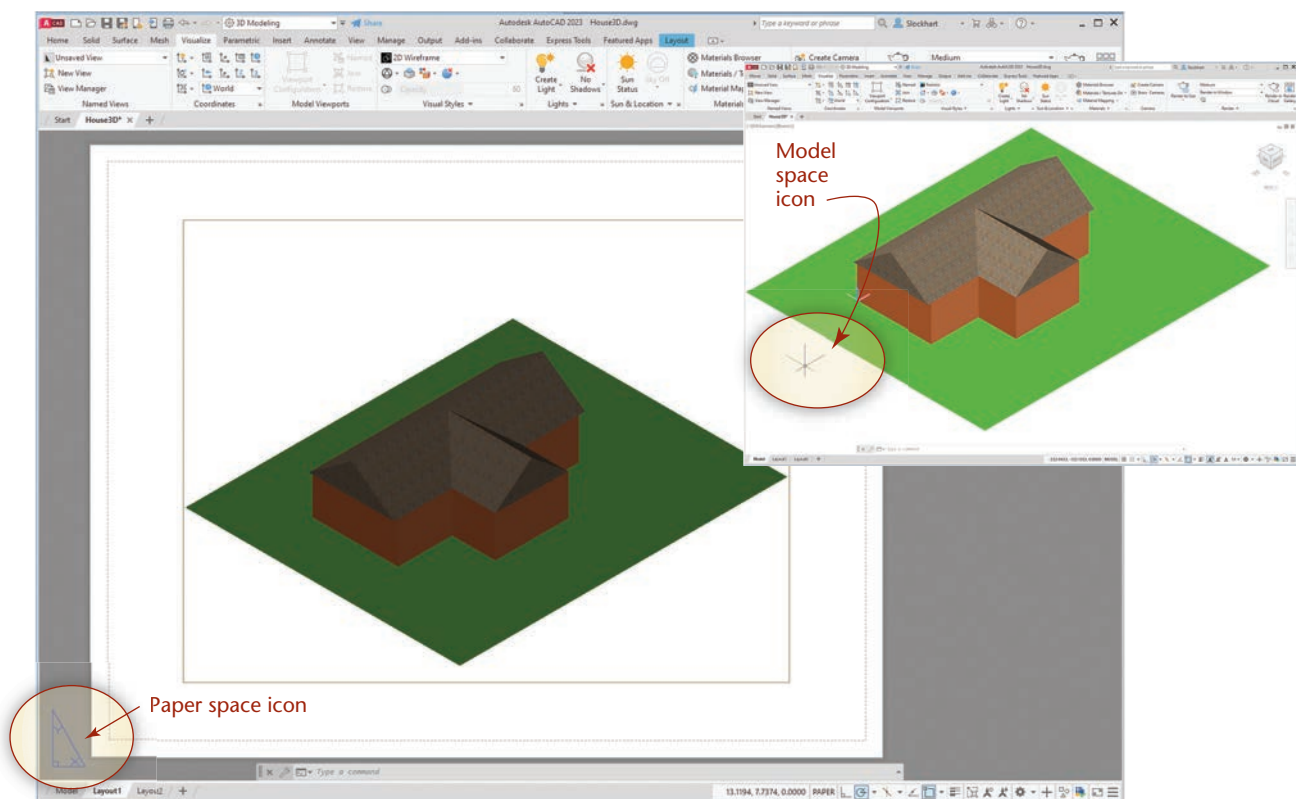
Using CAD, you can make an accurate model of the device or structure. To do this, you create the object at the actual size that it exists in the real world, using whatever system of measurement that you would use when constructing it.

On paper it is a different matter. You would have to have some really large sheets to print your building full size. AutoCAD software uses the concept of two “spaces,” model space and paper space, to describe how to transform the full-size CAD model to proportionate views that fit your sheet of paper.

Understanding scale as it relates to paper drawings or as it relates to creating layouts from a CAD drawing is an important concept for technical drawing because the ultimate goal is for drawings to be interpreted and used in the real world. Therefore, they must be easy to print and read.



(A) In AutoCAD, paper space allows you to see how various views of the full-size model can be shown on a sheet of paper.



(B) The window at left shows a paper space representation of the full-size CAD model in the smaller window at top right. Note that AutoCAD uses icons to help users differentiate the two “spaces.” (Autodesk screen shots reprinted courtesy of Autodesk, Inc.)

2.16 SKETCHING AND DRAWING MEDIA

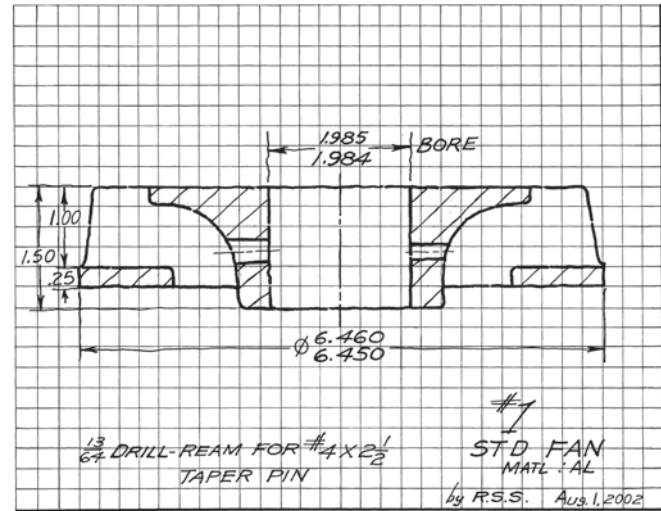
Many choices of *media* (paper and other) are available for particular sketching or drawing purposes. Whether you are sketching or are plotting a drawing from a CAD workstation, choose the type of sheet and size that suits your needs.

Small notebooks or sketch pads are useful when working at a site or when it is necessary to quickly record information. Many companies require bound notebooks of graph paper for recording engineering design notes so they are preserved for patent and documentation purposes. Graph paper can be helpful in making neat sketches like the one in Figure 2.31. Paper with 4, 5, 8, or 10 squares per inch is convenient for maintaining proportions.

A sketch pad of plain paper with a master grid sheet showing through underneath works well as a substitute for grid paper. You can create your own master grid sheets using CAD. Specially ruled isometric paper is available for isometric sketching, or you can use CAD to create masters.

The best drawing papers have up to 100% pure rag stock. Their strong fibers hold up well to erasing and folding, and they will not discolor or grow brittle with age. Good drafting paper should have a fine grain (or tooth) to pick up the graphite and produce clean, dense black lines. Paper that is too rough produces ragged, grainy lines, is harder to erase, and wears down pencils quickly. Look for paper that has a hard surface that will not groove too easily under pencil pressure.

Polyester film is a high-quality drafting material available in rolls and standard sized sheets. It is made by bonding a matte surface to one or both sides of a clear polyester sheet. Its transparency and printing qualities are good and it provides an



2.31 Sketch on Graph Paper

excellent matte drawing surface for pencil or ink. It is easy to erase without leaving ghost marks, and it has high dimensional stability. Its resistance to cracking, bending, and tearing makes it very durable. Many companies still plot their drawings in ink on polyester film for long-term storage and reproduction.

Even large coated sheets of aluminum (which provides a good dimensional stability) have been used in the aircraft and auto industry for full-scale layouts that were scribed into the coating with a steel point rather than a pencil.

2.17 STANDARD SHEETS

There are ANSI/ASME standards for international and U.S. *sheet sizes*. Table 2.2 describes the height and width of these standard sheets, the letters used to refer to them, and their margins and zones. Note that drawing sheet size is given as height $\times$ width. Most standard sheets use what is called a “landscape” orientation.

Table 2.2 Sheet Sizes

Nearest International Size (mm)	International Number of Zones	International Margin	Standard U.S. Size (in.)	U.S. Number of Zones (width)	U.S. Margin (in.)
A4 210 $\times$ 297	6	10	A* 8.5 $\times$ 11.0	2 (optional)	.50
A3 297 $\times$ 420	6	10	B 11.0 $\times$ 17.0	2 (optional)	.50
A2 420 $\times$ 594	8	10	C 17.0 $\times$ 22.0	4	.50
A1 594 $\times$ 841	12	20	D 22.0 $\times$ 34.0	4	.50
A0 841 $\times$ 1189	16	20	E 34.0 $\times$ 44.0	8	.50

* May also be used as a vertical sheet size at 11" tall by 8.5" wide.

2.18 STANDARD LAYOUT ELEMENTS

Margins and Borders

Each layout begins with a border drawn inside the sheet margin. Drawings in the U.S. use a .50" margin. Refer to Table 2.2 for international sheet sizes and margins. Some companies use slightly larger sheets to allow drawings to be bound into a set. This extra allowance should be added on to the standard sheet size so that the drawing border meets the size standards (see Figure 2.32). Figure 2.33 shows the alternative orientation of an A size drawing.

Zones

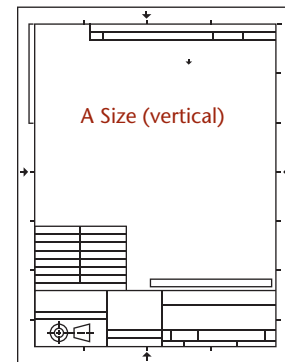
You have probably seen *zone numbers* on maps, where the margin is subdivided by letters along one side and by numbers along the other. These are also used along the outer edges of technical drawings so that you can refer to items by the area on the sheet where they are located. This is particularly useful when a client calls with a question. You can use zone numbers to make sure you are talking

about the same item. Zone numbers are also useful for locating revisions. You should provide zone numbers on all sheets larger than size B.

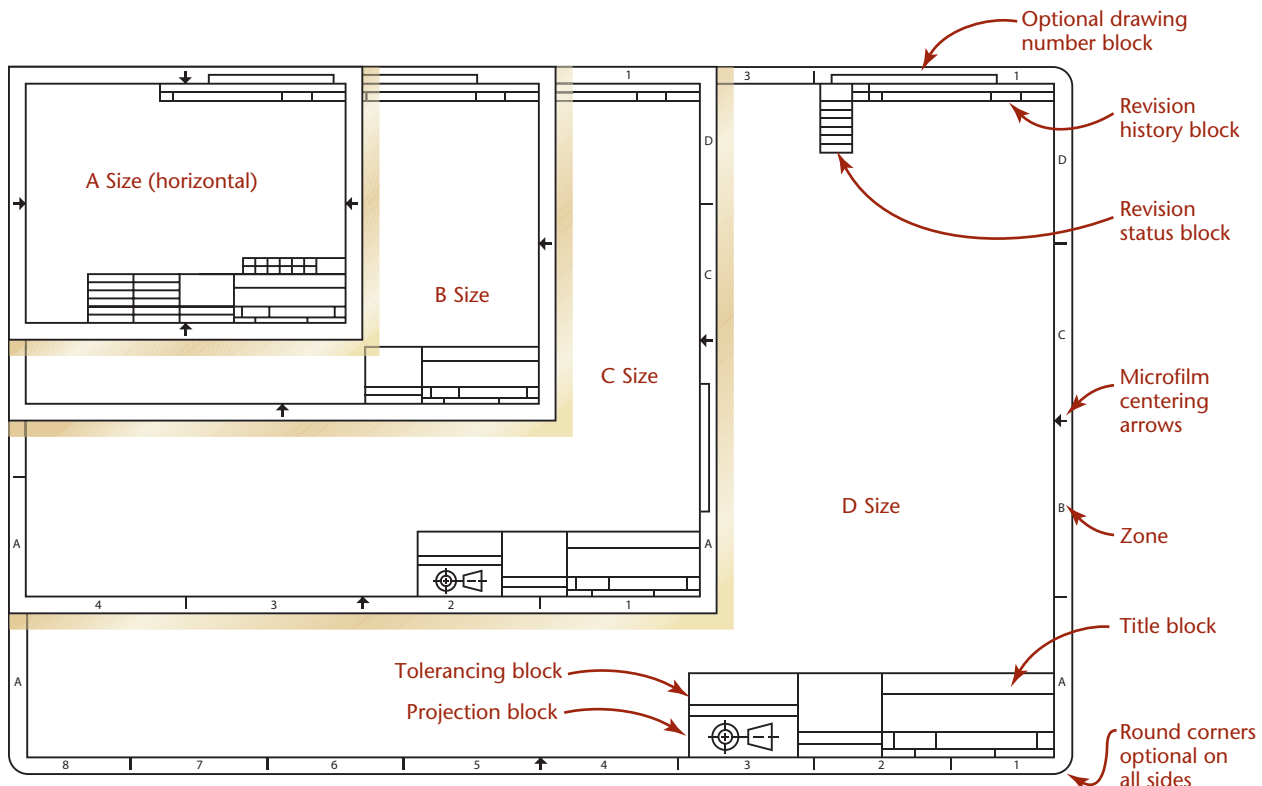
Typical Letter Sizes

Most lettering on drawings should be at least 3 mm or .12" (about 1/8") tall. Lettering is typically sized as follows:

Drawing Title, Drawing Size
6 mm (.24")
DAI
6 mm (.24")
Drawing Number, Revision Letter
6 mm (.24")
Section and View Letters
6 mm (.24")
Zone Letters and Numbers
6 mm (.24")
Drawing Block Headings
2.5 mm (.10")
All Others
3 mm (.12")



2.33 Vertical Orientation of A Size



2.32 Typical Sheet Sizes and Borders (See Table 2.2 for E-size and international standard sizes.)

Approval blocks

NAME ← Originating company name (and address if desired)

DRAWING TITLE

3.87

2.75

.38

2.00

1.38

.62

.25

4.25

3.25

1.75

6.25

Sheet size from Table 2.2

Predominant scale (use NONE when NTS)

SCALE

WEIGHT (actual or estimated)

SHEET 1 OF 2

DRAFTED BY: Name Date

ENGINEERED: Name Date

ADDITIONAL APPROVAL Name Date

ADDITIONAL APPROVAL Name Date

SIZE

DAI

DWG NO.

REV

2.34 Title Block for A-, B-, and C-Size Sheets

Title Block

The title block is located in the lower right corner of the format. Refer to Figure 2.34 for dimensions for a typical title block for A-, B-, and C-size sheets.* Standard areas in the title block provide the following information.

Name Show the name of the originating company or business (and address if desired).

Drawing Title Briefly describe the item using a singular noun or noun phrase and modifiers if necessary to distinguish it from similar items. Do not use the terms “for” or “or” in the title. For example, “Dust Cap” would be preferred over “Cap or Cover for Dust Protection,” which is too wordy.

Drawing Number Give each drawing a unique number, using the company’s numbering system.

Sheet Revision Block Track the drawing version using the number of the revision. The original release of the drawing typically shows revision 0.

Approval Block List the name(s) of the person(s) approving the drawing and the date it was approved. Additional areas of this block can be used for various design activities, if separate approval is required. For example, a company may use separate areas for structural design or manufacturing engineering approvals (Figure 2.35). Specify the dates numerically using a year-month-day format. For example, June 10, 1989, would be any of the following 1989-06-10, 19890610, or 1989/06/10. This is particularly helpful as items are easily sorted into date order by computer algorithms. It also avoids the ambiguity caused by some countries showing month before day, and others using the day, month, year format.

Scale List the predominant scale for the drawing. Drawings may include details at other scales, which should be noted below the detail. If the drawing is not made to a particular scale, note NONE in the scale area.

Drawing Size List the sheet size used for the drawing. This helps track the original size when the drawing is reproduced at a smaller size.

NAME	
DRAWING TITLE	
DRAFTED BY:	
ENGINEERED:	
ADDITIONAL APPROVAL	SIZE DAI DWG NO. REV
ADDITIONAL APPROVAL	SCALE WEIGHT SHEET 1 OF 2

2.35 Approval Block, Scale, Revision, and Drawing Size

NAME	
DRAWING TITLE	
DRAFTED BY:	
ENGINEERED:	
ADDITIONAL APPROVAL	SIZE DAI DWG NO. REV
ADDITIONAL APPROVAL	SCALE WEIGHT SHEET 1 OF 2

2.36 DAI and Weight May Be Listed

Sheet Number List the number of the sheet in the set, using whole numbers starting at 1. A format that lists this sheet out of the total number helps keep track of the entire set, for example, 1 OF 2.

DAI List the Design Activity Identification in this area when it is required. This block may be left blank or the block removed if it is not needed. Examples of DAI include: design activity name, activity name and address, and Commercial and Government Entity (CAGE) codes if applicable. CAGE codes are numbers assigned to entities that manufacture items for the government, based on the DAI.

Weight List the actual or estimated weight of the part if required (Figure 2.36).

*For more formats, title blocks, revision blocks, and materials blocks, see inside the front cover of this book.

2.19 LAYOUTS

A particular size sheet with a drawing border is called a *layout*. Using a CAD system, you may often be able to select from standard layouts or templates that set the sheet size limits, the border, and even the title block as the starting point for your drawing. Regardless of whether you draw by hand or use CAD or 3D modeling methods, you need to plan your sheet so that the information will fit and show clearly.

When you are sketching, your layout may be a simple border and title strip along the bottom of the sheet (or you may be using preprinted tablets that have space to record the sketch title, date, and other pertinent data).

When creating a 2D CAD drawing, you may use a drawing template showing the sheet and border and title block, perhaps using different templates or even software interface settings for different types of drawings, such as mechanical/manufacturing, architectural, or civil.

When creating a 2D drawing from a 3D solid model, you may use a layout space that contains different viewports that allow you to show different views of the same 3D model with a border and title block.

2.20 PLANNING YOUR DRAWING OR SKETCH

When laying out a drawing sheet, you will need to consider:

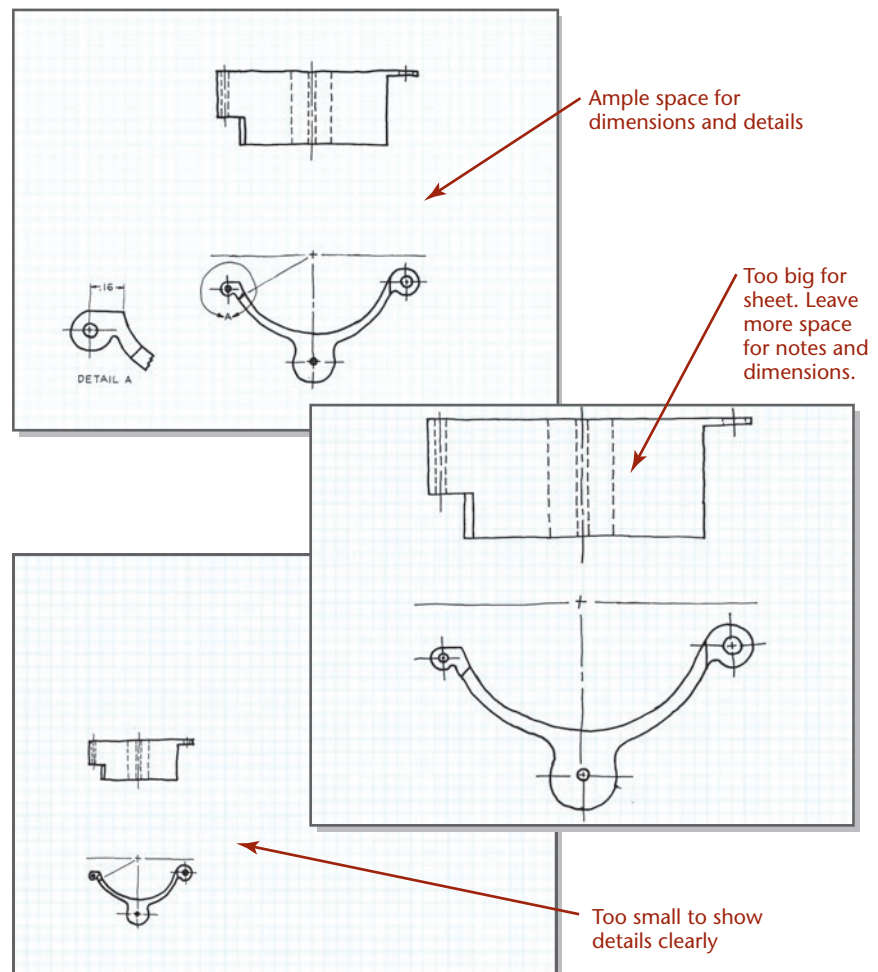
- the size and scale of the object you will show;
- the sheet size;
- the measurement system (units) for the drawing; and
- the space necessary for standard notes and title block.

The object you are drawing is the “star” of the sketch. Keep the object near the center of the sheet. It should be boldly drawn, using thick visible lines. Make it large enough to fill most of the sheet and so that details show clearly (Figure 2.37).

Show Details Clearly

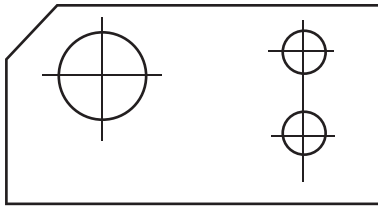
Show small objects larger than their actual size to represent the details clearly. If the details are too small, switch to a larger sheet size and use a larger scale.

You can also add details at a larger scale if necessary to show features that are smaller than the typical features of the drawing. If you add details at a different scale, label the view, for example, DETAIL A, and note the scale for the detail below it.



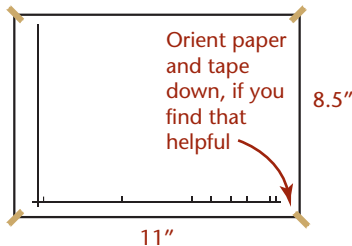
2.37 Show details clearly by selecting the appropriate scale and sheet size.

SHEET LAYOUT

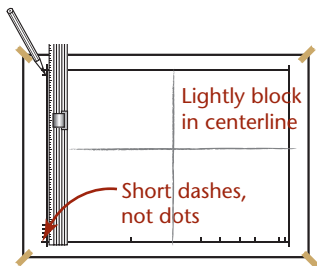


Given drawing

- 1 To draw the part shown in the given figure, select the sheet size, keeping in mind the size of the objects. Show the part large enough to represent features clearly. Use larger sheets for larger or more detailed objects. ($8.5'' \times 11''$ will be large enough for the part shown.) Add the border and title block to the sheet using the margin sizes specified in the standards. Refer to Table 2.2.



- 2 Determine the units for the drawing. Will it be metric or U.S. Customary (inches, feet and inches)? What system will be used to construct, manufacture, and inspect the actual object? Use that system of measurement for the drawing. This part is in inches.



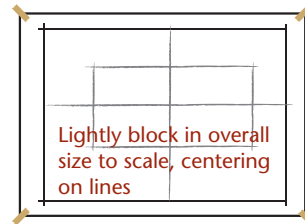
Before you begin drawing, determine the scale at which the object will best fit on the sheet.

First, figure the available space within the drawing border. For example, the horizontal $8.5'' \times 11''$ sheet with a $.5''$ margin leaves $7.5'' \times 10''$. If you subtract space for a $.375''$ title strip across the bottom, it leaves $7.125'' \times 10''$ for the drawing.

Now, consider the size of the object. Will it fit on the sheet at full size? Half size? Do you need to enlarge it to show small features larger than actual size? The $12''$ gasket shown in the example will fit well at half size on the $8.5'' \times 11''$ sheet and still show the details clearly. Use typical scales when possible. Refer to Section 2.5.

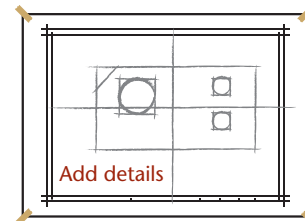
Approximately center the object on the sheet. To do this, subtract the size of the scaled drawing from the available sheet space and use half of the difference on each side of the object.

- 3 One quick technique is to find the center of the available space and lay out the drawing on each side of that centerline. Using CAD, you can easily move the drawing to the center of the sheet visually.

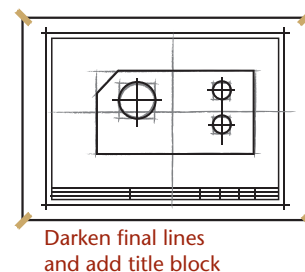


Sketches do not have to be perfectly centered, but plan ahead so the drawing isn't crammed in one corner of the sheet. Let your drawing be the "star" of the page. Remember to leave enough space around your drawing for notes and dimensions. If you don't, you will run out of room and your layout will look crowded.

- 4 Lightly add details of the drawing.



- 5 Darken final drawing lines.



TIP Scale When Using CAD

Keep in mind that when using CAD you will create the object the size that it actually exists in real life. On the plotted sheet, when showing the drawing to scale, it is easy to try a few different scales and see which fits. You can always change the scale later if needed.

SCALING ANNOTATIONS AUTOMATICALLY USING AUTOCAD

You might think that displaying text in a CAD drawing is one of the easiest things to do. You can quickly type in the text you want to display and select the font, height, slant, and rotation angle. That part is easy, but annotations are useless if nobody can read them.

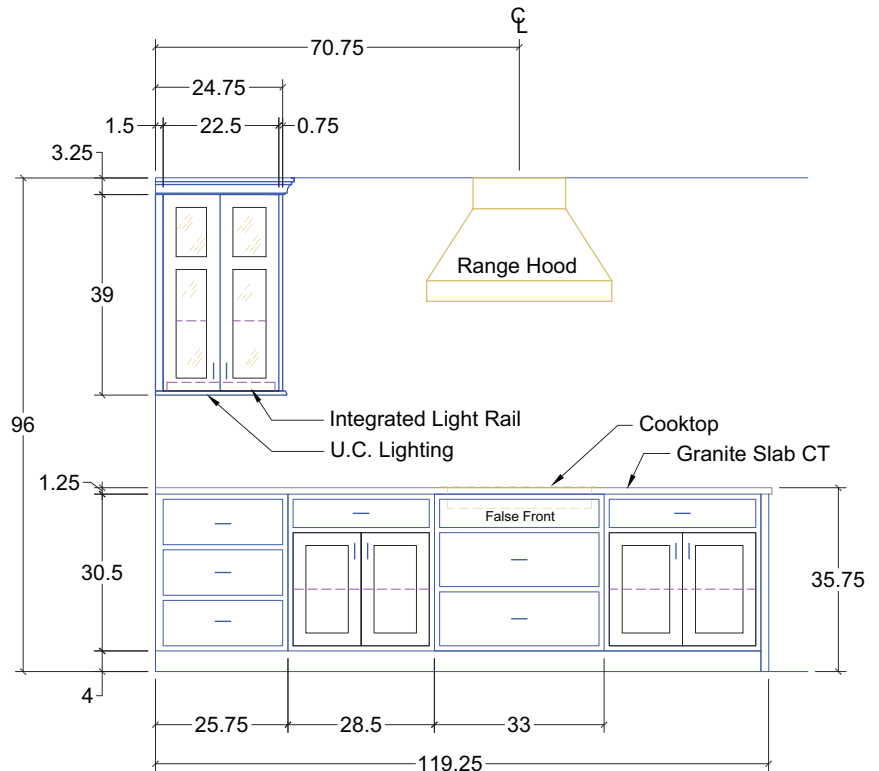
When you create drawings that will be plotted on different sized sheets or at different scales, sizing the text can require a lot of planning. Take the architectural plan drawing shown in Figure A for example. When plotted to scale $1/2" = 1'$, the text showing dimensions is clearly visible. But when shown at one tenth of that size, scale $1/2" = 10'$, that same text becomes illegible.

The ability to reuse the same CAD data at different scales without having to recreate the drawing is one of its big advantages over pen and paper drawings. Yet, cumbersome workarounds were once necessary to make legible text at different scales. One workaround was to have several different sizes of the same text, which the user would turn on or off depending on what drawing scale was used.

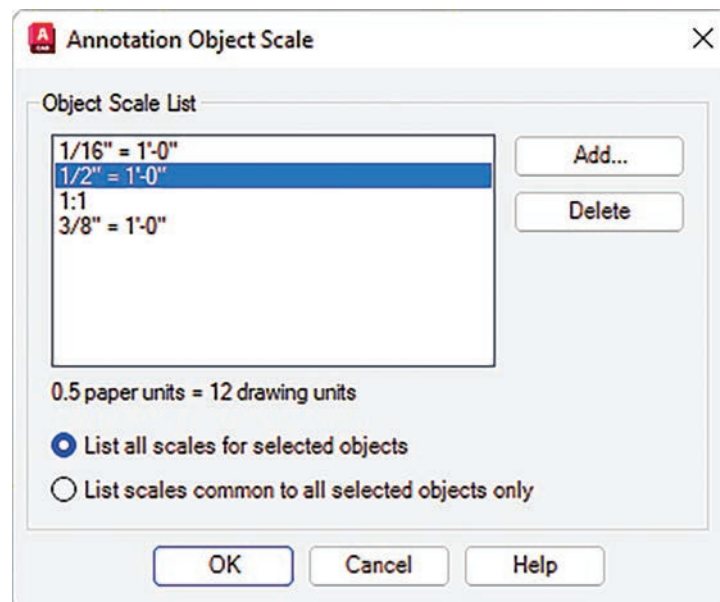
Now, AutoCAD software provides a feature called annotation scaling. Here is how it works: Drawing objects that are commonly used to annotate drawings (provide text information) can have their annotation property turned on. This allows you to create one annotative object that displays at different sizes, based on scale properties.

In the AutoCAD software, object types that can have annotative object properties include text, mtext, dimensions, hatches, tolerances, multileaders, leaders, blocks, and attributes.

(B) The annotation scaling feature of AutoCAD software allows annotative text to be made legible at various scales. (Autodesk screen shots reprinted courtesy of Autodesk, Inc.)



(A) When plotted to a scale of $1/2" = 1'$ on an $11" \times 8.5"$ sheet, the text showing the dimensions is clearly visible. (Courtesy of Centermark Industries, Inc.)



NOTES:

- UNLESS OTHERWISE SPECIFIED
- ALL DIMENSIONS ARE IN DECIMAL INCHES.
- REMOVE ALL BURRS & SHARP EDGES.
- DIMENSIONS & TOLERANCES SHALL BE HELD AFTER FINISHING OF PARTS.

THIS PART IS PRODUCED FROM AN ELECTRONIC DATA FILE PROVIDED TO THE VENDOR. DIMENSIONS SHOWN ARE FOR REFERENCE. ANY OTHER DIMENSIONS SHOULD BE OBTAINED FROM THE MODEL.

SEE ASSEMBLY DIRECTIONS FOR INSTRUCTIONS ON SQUARING CHASSIS ASSEMBLY.

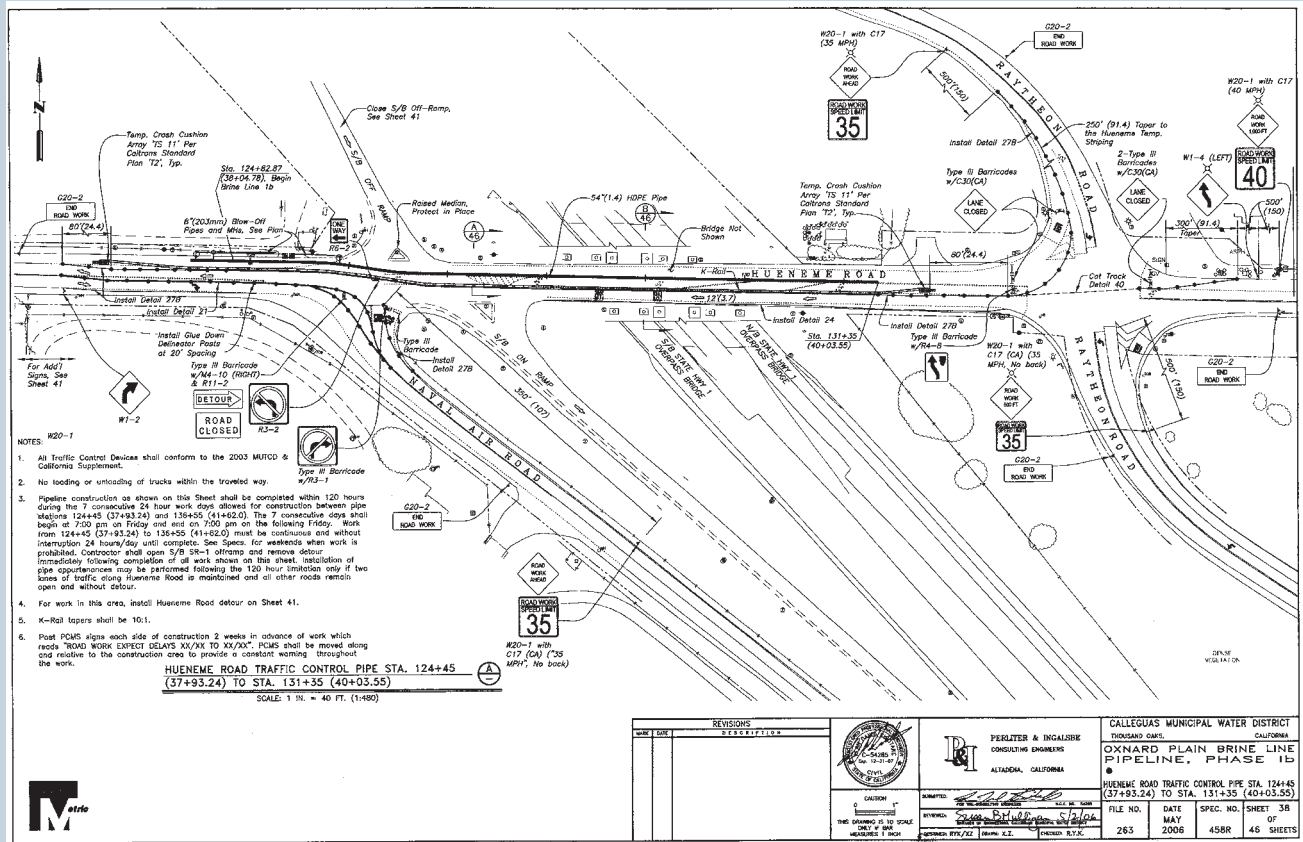
CHECK MRP SYSTEM FOR UPDATED BOM AND/OR UNLISTED PARTS & QUANTITIES.

ITEM NO.	PART NUMBER	DESCRIPTION
1	63224100	DRUM MODULE
2	61124290	CARRIAGE RUNNER-RIGHT, M/C3
3	61124291	CARRIAGE RUNNER-LEFT, M/C3
4	21620101	ANGLE, RAIL MOUNT-LEFT
5	21620102	ANGLE, RAIL MOUNT-RIGHT
6	21224200	BULKHEAD - DRUM MODULE
7	21224200	CARRIAGE MIDDLE BULKHEAD
8	61124820	FRONT CARRIAGE ASSY
9	21195301	3G SUPPORT BOX
10	61124690	CARRIAGE FRONT PANEL ASSY
11	61129170	HOOD WLDNMT-TOP-DRUM
12	21224501	HOOD, DRUM MODULE
13	61124691	CARRIAGE RIGHT TOP COVER ASSY
14	61124692	CARRIAGE LEFT TOP COVER ASSY
15	21224501	WIRING Z-BRACKET
16	21224502	HOOD-SIDE, DRUM MODULE
17	21224303	CARRIAGE LEFT SIDE PANEL
18	21224300	CARRIAGE RIGHT SIDE PANEL
19	21224302	CARRIAGE TOP CENTER PANEL V2
20	21224511	NEW DYNO WIRING BRACKET
21	36582034	SCREW, 3/8"-16x1-1/4", BH-FUNG
22	36923100	WASHER, 3/8", HARDENED, FLAT, STL
23	36488100	NUT, 3/8"-16, NYLOCK
24	36561045	SCREW, 1/4"-20x5/8", PH,TORX
25	DM150-011-004	3/8 NUT

**UNRELEASED!
FOR QUOTING OR
PROTOTYPE ONLY**

MATERIAL	FIGURE	TOLERANCES	FRACTIONAL	DECIMAL	ANGULAR	DO NOT SCALE THE DRAWING
		UNLESS OTHERWISE SPECIFIED:				
		TOLERANCES	FRACTIONAL	DECIMAL	ANGULAR	
			1/16"	± 0.005"	± 0.005°	
			1/32"	± 0.002"	± 0.002°	
			1/64"	± 0.001"	± 0.001°	
			1/128"	± 0.0005"	± 0.0005°	
			1/256"	± 0.0002"	± 0.0002°	
			1/512"	± 0.0001"	± 0.0001°	
			1/1024"	± 0.00005"	± 0.00005°	
			1/2048"	± 0.00002"	± 0.00002°	
			1/4096"	± 0.00001"	± 0.00001°	
			1/8192"	± 0.000005"	± 0.000005°	
			1/16384"	± 0.000002"	± 0.000002°	
			1/32768"	± 0.000001"	± 0.000001°	
			1/65536"	± 0.0000005"	± 0.0000005°	
			1/131072"	± 0.0000002"	± 0.0000002°	
			1/262144"	± 0.0000001"	± 0.0000001°	
			1/524288"	± 0.00000005"	± 0.00000005°	
			1/1048576"	± 0.00000002"	± 0.00000002°	
			1/2097152"	± 0.00000001"	± 0.00000001°	
			1/4194304"	± 0.000000005"	± 0.000000005°	
			1/8388608"	± 0.000000002"	± 0.000000002°	
			1/16777216"	± 0.000000001"	± 0.000000001°	
			1/33554432"	± 0.0000000005"	± 0.0000000005°	
			1/67108864"	± 0.0000000002"	± 0.0000000002°	
			1/134217728"	± 0.0000000001"	± 0.0000000001°	
			1/268435456"	± 0.00000000005"	± 0.00000000005°	
			1/536870912"	± 0.00000000002"	± 0.00000000002°	
			1/1073741824"	± 0.00000000001"	± 0.00000000001°	
			1/2147483648"	± 0.000000000005"	± 0.000000000005°	
			1/4294967296"	± 0.000000000002"	± 0.0000	

Assembly Drawing. An assembly drawing showing a revision block and a standard title block. (Courtesy of Dynojet Research, Inc.)



A Civil Drawing. The drawing shows approval blocks and engineers' stamp. (Courtesy of Perliter and Ingalsbe Consulting Engineers and Calleguas Municipal Water District.)

WINDOW SCHEDULE					
MK	MANUFACTURER/LINE	TYPE	UNIT SIZE / MODEL #	R.O.	NOTES
A	PITTSBURG CORNING "LIGHTWEE"	GLASS BLOCK	32X80	33 3/16" X 79 7/8"	6X6 GLASS BLOCK PANEL W/ WHITE VINYL FRAME PATTERN TO BE DETERMINED
B1	SUNNIT VINYL	AWNING	24X24	24" X 24"	PROVIDE AUTOMATIC OPENERS FOR CLERESTORY WINDOWS
B2	SUNNIT VINYL	PICTURE	24X24	24" X 24"	
C	PITTSBURG CORNING "LIGHTWEE"	GLASS BLOCK	24X24	25 3/8" X 25 3/8"	6X6 GLASS BLOCK PANEL W/ WHITE VINYL FRAME PATTERN TO BE DETERMINED
D	KOLBE & KOLBE "ULTINUM"	SPECIALTY	48 5/8"	48 1/2" X 54 1/2"	WOOD / CLAD EXTERIOR
E	SUNNIT VINYL				
F	PITTSBURG "LIGHTWEE"				
G	PITTSBURG				

DOOR SCHEDULE							
MK	SIZE	DOOR TYPE	MANUFACTURER/LINE	STYLE	MATERIAL	SPECIAL HARDWARE	NOTE
1	10070	OVERHEAD DOOR		FLUSH PANEL		AUTOMATIC OPENER	(4) WINDOWS IN UPPER PANEL
2	3068	EXTERIOR DOOR	SOUTHWEST DOOR / CLASSIC	18 LITE	ALDER		FINISH & TEXTURE TO BE DETERMINED
3	6080	EXTERIOR DOUBLE FRENCH DOORS	WEATHERSHIELD / PROSHIELD	SINGLE PANE	VINYL		WHITE FRAME, TEMPERED GLASS
4	5068	EXTERIOR DOUBLE FRENCH DOORS	WEATHERSHIELD / PROSHIELD	SINGLE PANE	VINYL		WHITE FRAME, TEMPERED GLASS
5	2668	INTERIOR DOOR		SINGLE PANE			TEMPERED GLASS
6	2468	POCKET DOOR		FLUSH HOLLOW CORE	WOOD		FINISH TO BE DETERMINED
7	2468	POCKET DOOR		FLUSH HOLLOW CORE	WOOD		FINISH TO BE DETERMINED

Window and door schedules are used in architectural drawings to specify the type of window or door, rough opening size, manufacturer, and other information. (Courtesy of Frog Rock Design, LLP.)

KEY WORDS

Approval Block
Architects' Scale
Construction Lines
Decimal Inch
Decimal-Inch Scale
Drawing Lines
Drawing Number
Drawing Scale
Drawing Size
Drawing Title
Engineers' Scale
Font
Freehand Line
Guidelines
Inclined
Italic
Kerned Pairs
Layout
Lettering
Measurement Systems
Mechanical Engineers' Scale
Media
Multiview Projection
Name
Oblique Projection
Orthographic Projections
Parallel Projections
Perspective Projections
Piercing Points
Plane of Projection
Projectors
Roman
Sans Serif
Scale
Scales
Serif
Sheet Number
Sheet Revision Block
Sheet Sizes
Station Point
Thick Lines
Thin Lines
Title Blocks
Vertical
Weight
Zone Numbers

CHAPTER SUMMARY

Now that you have completed this chapter you should be able to:

- Understand the basic principles of projection used in drawings.
- Demonstrate the lineweights (thickness) and types (dashed or solid) of lines used in the alphabet of lines that specify meaning in technical drawings.
- List the two main systems of measurement used on drawings.
- Use different types of scales to make measurements.
- Note the scale for a drawing in the title block. Paper drawings are scaled before they are drawn. CAD drawings are scaled when they are to be printed.
- List the advantages of several different drawing media and the qualities that distinguish them.
- Add legible and quick notes and dimensions to sketches using uppercase letters drawn by hand.
- Lay out a sheet and fill in the information in the title block using standard letter shapes.

REVIEW QUESTIONS

1. Draw the alphabet of lines and label each line.
2. What are the main advantages of polyester film as a drawing media?
3. What are the four standard types of projection?
4. Which drawing lines are thick? Which are thin? Which are very light and should not reproduce when copied?
5. What font provides the shape of standard engineering lettering?
6. Describe the characteristics of good freehand lettering.
7. Why should guidelines be used for lettering?
8. List the standard items found in a title block.

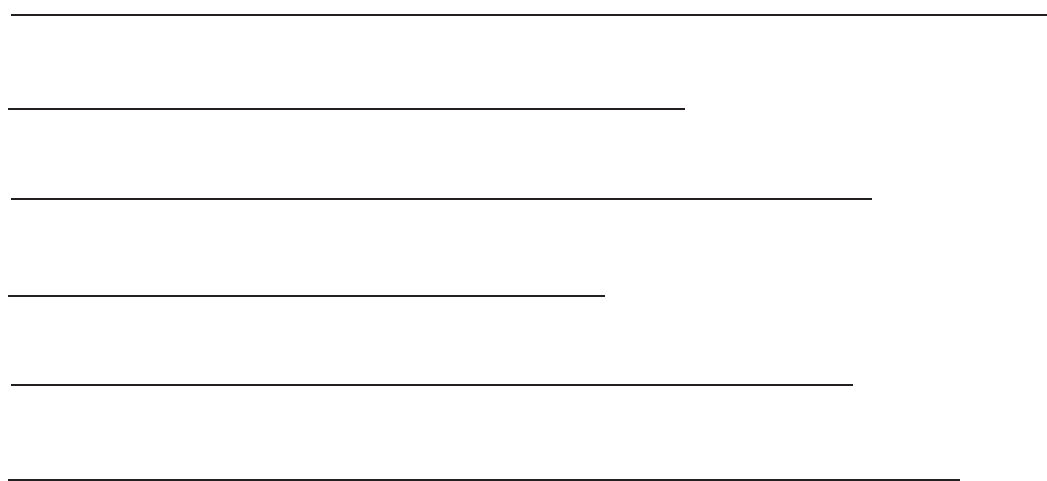
CHAPTER EXERCISES

Drawing Exercises

Practice your skills for making measurements, laying out drawing sheets, and forming neat standard lettering with these drawing exercises.

These problems are designed to fit easily on a sheet. (See the inside front cover of this book). Draw all construction lines lightly, using a hard lead (4H to 6H), and all required lines dense black with a softer lead (F to H). Draw your construction lines lightly so that they do not need to be erased.

In Exercises 2.1–2.3 you will practice measuring, and in Exercises 2.4–2.6 you will practice drawing layouts.



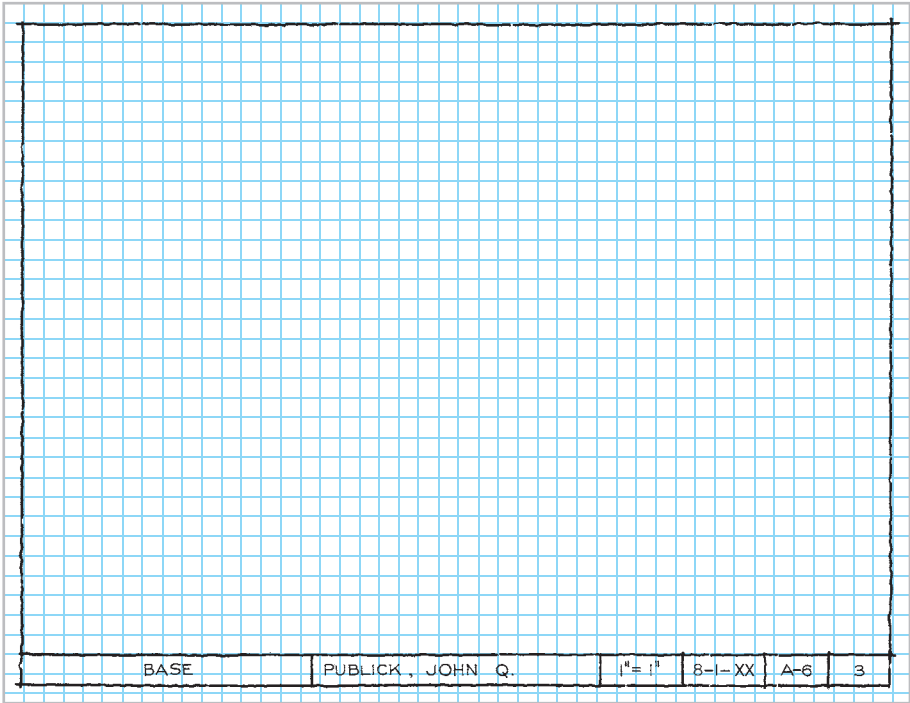
Exercise 2.1 Measure the lines shown above and list their lengths using millimeters. List the inch measurements for each in square brackets [] to the right of the millimeter measurement.

Exercise 2.2 Measure the lines shown above and draw them at Scale 1:2, Scale 2:1, and list their scales below them using the form Scale: X:X.

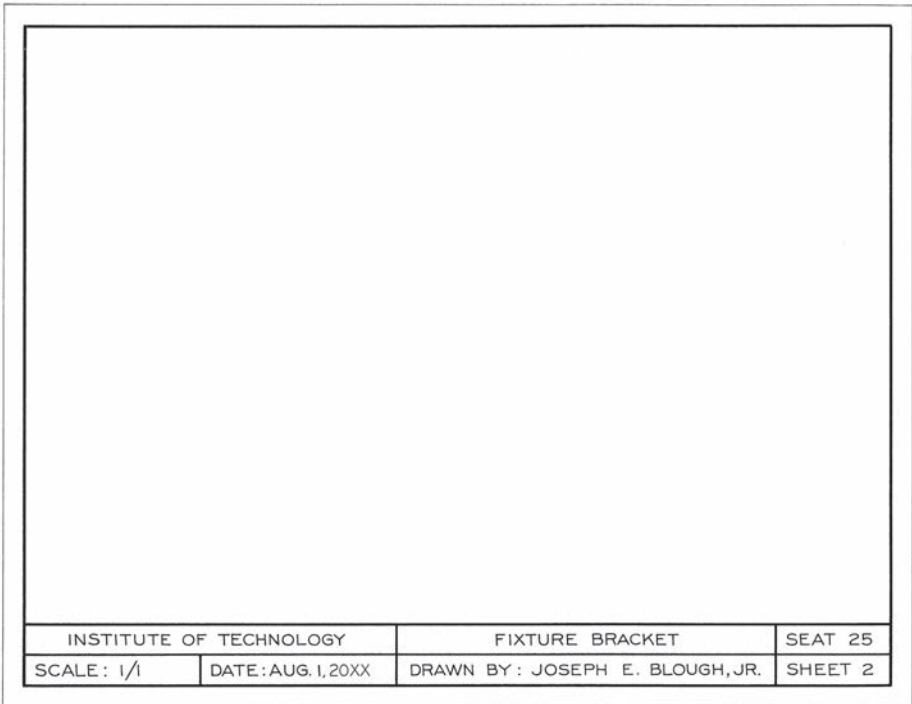
Exercise 2.3 Measure the overall interior dimensions of your room. Letter the measured length neatly in the first column as shown in the example. In the second column list how long you would draw that line at a scale of $1/4'' = 1'$, third column at $3/8'' = 1'$, fourth column at $1'' = 1'$, fifth column at 1:100 metric scale (10 mm = 1 meter).

Measurement	$1/4'' = 1'$	$3/8'' = 1'$	$1'' = 1'$	1:100 Metric
10'-6"	2.625"	3.9375"	10.5"	32mm

Exercise 2.4 Create the layout for an $8.5'' \times 11''$ sheet as shown at right.



Exercise 2.5 Create the layout for the 210 mm $\times$ 297 mm sheet shown at right.

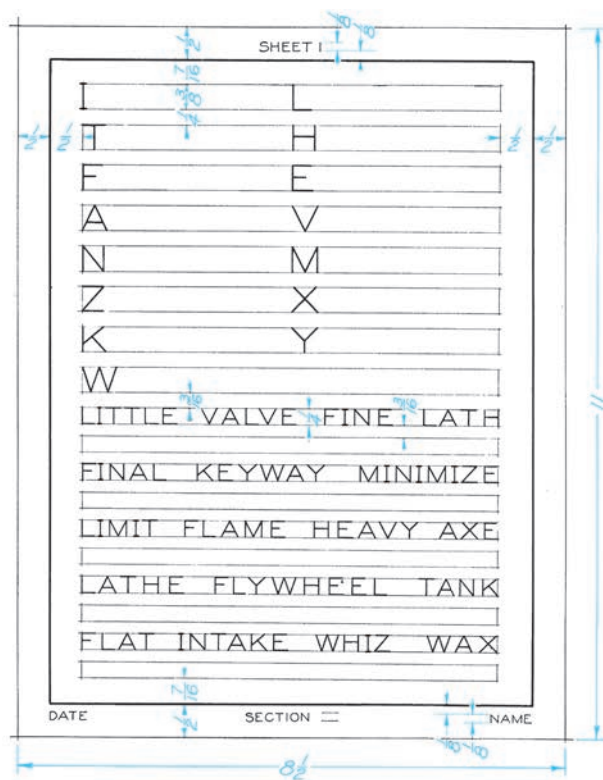


Exercise 2.6 Design a title block and layout for a C-size sheet. Create a name and logo for your company. Use an attractive but legible font for the titles on your layout. If assigned, design a special north arrow to be used on your drawings.

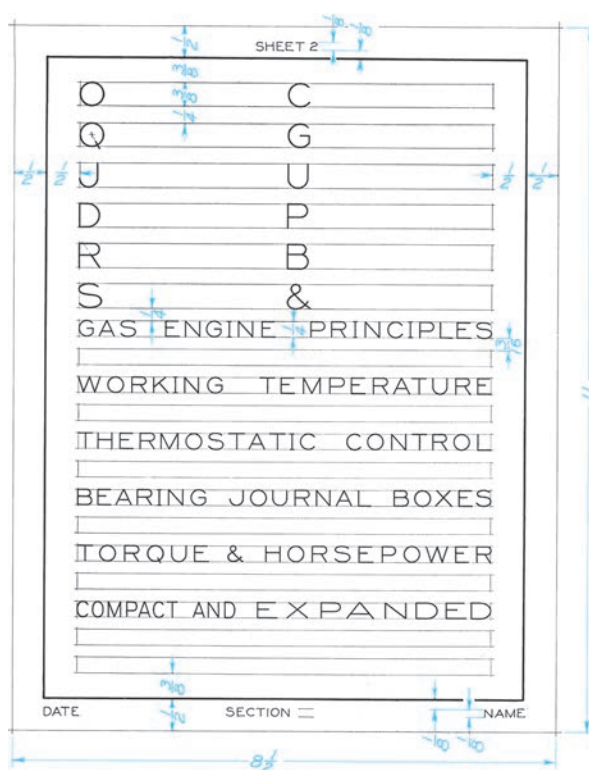
Lettering Exercises

Layouts for lettering problems are given in Exercises 2.7–2.11. Draw complete horizontal and vertical or inclined guidelines very lightly. Draw the vertical or inclined guidelines through the full height of the lettered area of the sheet. For practice in ink lettering, the last two lines and the title strip on each sheet may be lettered in ink, if assigned by the instructor. Omit all dimensions.

Exercise 2.7 Letter the words to your favorite song, joke, or inspirational quote of 50 words or more. Use 1/8" tall UPPERCASE engineering lettering. Center the words near the middle of the sheet. Make sure to leave a row of space between each row of lettering. Make sure that the subject you choose is professional and appropriate.



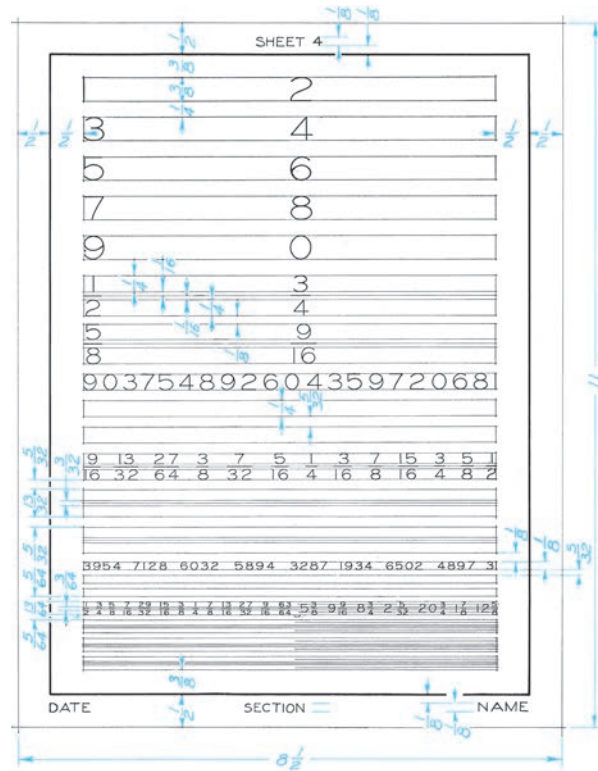
Exercise 2.8 Lay out sheet as shown. Add vertical or inclined guidelines and fill in vertical or inclined capital letters as assigned. For decimal-inch and millimeter equivalents of given dimensions, see the inside back cover.



Exercise 2.9 Lay out sheet as shown. Add vertical or inclined guidelines and fill in vertical or inclined capital letters as assigned. For decimal-inch and millimeter equivalents of given dimensions, see the inside back cover.



Exercise 2.10 Lay out sheet as shown. Add vertical or inclined guidelines and fill in vertical or inclined lowercase letters as assigned. For decimal-inch and millimeter equivalents of given dimensions, see the inside back cover.



Exercise 2.11 Lay out sheet as shown. Add vertical or inclined guidelines and fill in vertical or inclined numerals and fractions as assigned. For decimal-inch and millimeter equivalents of given dimensions, see the inside back cover.

CHAPTER THREE

VISUALIZATION AND SKETCHING

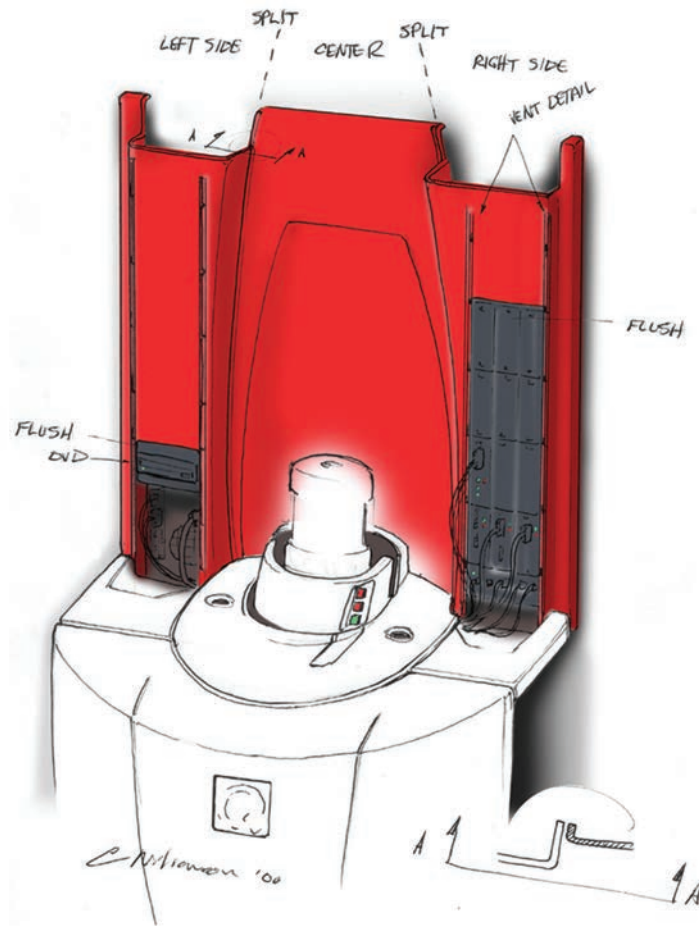
OBJECTIVES

After studying the material in this chapter, you should be able to:

1. Define the terms vertex, edge, plane, surface, and solid.
2. Identify four types of surfaces.
3. Identify five regular solids.
4. Draw points, lines, angled lines, arcs, circles, and ellipses.
5. Apply techniques that aid in creating legible well-proportioned freehand sketches.
6. Apply techniques to draw irregular curves.
7. Create a single-view sketch.
8. Create an oblique sketch.
9. Create perspective sketches.
10. Create an isometric sketch of an object.

Refer to the following standard:

- ANSI/ASME Y14.3 *Orthographic and Pictorial Views*



Shaded Sketch Showing Details of Wire Placement (Courtesy of Quantum Design.)

OVERVIEW

The ability to envision objects in three dimensions is one of the most important skills for scientists, designers, engineers, and technicians. Learning to visualize objects in space, to imagine constructively, is something you can learn by studying technical drawing. People who are extraordinarily creative often possess outstanding ability to visualize, but with practice anyone can improve his or her ability.

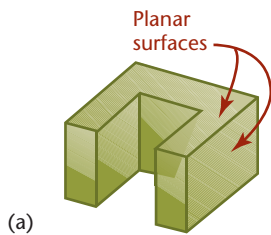
In addition to its role in developing spatial thinking skills, sketching is a valuable tool that allows you

to communicate your ideas quickly and accurately. During the development stage of an idea, a picture is often worth a thousand words.

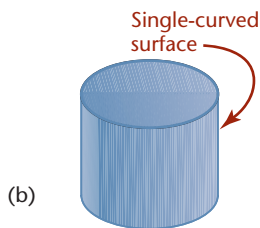
Sketching is also an efficient way to plan your drawing and record notes needed to create a complex object. When you sketch basic ideas ahead of time, you can often complete a final CAD drawing sooner and with fewer errors. Using good technique makes sketching faster, easier, and more legible.



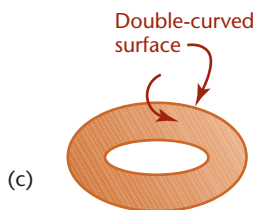
Complex Surface Models Created Using 3D CAD (Courtesy of Professor Richard Palais, University of California, Irvine, and Luc Benard.)



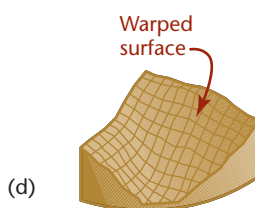
(a)



(b)



(c)



(d)

3.1 Types of Surfaces

UNDERSTANDING SOLID OBJECTS

Sketches and drawings are used to communicate or record ideas about the shape of 3D objects. Before starting to sketch, it helps to develop a vocabulary for understanding and discussing 3D shapes.

Three-dimensional figures are referred to as **solids**. Solids are bounded by the **surfaces** that contain them. These surfaces can be one of the following four types:

- **Planar**
- **Single curved**
- **Double curved**
- **Warped**

Regardless of how complex a solid may be, it is composed of combinations of these basic types of surfaces. Figure 3.1 shows examples of the four basic types of surfaces.

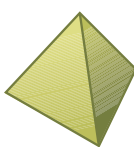
Types of Solids

Polyhedra

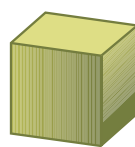
Solids that are bounded by plane surfaces are called **polyhedra** (Figures 3.2–3.4). These planar surfaces are also referred to as **faces** of the object. A polygon is a planar area that is enclosed by straight lines.

Regular Polyhedra

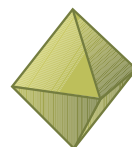
If the faces of a solid are equal regular polygons, it is called a **regular polyhedron**. There are five regular polyhedra: the tetrahedron, hexahedron, octahedron, dodecahedron, and icosahedron (Figure 3.2).



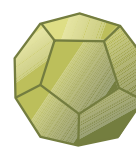
Tetrahedron
(4 triangles)



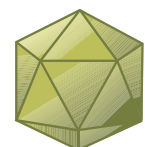
Hexahedron
(cube)



Octahedron
(8 triangles)



Dodecahedron
(12 pentagons)



Icosahedron
(20 triangles)

3.2 Regular Polyhedra

Prisms

A **prism** has two bases, which are parallel equal polygons, and three or more additional faces, which are parallelograms (Figure 3.3). A triangular prism has triangular bases, a rectangular prism has rectangular bases, and so on. (If a prism's bases happen to be parallelograms, the prism is called a parallelepiped, a word rarely heard in everyday conversation.)

A right prism has faces and lateral (side) edges that are perpendicular to the bases; an oblique prism has faces and lateral edges that are angled to the bases. If one end is cut off to form an end that is not parallel to the bases, the prism is said to be truncated (a word that simply means “shortened by having a part cut off”).

Pyramids

A **pyramid** has a polygon for a base and triangular lateral faces that intersect at a common point called the vertex (Figure 3.4). The line from the center of the base to the vertex is called the axis. If the axis is perpendicular to the base, the pyramid is called a right pyramid; otherwise, it is an oblique pyramid. A triangular pyramid has a triangular base, a square pyramid has a square base, and so on. If a portion near the vertex has been cut off, the pyramid is truncated; if the face formed by truncation is parallel to the base, the shape is referred to as a *frustum*.

Cylinders

A **cylinder** has a single-curved exterior surface (Figure 3.5). You can think of a cylinder as being formed by taking a straight line and moving it in a circular path to enclose a volume. Each position of this imaginary straight line in its path around the axis is called an *element* of the cylinder.

Cones

A **cone** has a single-curved exterior surface (Figure 3.6). You can think of it as being formed by moving one end of a straight line around a circle while keeping the other end fixed at a point, the vertex of the cone. An element of the cone is any position of this imaginary straight line.

Spheres

A **sphere** has a double-curved exterior surface (Figure 3.7). You can think of it as being formed by revolving a circle about one of its diameters, somewhat like spinning a coin. The poles of the sphere are the points at the top and bottom of the sphere that would not move while it was spinning. The axis of the sphere is the term for the line between its poles.

Tori

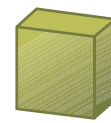
A **torus** is shaped like a doughnut (Figure 3.8). Its boundary surface is double curved. You can think of it as being formed by revolving a circle (or other closed curve) around an axis that is positioned away from (outside) the curve.

Ellipsoids

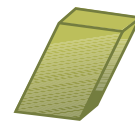
An oblate or prolate **ellipsoid** is shaped like an egg (Figure 3.9). You can think of it as being formed by revolving an ellipse about its minor or major axis, respectively.



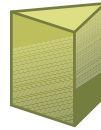
Right square



Right rectangular



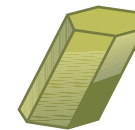
Oblique rectangular



Right triangular



Right pentagonal

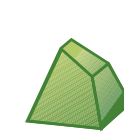


Oblique hexagonal

3.3 Right Prisms and Oblique Prisms



Right rectangular



Right square (truncated)

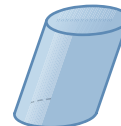


Oblique pentagonal

3.4 Pyramids



Right circular



Oblique circular

3.5 Cylinder and Oblique Cylinder



Right circular



Oblique circular (frustum)



Oblique circular (truncated)

3.6 Cones



Sphere



Torus

3.7 Sphere

3.8 Torus



Oblate Ellipsoid



Prolate Ellipsoid

3.9 Ellipsoids

UNDERSTANDING SKETCHING TECHNIQUES

Analyzing Complex Objects

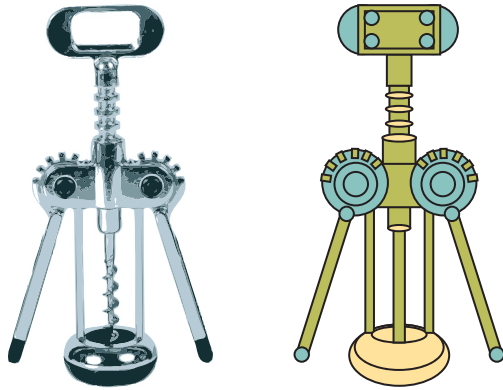
The ability to break down complex shapes into simpler geometric primitives is an essential skill for sketching and modeling objects.

Before you begin to draw the outline of an object, consider its overall shape and the relationships between its parts. Construction lines can help you preserve the overall dimensions of the object as you sketch.

Bear in mind that you should be thinking in terms of basic shapes whether you are sketching by hand or using a CAD program. Because basic curves and straight lines are the basis of many of the objects that people create, practice in creating the basic elements of a drawing will help you sketch with ease.



3.10 Identifying Essential Shapes



3.11 The essential shapes of the corkscrew in the photo are sketched at right.

Essential Shapes

Look for the essential shapes of objects. If you were to make a clay model of an object, what basic shape would you start with? A ball? A box?

Try squinting your eyes and looking at familiar objects. Do you see their shape as a rectangle? A circle? What other basic shapes do you notice when you look at objects this way?

Think about breaking down more complex objects into their simpler geometric shapes as shown in Figure 3.10. You can block in these shapes using construction lines to show their relationships to one another. Then, add details, continuing to pay attention to the spatial relationships between them.

Construction Lines

Artists often begin a sketch by blocking in light guidelines to help them preserve basic shapes and proportions. In technical drawing these are called **construction lines** (Figure 3.12).

It is often helpful to begin a sketch by describing the object's main shapes with construction lines, taking some care to accurately represent the relative size and placement of features.

Use the basic shapes as a guide to place key features, then use those main features as a “reference map” to place smaller details. For example, the sixth fret line is about halfway up the rectangular guitar neck.

Throughout this chapter you will use light construction lines to draw circles, arcs, and ellipses. Section 3.4 discusses the process of estimating and maintaining the proportions of an object in further detail.

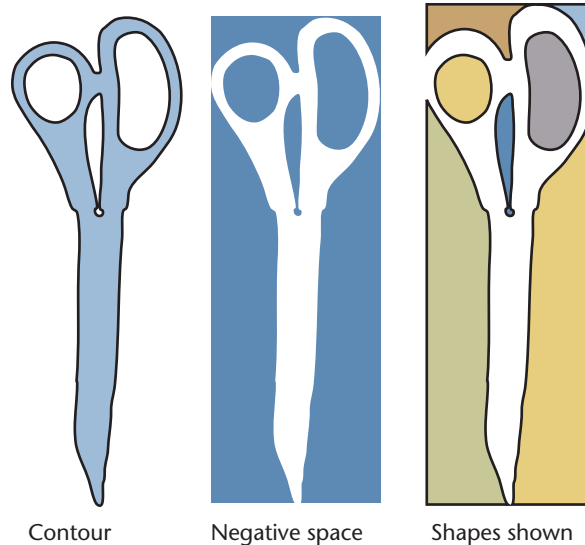


3.12 Using Construction Lines

Contours and Negative Space

The **contours** of an object are the main outlines that separate it from the surrounding space. One way to think about the contours of objects is to look at the contrast between the positive and negative space. Positive space is the space occupied by the object. **Negative space** is the unoccupied space around it.

In Figure 3.13 the space occupied by the contour of a pair of scissors is shown. Note how you can identify specific shapes by looking at the negative space. The individual shapes that make up the negative space are shown in different colors to make them easier for you to see. Some people sketch more accurately when they try to draw the negative space that surrounds the object.



3.13 Negative Space

TIP



Practice Drawing Contours

Try sketching the negative spaces that define the shape of a chair. Look at each space as an individual shape. What is the shape of the space between the legs? What is the shape of the space between the rungs and the seat?

Make a sketch of a chair, paying careful attention to sketching the negative spaces of the chair as they really appear. The positive and negative spaces should add up to define the chair.

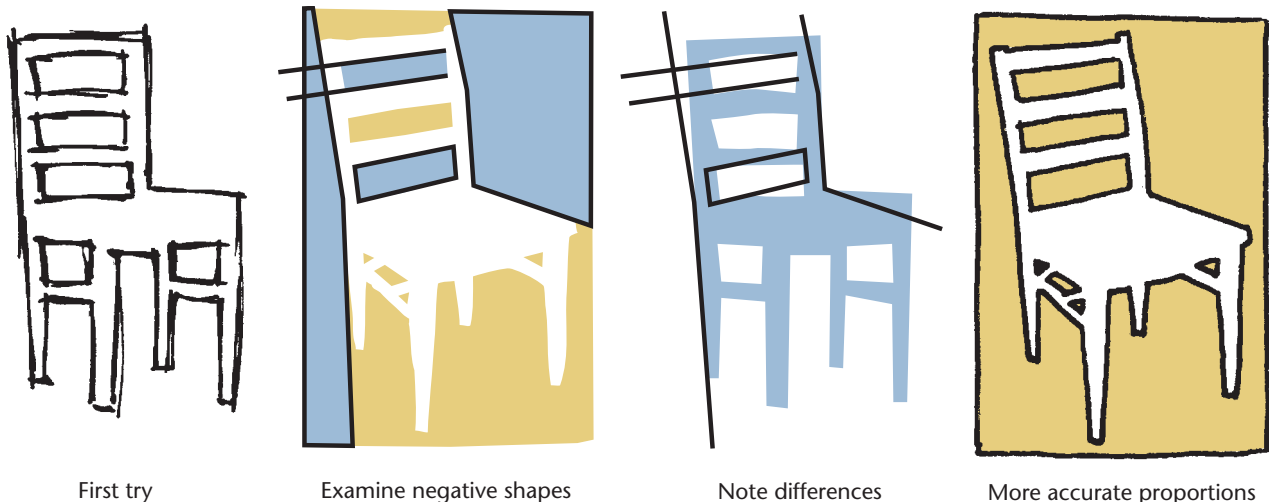
If you have difficulty, make corrections to your sketch by defining the positive shapes and then check to see if the negative shapes match.

An 8.5" × 11" sheet of Plexiglas® (available at most glass stores) is an excellent

tool for developing sketching ability. Using a dry-erase marker, hold the Plexiglas up in front of an object and trace the object's contours on the Plexiglas. If you don't move, the outline should match the object's outline exactly. Lower the Plexiglas and look at the orientation of the lines. Are they what you expected?

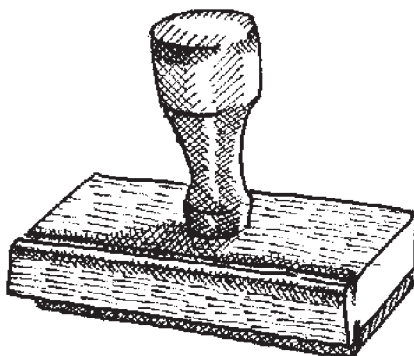
Try looking at the object and drawing the sketch with the Plexiglas lying on your desktop or knees. Then, raise it and see if your drawing matches the object.

To develop sketching ability, try drawing everyday objects like your toaster, printer, or lamp, as well as exterior and interior views of buildings and equipment.

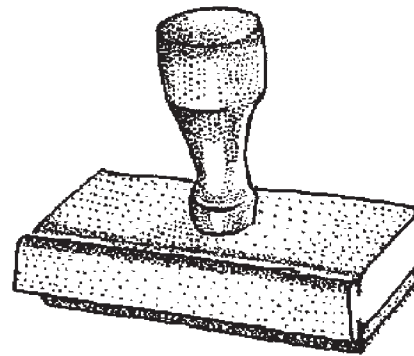




3.14 Rubber Stamp



3.15 Hatching



3.16 Stippling

Viewpoint

As you sketch objects, keep in mind that you want to maintain a consistent **viewpoint**, as a camera does. This is easier when you are sketching a picture from a book, because you can't move around the object. When you move, you see a different view of the object depending on where you stand.

Sometimes people have difficulty sketching because they want to show parts of the object that cannot really be seen from a single viewpoint. For example, knowing that the handle of the rubber stamp in Figure 3.14 appears circular from the top, you may be tempted to show it as round, even though it may appear elliptical from your viewpoint.

When you are sketching an object pictorially, temporarily set aside your knowledge of the shapes the object is actually made of and carefully examine the shapes you see from a single, static viewpoint. In this type of sketching, instead of trying to envision the object as it is, try only to see it as it looks.

Shading

Adding **shading** to your sketch can give it a more realistic appearance because it represents the way the actual object would reflect light. Shading doesn't mean "coloring in." You may want to shade only the most prominently shadowed areas. First, identify the darkest and lightest areas on an object. If you want, you can shade various middle tones, placed exactly as they look on the object.

In some ways, shading is like doing a drawing within a drawing, because it is a matter of identifying shapes. When you are shading, instead of identifying the shapes of the object's contours, you are identifying the shape and relative darkness of the shadows.

Hatching lines, shown in Figure 3.15, and stippling, shown in Figure 3.16, are commonly used methods to add shading because they are easier to reproduce with a photocopier than continuous-tone pencil shading. In the illustration, you can see that shadowed areas are darkened simply by adding more hatching lines or stippling dots.

It is not uncommon for people to draw outlines by hand and add digital shaded fills to a scan of the outline. Another way to make a subject the clear focal point of a drawing is to stylize the shadows. Industrial designers often use markers to add a stylized shadow to their sketches. You can use a straight

edge to protect the original sketch from the marker and quickly sketch the shadow (Figure 3.17).

Regardless of how you apply shading, darken the outline to define the shape clearly and boldly. Remember that when you are communicating by using a sketch, its subject should be clear. To make the subject—in this case, a rubber stamp—clear, make it stand out with thick bold contour lines.



3.17 Marker Shading in a Concept Sketch (Courtesy of Douglas Wintin.)

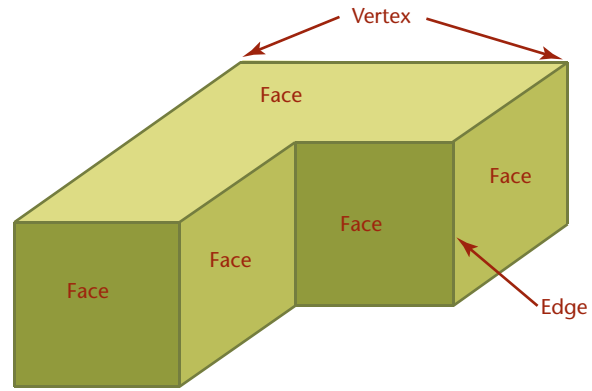
Edges and Vertices

Edges

An **edge** of a solid is formed where two surfaces intersect. Edges are represented in drawings by visible or hidden lines (Figure 3.18).

Vertices

A **vertex** (plural, vertices) of a solid is formed where three or more surfaces intersect. The end of an edge is a vertex. These vertices or “points” are very useful in defining the locations of the solid object feature that you will sketch (Figure 3.18).

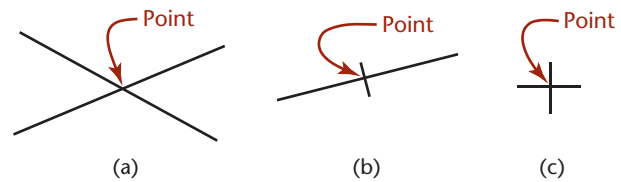


3.18 Edges and Vertices of a Solid

Points and Lines

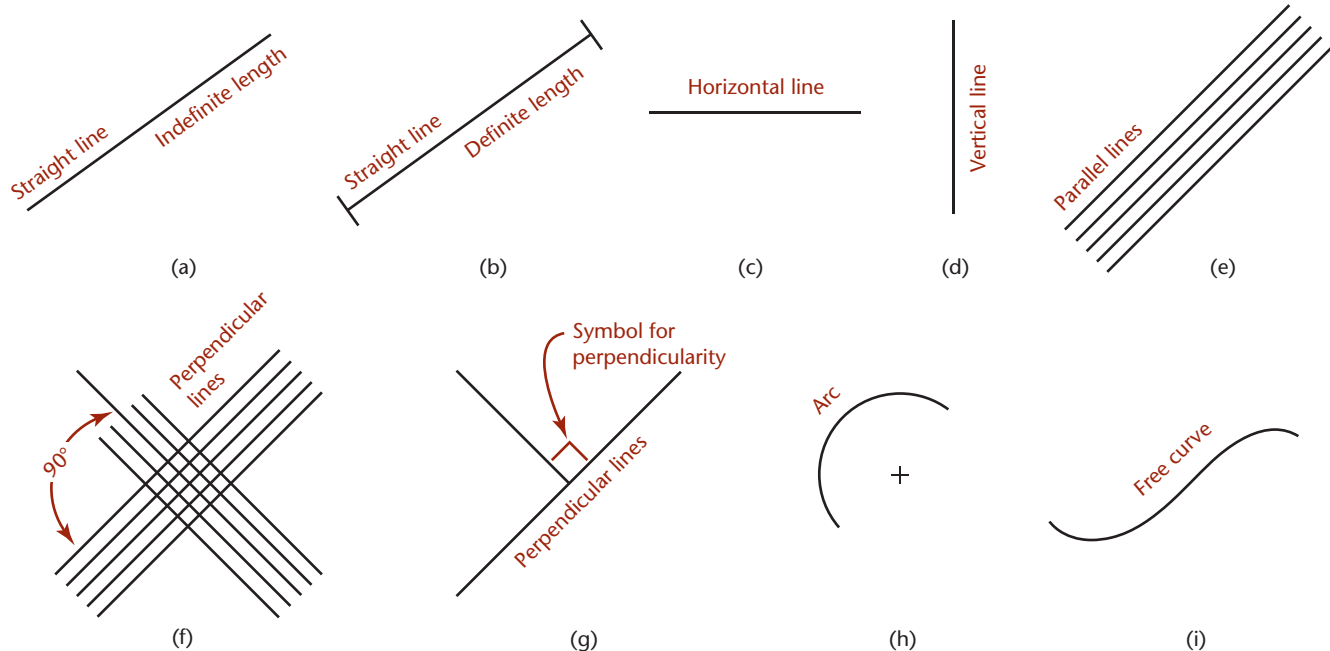
A **point** is used to represent a location in space but has no width, height, or depth (Figure 3.19). A point in a drawing is represented by the intersection of two lines (Figure 3.19a), by a short crossbar on a line (Figure 3.19b), or by a small cross (Figure 3.19c). Do not represent points by simple dots on the paper. This makes the drawing look “blobby” and is not as accurate.

A **line** is used in drawings to represent the edge of a solid object. A straight line is the shortest distance between two points and is commonly referred to simply as a “line.” If the line is indefinite in extent, in a drawing the length is a matter of convenience, and the endpoints are not marked (Figure 3.20a). If the endpoints of the line are significant, they are marked by small drawn crossbars (Figure 3.20b). Other common terms are illustrated in Figures 3.20c to i. Either straight lines or curved lines are parallel if the shortest distance between them remains

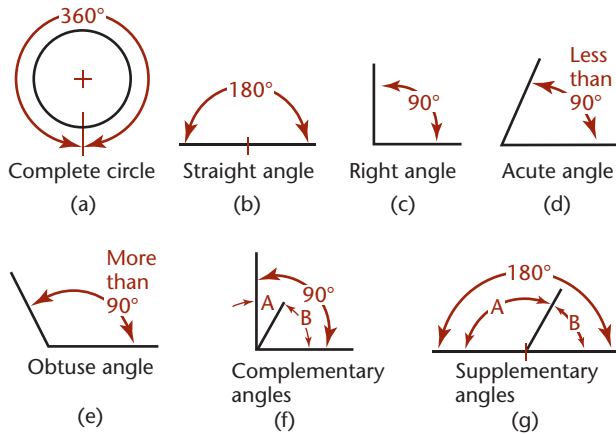


3.19 Showing Points

constant. The common symbol for parallel lines is $\parallel$, and for perpendicular lines it is $\perp$. Two perpendicular lines may be marked with a “box” as shown in Figure 3.20g. Such symbols may be used on sketches, but not on production drawings.



3.20 Showing Lines



3.21 Showing Angles

Angles

An **angle** is formed by two intersecting lines. A common symbol for angle is $\angle$.

There are 360 degrees (360°) in a full circle, as shown in Figure 3.21a. A degree is divided into 60 minutes ($60'$), and a minute is divided into 60 seconds ($60''$). The angle value $37^\circ 26' 10''$ is read 37 degrees, 26 minutes, and 10 seconds. When minutes alone are indicated, the number of minutes should be preceded by $0'$, as in $0^\circ 20'$. If the minutes value

were listed alone without showing the zero value, it might be mistaken for a distance measurement. For example, 20 minutes, 10 seconds written as $20'10''$ might be mistaken for 20 feet 10 inches.

The different kinds of angles are illustrated in Figure 3.21. Two angles are complementary if they total 90° (Figure 3.21f) and are supplementary if they total 180° (Figure 3.21g).

In sketching, most angles can be estimated. Use a protractor if necessary when drawing odd angles.

Drawings and Sketches

The following are important skills to keep in mind for sketches and drawings:

1. Accuracy. No drawing is useful unless it shows the information correctly.
2. Speed. Time is money in industry. Work smarter and learn to use techniques to speed up your sketching and CAD drawings while still producing neat accurate results.
3. Legibility. A drawing is a means of communicating with others, so it must be clear and legible. Give attention to details. Things that may seem picky and small as you are drawing may be significant and save money or even lives when the product is built.
4. Neatness. If a drawing is to be accurate and legible, it must also be clean.



Accuracy and legibility in drawings is serious business, as is demonstrated in this article from New Scientist magazine about the September 2004 Genesis dizzy probe crash.

(Photo courtesy of NASA.)

Dizzy Probe Crash

You would think that by now, NASA should be able to tell up from down. Not so in the case of their Genesis space capsule, which crashed into the desert in Utah instead of parachuting gently down so that helicopter stunt pilots could pluck it to safety.

Since its launch in August 2001, the capsule had been collecting precious particles from the solar wind that would have told us something about the composition of the solar system.

But after reentering Earth's atmosphere on September 8, 2004, Genesis plunged into the ground, and much of its payload was lost.

On October 15, investigators released their preliminary conclusions, blaming the crash on "a design error that involves the orientation of gravity-switch devices." Huh?

The four small cylindrical switches were designed to sense the reentry and trigger the parachute. But they

were drawn upside down in Lockheed Martin's technical drawings, so they were installed upside down—although NASA's Michael Ryschkewitsch, who led the investigation, is reluctant to use those exact words. The switches never detected the reentry. Similar devices are installed on another sample-collecting mission called Stardust. Ryschkewitsch believes these were installed the right way up.

(Courtesy of New Scientist magazine.)

ACCURACY COUNTS

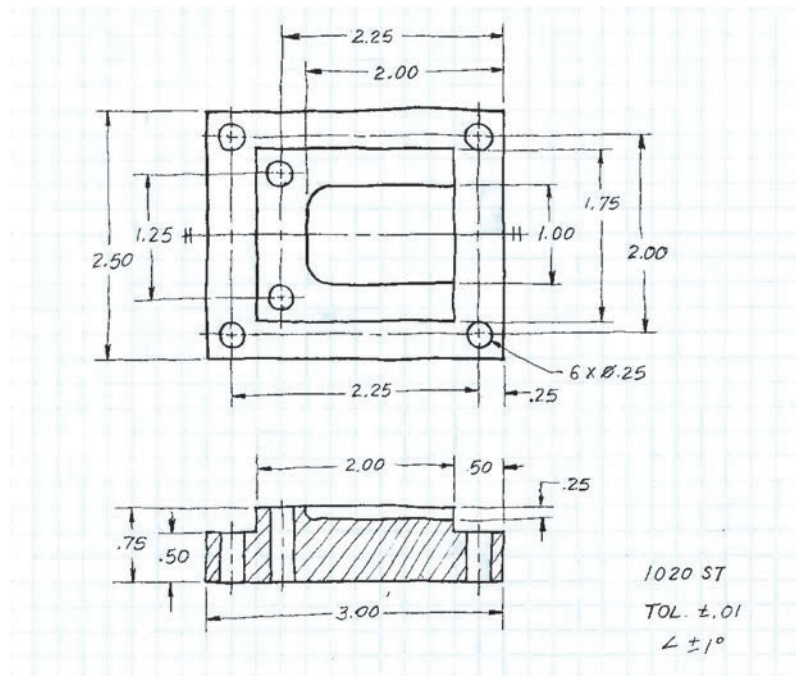
Freehand Sketching

Freehand sketches are a helpful way to organize your thoughts and record ideas. They provide a quick, low-cost way to explore various solutions to design problems so that the best choices can be made. Investing too much time in creating a detailed layout before exploring your options through sketches can be costly.

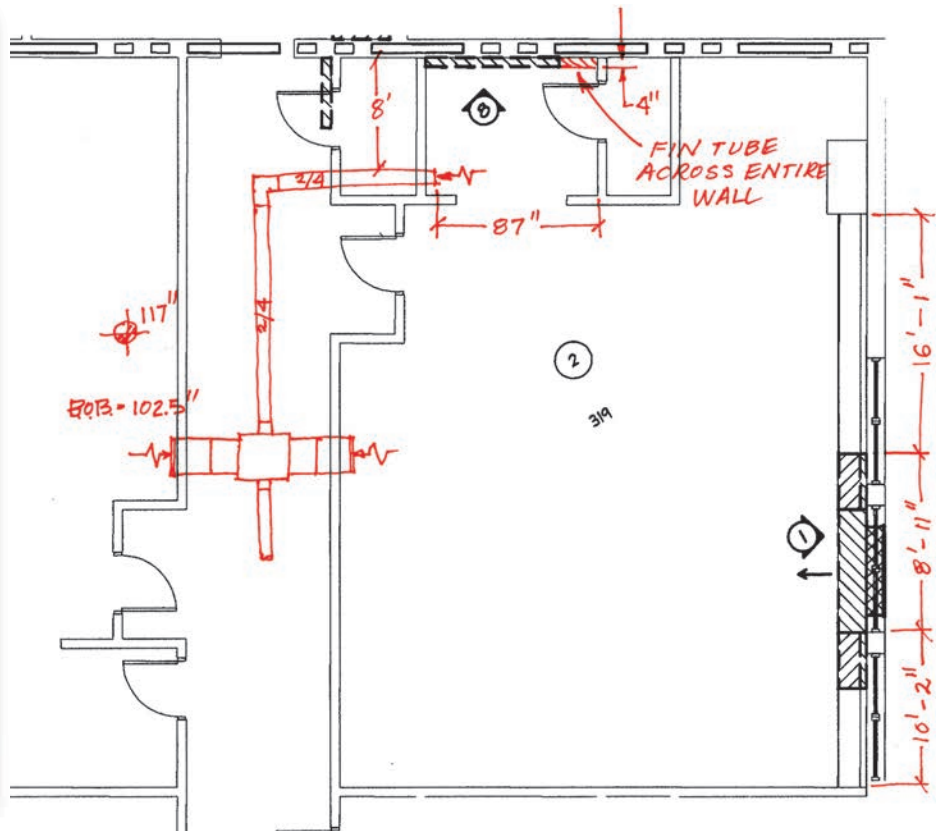
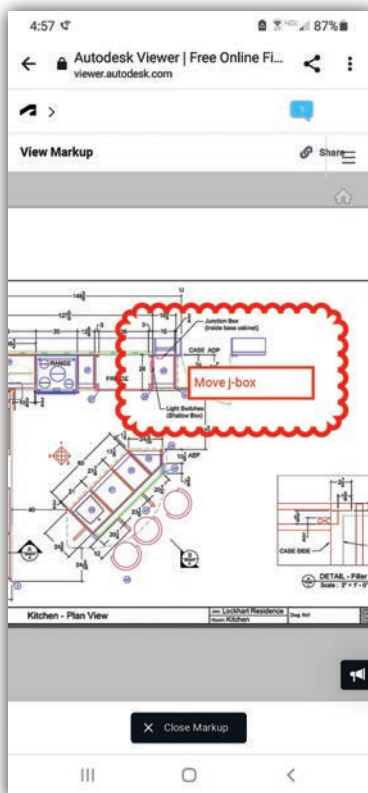
The degree of precision needed in a given sketch depends on its use. Quick sketches to supplement verbal descriptions may be rough and incomplete. Sketches can be used to convey important and precise information when they are clearly drawn and annotated.

Freehand sketching requires only pencil, paper, and eraser. Mastering the techniques in this chapter for showing quick single-view, oblique, perspective, and isometric drawings using good freehand line technique will give you a valuable tool for communicating your ideas.

The term *freehand sketch* does not mean a sloppy drawing. As shown in Figure 3.22, a freehand sketch shows attention to proportion, clarity, and correct line widths. Figure 3.23 shows an as-built drawing with corrected items sketched on the printed CAD drawing.



3.22 Sketch on Graph Paper. Sketches are also used to clarify information about changes in design or to provide information on repairing existing equipment.

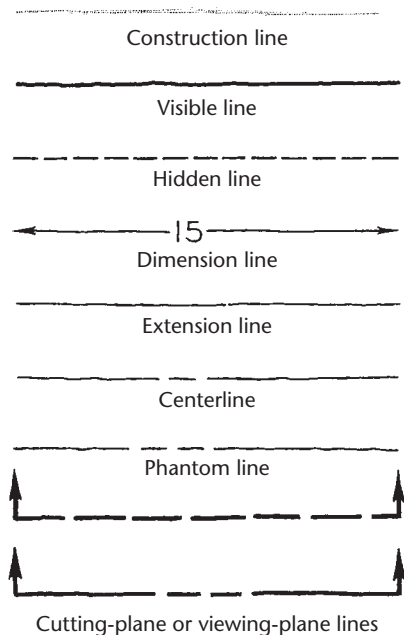


3.23 An As-Built Drawing with Corrected Items Sketched on the Printed CAD Drawing. Computers and even cell-phones can run apps allowing you to mark up drawings, as shown in inset at left. (Autodesk screen shots reprinted courtesy of Autodesk, Inc.)

3.1 TECHNIQUE OF LINES

The chief difference between a drawing and a freehand sketch lies in the character or technique of the lines. A good freehand line is not expected to be as rigidly straight or exactly uniform. A good freehand line shows freedom and variety, whereas a line drawn using CAD or instruments should be exact. Still, it is important to distinguish between line patterns to make your drawing legible.

The **line patterns** in Figure 3.24 are examples of good freehand quality. Figure 3.25 shows examples of good and poor technique.



3.24 Freehand Alphabet of Lines (Full Size)

Lineweights

- Make dimension, extension, and centerlines thin, sharp, and black.
- Make hidden lines medium and black.
- Make visible and cutting-plane lines thick and black.
- Make construction lines thick and light.

TIP

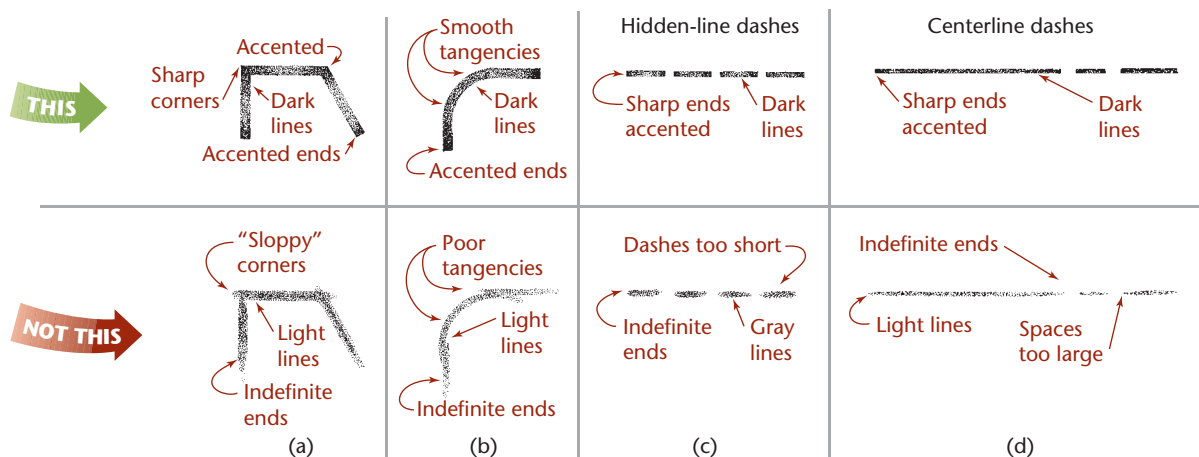
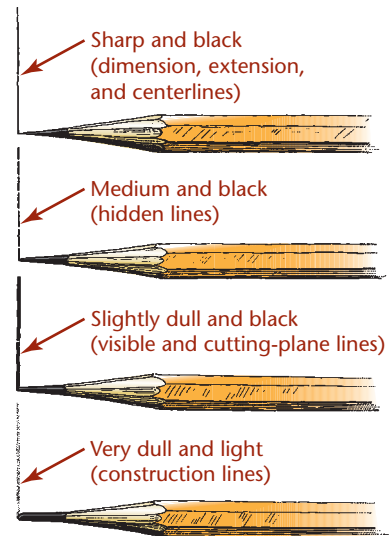
Even in freehand drawings, thick lines should be twice the width of thin lines.

Thicknesses do not have to be exact, but there should be an obvious difference between thick and thin lines.

Because visible lines and cutting-plane lines are the two thick line patterns, other lines should be distinctly thinner in comparison.

To draw thick and thin lines freehand, you might like to keep two pencils handy, one that is razor sharp for thin lines and another that is dull, to create thicker lines.

As the sharp point becomes dull, switch it with the dull pencil, and sharpen the other, so that there is always one sharp and one dull point ready to use.



3.25 Technique of Lines (Enlarged)

3.2 SKETCHING STRAIGHT LINES

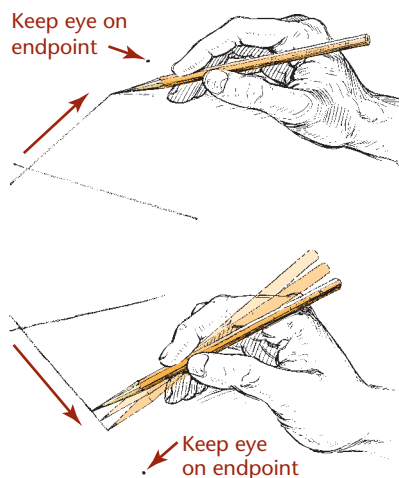
Most of the lines in an average sketch are straight lines. With practice, your straight lines will naturally improve, but these basics may help you improve quickly.

- Hold your pencil naturally, about 1" back from the point, and approximately at a right angle to the line to be drawn.
- Draw horizontal lines from left to right with a free and easy wrist and arm movement.
- Draw vertical lines downward with a wrist and arm movement.
- Draw curved lines using finger and wrist movements.

TIPS

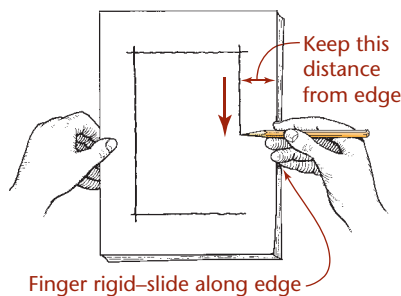
Drawing Long Freehand Lines

For long freehand lines, make light end marks and lightly sweep your pencil between them, keeping your eye on the mark toward which you are moving. When you are satisfied with the accuracy of your strokes, apply more pressure to make a dark line.



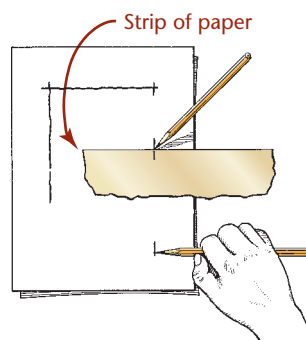
Blocking in a Border Freehand

Hold your hand and pencil rigidly and glide your fingertips along the edge of the paper to maintain a uniform border.



Blocking in a Border Using a Strip of Paper

Mark the distance on the edge of a card or a strip of paper and use it like a ruler to mark at intervals, then draw a final line through the points.



Blocking in a Freehand Drawing

Over the years, freehand sketchers have developed all sorts of tricks to improve speed and accuracy. Methods for finding midpoints or quickly blocking in straight vertical and horizontal lines are just a few secrets of the technical sketching craft that can come in handy, even today. When a great idea hits, or you need to sketch quickly at a meeting or on a job site, you might not have access to a CAD system, or even a ruler.

If your line looks like this, you may be gripping your pencil too tightly or trying too hard to imitate mechanical lines.



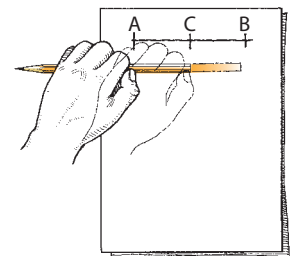
Slight wiggles are OK as long as the line continues on a straight path.



Occasional very slight gaps are fine and make it easier to draw straight.

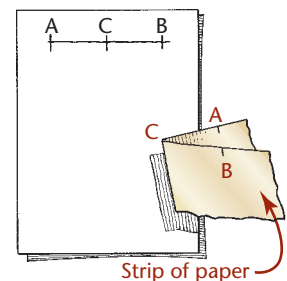
Finding a Midpoint Freehand

Use your thumb on your pencil to guess half the distance. Try this distance on the other half. Continue adjusting until you locate the center, then mark it.



Folding a Paper to Find a Midpoint

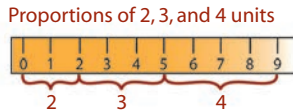
Mark the total distance on the edge of a strip of paper, then fold the paper to locate its center at the crease. You can fold one half to find quarter points, and so on.



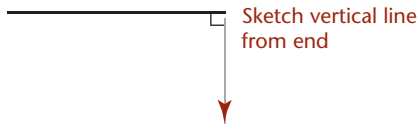
DIVIDING LINES INTO EQUAL OR PROPORTIONAL PARTS

Proportional Parts

To divide the given line shown into proportions of (for example) 2, 3, and 4 units:

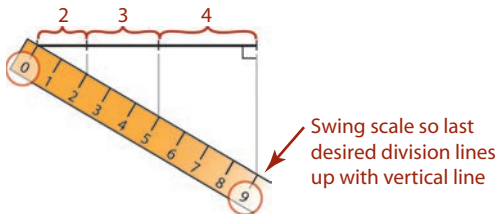


- 1 Draw a vertical construction line at one end of the line you are dividing.



- 2 Set the zero point of your scale at the other end of the line.

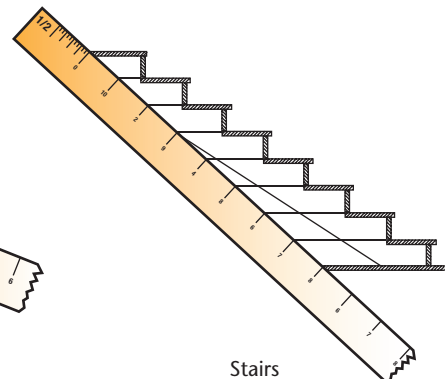
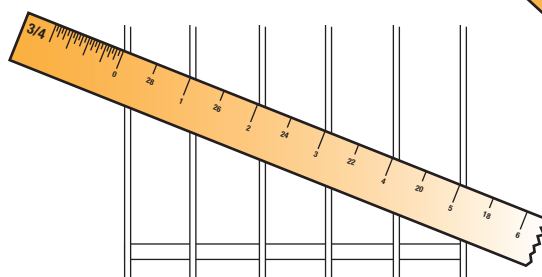
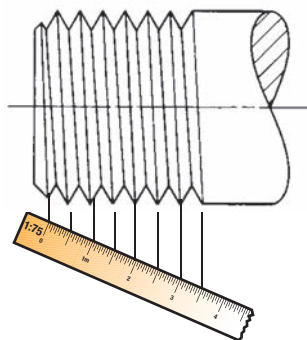
- 3 Swing the scale so the desired unit falls on the vertical line. In this case it will be the 9th unit, because $2 + 3 + 4 = 9$.



- 4 Draw vertical lines upward from the corresponding scale divisions and mark tiny crossbars on the line as shown.

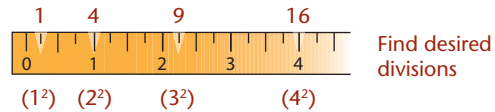
Equal Parts

If you use uniform divisions for the steps above (every third division, for instance) you will get equal parts. Examples of practical applications for dividing lines equally are shown below.

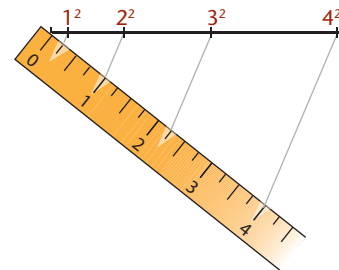


Calculated Proportions

To divide a line into proportions equal to the square of 1, 2, 3, and 4 (1, 4, 9, and 16), find 16 divisions on your scale.



- 1 Set the zero point of your scale at the end of the line and draw a light construction line at any convenient angle from one end of the line you are dividing to the appropriate division on the scale. In this case the 4 mark is 16 equal divisions from the 0.
- 2 Draw construction lines parallel to the end line through each proportionate scale division.



TIP

Exaggerating Closely Spaced Parallel Lines

Sometimes it is helpful to exaggerate the distance between closely spaced parallel lines so there is no fill-in when the drawing is reproduced.

Usually this is done to a maximum of 3 mm or .120". When using CAD it is better to draw the features the actual size and include a detail showing the actual spacing.



3.26 Many objects have rounded features that circles, arcs, and ellipses are used to represent. (Dennis MacDonald/Alamy Stock Photo.)

3.3 SKETCHING CIRCLES, ARCS, AND ELLIPSES

Circles

Small circles can be sketched using one or two strokes, without blocking in any construction lines. Circle templates also make it easy to sketch circles of various sizes.

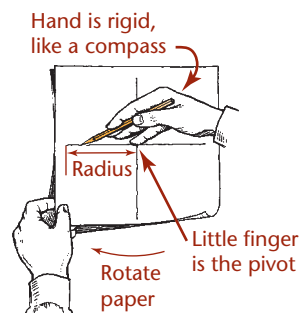
For better looking freehand sketched circles of larger sizes, try the construction methods shown here. Figure 3.26 shows an object with rounded features to sketch using circles, arcs, and ellipses.

TIP

The Freehand Compass

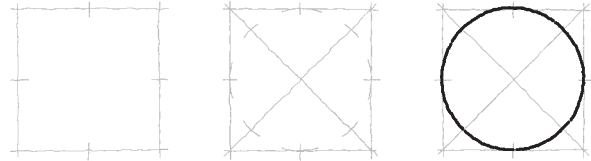
Using your hand like a compass, you can create circles and arcs with surprising accuracy after a few minutes of practice.

1. Place the tip of your little finger or the knuckle joint of your little finger at the center.
2. "Feed" the pencil out to the radius you want as you would do with a compass.
3. Hold this position rigidly and rotate the paper with your free hand.



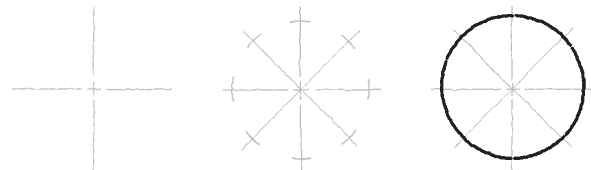
METHODS FOR SKETCHING CIRCLES

Enclosing Square Method



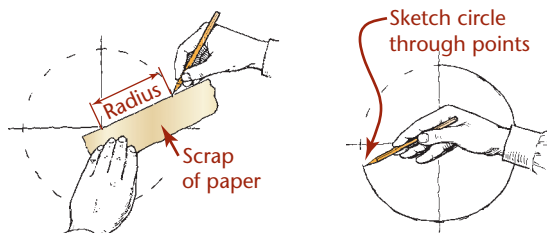
- 1 Lightly sketch an enclosing square and mark the midpoint of each side.
- 2 Lightly draw in arcs to connect the midpoints.
- 3 Darken the final circle.

Centerline Method



- 1 Sketch the two centerlines of the circle.
- 2 Add light 45° radial lines and sketch light arcs across them at an estimated radius distance from the center.
- 3 Darken the final circle.

Paper Method



- 1 Mark the estimated radius on the edge of a card or scrap of paper and set off from the center as many points as desired.
- 2 Sketch the final circle through these points.

METHODS FOR SKETCHING ARCS

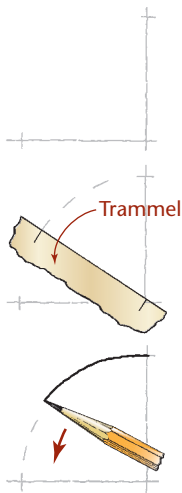
Radius Method

- 1 Locate the center of the arc and lightly block in perpendicular lines. Mark off the radius distance along the lines.
- 2 Draw a 45° line through the center point and mark off the radius distance along it.
- 3 Lightly sketch in the arc as shown. Darken the final arc.



Trammel Method

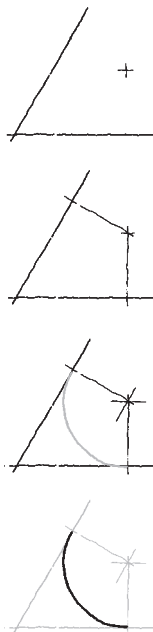
- 1 Locate the center of the arc and lightly block in perpendicular lines. Mark off the radius distance along the lines.
- 2 Mark the radius distance on a strip of paper and use it as a trammel.
- 3 Lightly sketch in the arc, then darken the final arc.



Tangent Method

Use these steps to draw arcs sketched to points of tangency.

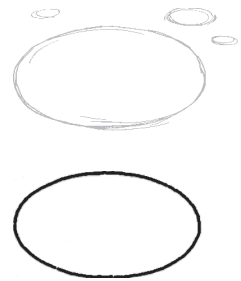
- 1 Locate the center of the arc and sketch in the lines to which the arc is tangent.
- 2 Draw perpendiculars from the center to the tangent lines.
- 3 Draw in the arc tangent to the lines ending at the perpendicular lines.
- 4 Darken in the arc and then darken the lines from the points of tangency.



METHODS FOR SKETCHING ELLIPSES

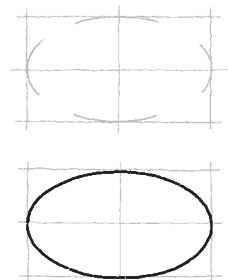
Freehand Method

- 1 Rest your weight on your upper forearm and move the pencil rapidly above the paper in an elliptical path.
- 2 Lower the pencil to draw very light ellipses.
- 3 Darken the final ellipse.



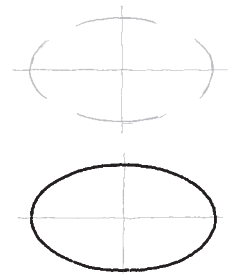
Rectangle Method

- 1 Lightly sketch an enclosing rectangle.
- 2 Mark the midpoint of each side and sketch light tangent arcs.
- 3 Darken in the final ellipse.



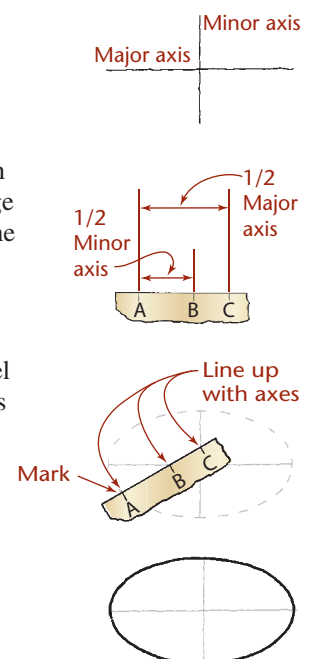
Axes Method

- 1 Lightly sketch in the major and minor axes of the ellipse.
- 2 Mark the distance along the axes and lightly block in the ellipse.
- 3 Darken the final ellipse.



Trammel Method

- 1 To sketch accurate ellipses, you can make a trammel.
- 2 Mark half the desired length of the minor axis on the edge of a strip of paper (A-B). Using the same starting point, mark half the length of the major axis (A-C). (The measurements will overlap.)
- 3 Line up the last two trammel points (B and C) on the axes and mark a small dot at the location of the first point (A).
- 4 Move the trammel to different positions, keeping B and C on the axes, and mark more points at A. Sketch the final ellipse through the points.



Sketching Arcs

Sketching arcs is similar to sketching circles. In general, it is easier to hold your pencil on the inside of the curve. Look closely at the actual geometric constructions and carefully approximate points of tangency so that the arc touches a line or other entity at the right point.



3.27 A Circle Seen as an Ellipse

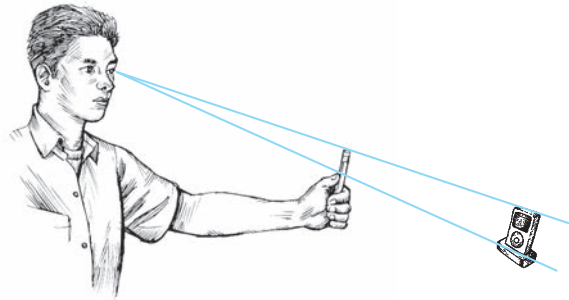
Sketching Ellipses

If a circle is tipped away from your view, it appears as an ellipse. Figure 3.27 shows a coin viewed so that it appears as an ellipse. You can learn to sketch small ellipses with a free arm movement similar to the way you sketch circles, or you can use ellipse templates to help you easily sketch ellipses. These templates are usually grouped according to the amount a circular shape would be rotated to form the ellipse. They provide a number of sizes of ellipses on each template but usually include only one or two typical rotations.

3.4 MAINTAINING PROPORTIONS

Sketches are not usually made to a specific scale, although it can be handy to do so at times. The size of the sketch depends on its complexity and the size of the paper available. The most important rule in freehand sketching is to keep the sketch in **proportion**, which means to accurately represent the size and position of each part in relation to the whole. No matter how brilliant the technique or how well drawn the details, if the proportions are off, the sketch will not look right.

To maintain proportions, first determine the relative proportions of height to width and lightly block them in. You can mark a unit on the edge of a strip of paper or use your pencil (as in Figure 3.28) to gauge how many units wide and high the object is. Grid paper can help you maintain proportions by providing a ready-made scale (by counting squares). As you block in the medium-size areas, and later as you add small details, compare each new distance with those already established.

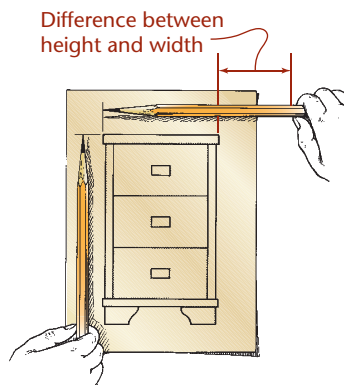


3.28 Estimating Dimensions

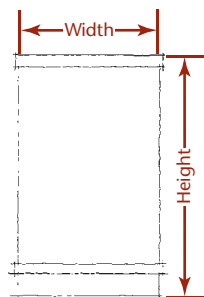
MAINTAINING PROPORTIONS IN A SKETCH

STEP by STEP

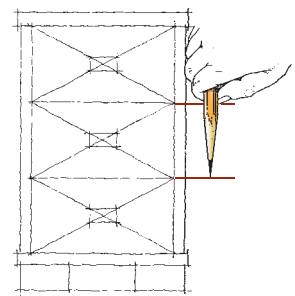
- 1 If you are working from a given picture, such as this utility cabinet, first establish the relative width compared to the height. One way is to use the pencil as a measuring stick. In this case, the height is about 1-3/4 times the width.



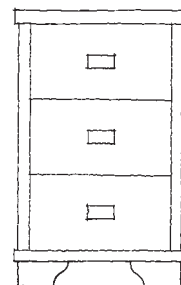
- 2 Sketch the enclosing rectangle in the correct proportion. This sketch is to be slightly larger than the given picture.



- 3 Divide the available drawer space into three parts with the pencil by trial. Hold your pencil about where you think one third will be and then try that measurement. If it is too short or long, adjust the measurement and try again. Sketch light diagonals to locate centers of the drawers and block in drawer handles. Sketch all remaining details.

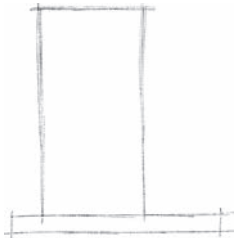


- 4 Darken all final lines, making them clean, thick, and black.

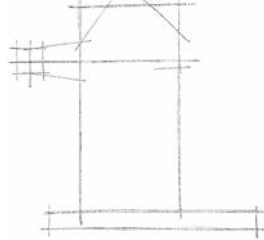


HOW TO BLOCK IN AN IRREGULAR OBJECT

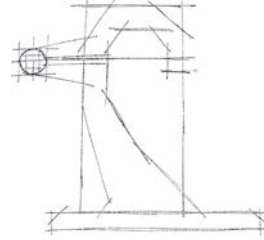
- 1 Capture the main proportions with simple lines.



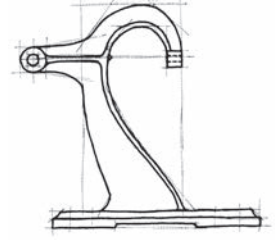
- 2 Block in the general sizes and direction of flow of curved shapes.



- 3 Lightly block in additional details.



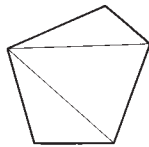
- 4 Darken the lines of the completed sketch.



GEOMETRIC METHODS FOR SKETCHING PLANE FIGURES

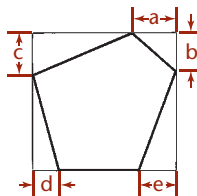
Sketching a Polygon by the Triangle Method

- 1 Divide the polygon into triangles as shown. Use the triangles as a visual aid to sketch the shape.



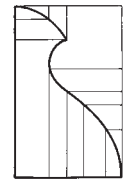
Sketching a Polygon by the Rectangle Method

- 1 Imagine a rectangle drawn around the polygon as shown.
- 2 Sketch the rectangle and then locate the vertices of the polygon (points a, b, c, and so on) along the sides of the rectangle.
- 3 Join the points to complete the shape.



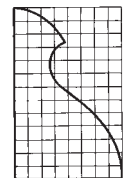
Creating Irregular Shapes by Offset Measurements

- 1 Enclose the shape in a rectangle.
- 2 Use the sides of the rectangle as a reference to make measurements that locate points along the curve.



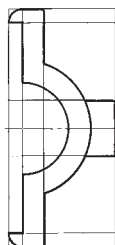
Enlarging Shapes Using a Grid of Squares

- 1 Complex curved shapes can be copied, enlarged, or reduced by hand, if necessary.
- 2 Draw or overlay a grid of squares on the original drawing.
- 3 To enlarge, draw the containing rectangle and grid of squares at the desired percentage and transfer the lines of the shape through the corresponding points in the new set of squares.



Visual Aids for Sketching Irregular Figures

- 1 Visualize shapes made up of rectangular and circular forms by enclosing those features in rectangles.
- 2 Determine where the centers of arcs and circles are located relative to the rectangles as shown.
- 3 Sketch the features inside the rectangular shapes you have lightly blocked in and darken the final lines.

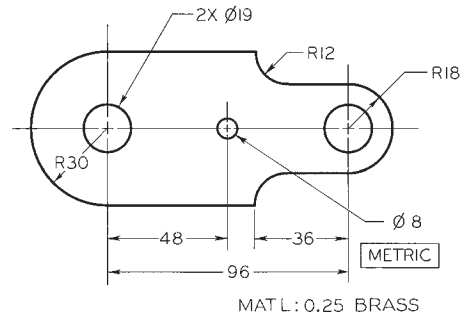


3.5 ONE-VIEW DRAWINGS

Frequently, a single view supplemented by notes and dimensions is enough information to describe the shape of a relatively simple object.

In Figure 3.29, one view of the shim plus a note indicating the thickness as 0.25 mm is sufficient.

Nearly all shafts, bolts, screws, and similar parts should be represented by single views in this way.

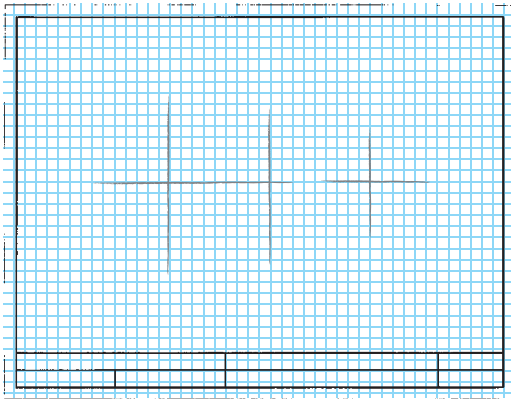


3.29 One-View Drawing of a Shim

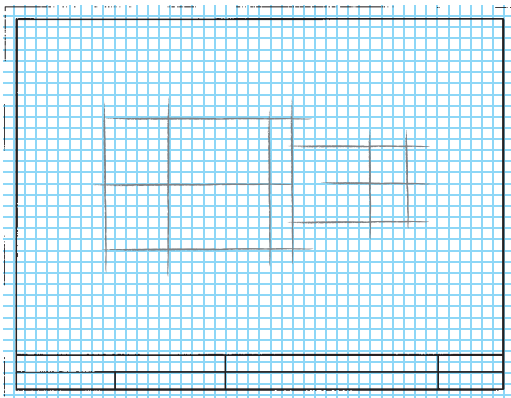
SKETCHING A SINGLE-VIEW DRAWING

Follow the steps to sketch the single-view drawing of the shim shown in Figure 3.29.

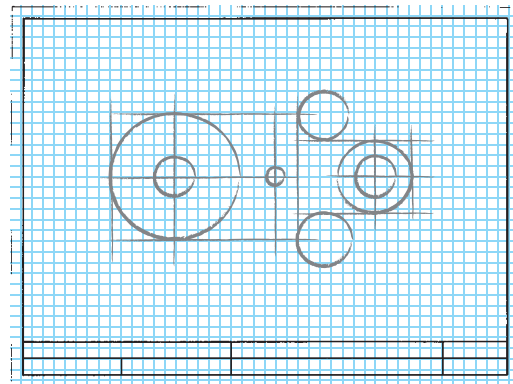
- 1 Lightly sketch the centerlines for the overall width and height of the part. Estimate overall proportions by eye, or if you know the dimensions, use a measuring scale to sketch accurately sized views. Space the enclosing rectangle equally from the margins of the sheet.



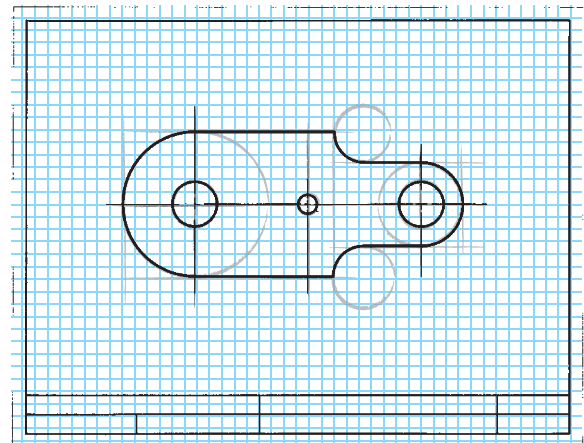
- 2 Block in all details lightly, keeping the drawing proportions in mind. Use techniques introduced in this chapter to help you.



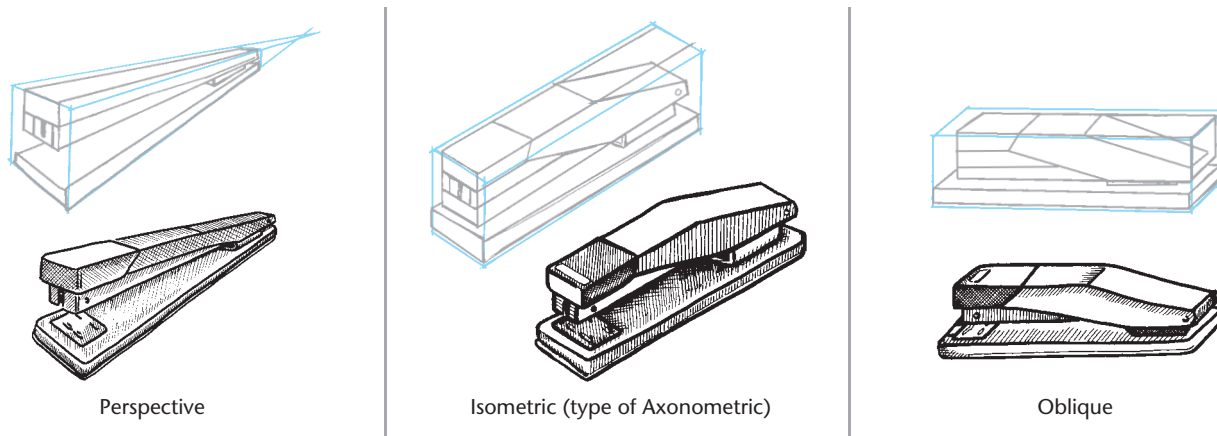
- 3 Locate the centers of circles and arcs. Block in where they will fit using rectangles. Then, sketch all arcs and circles lightly.



- 4 Darken your final lines.



- 5 Add annotations to the drawing using neat lettering. Fill in the title block or title strip. Note the scale for the sketch if applicable. If not, letter NONE in the Scale area of the title block.



3.30 Three Types of Pictorial Sketches

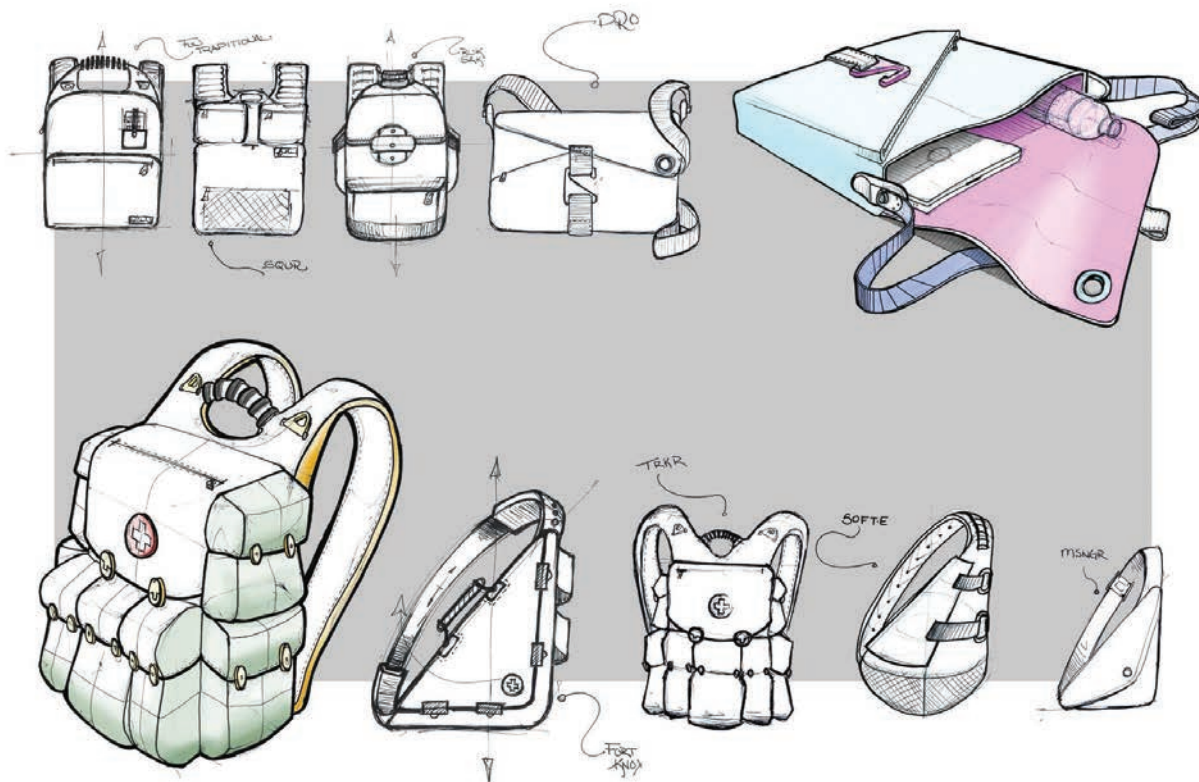
3.6 PICTORIAL SKETCHING

A *pictorial sketch* represents a 3D object on a 2D sheet of paper by orienting the object so you can see its width, height, and depth in a single view.

Pictorial sketches are used frequently during the ideation phase of engineering design to record ideas quickly and communicate them to others. Their similarity to how the object is viewed in the real world makes them useful for communicating engineering designs to nonengineers. Later in the design process, pictorial drawings are often used to show how parts

fit together in an assembly and, in part catalogs and manuals, to make it easy to identify the objects.

This chapter examines three common methods used to sketch pictorials: isometric sketching, which is a subtype of the general category of axonometric projection, oblique sketching, and perspective sketching. Figure 3.30 shows perspective, isometric, and oblique sketches of a stapler. Figure 3.31 shows pictorial sketches for backpack concepts.



3.31 Pictorial sketching is used frequently to convey preliminary design ideas, as in these backpack concept sketches. (Courtesy of André Côté.)

Each of the pictorial methods differs in the way points on the object are located on the 2D viewing plane (the piece of paper).

A perspective sketch presents the most realistic looking view. It shows the object much as it would appear in a photograph—portions of the object that are farther from the viewer appear smaller, and lines recede into the distance.

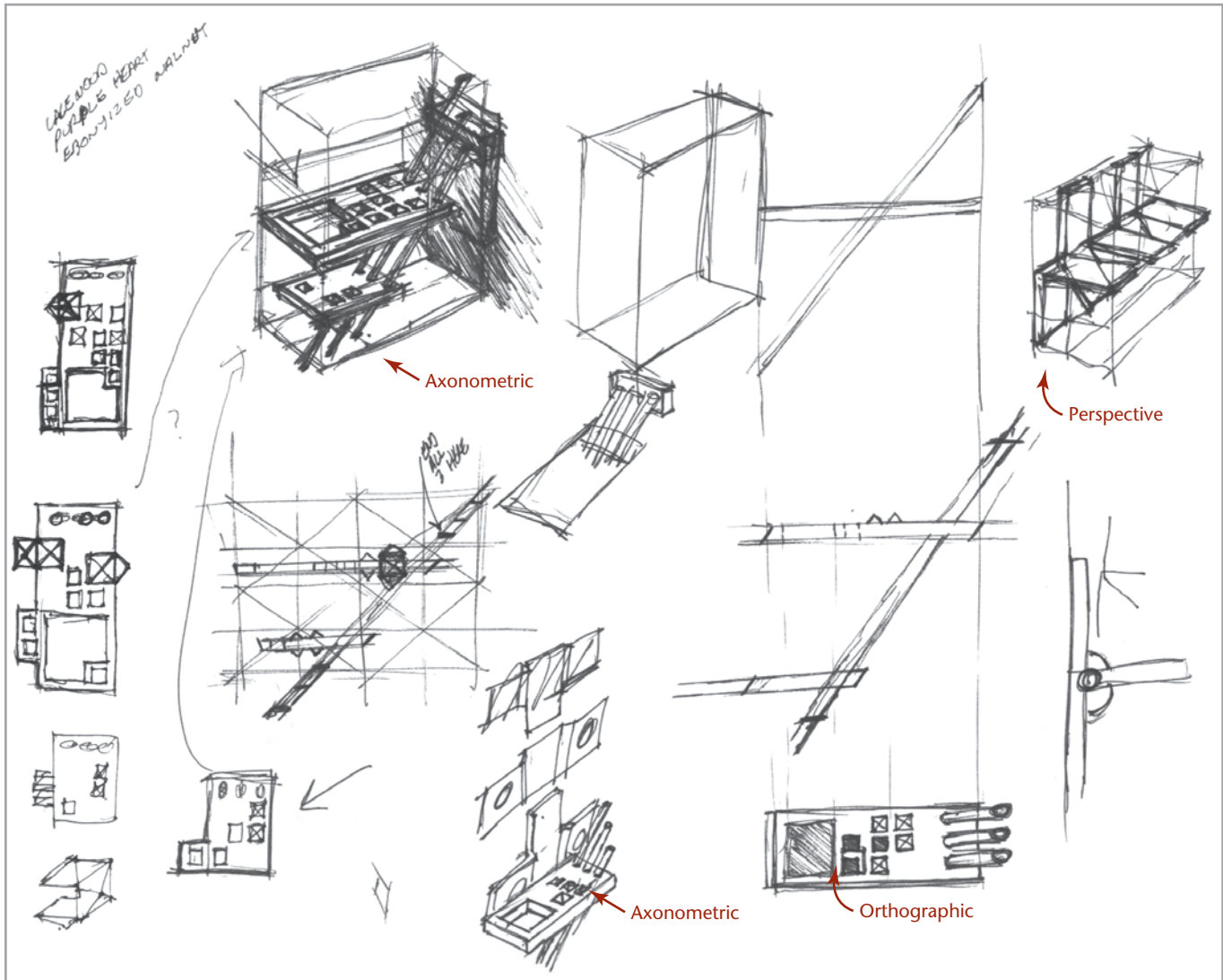
An axonometric sketch is drawn so that lines do not recede into the distance but remain parallel. This makes isometric views easy to sketch but takes away somewhat from the realistic appearance.

An oblique sketch shows the front surface of the object looking straight on and is easy to create, but it presents the least realistic representation because the depth of the object appears to be out of proportion.

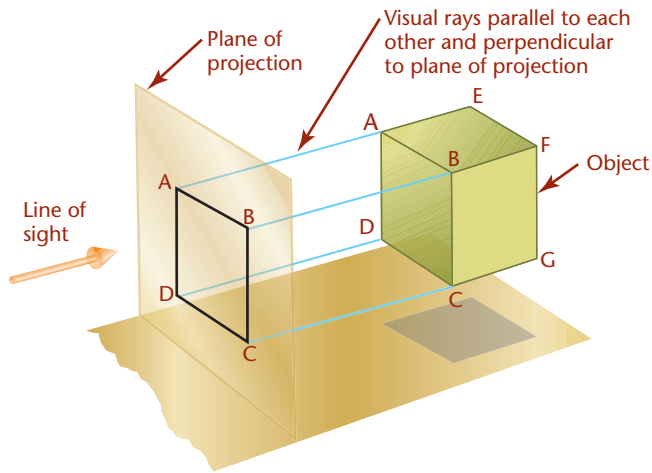
Various types of pictorial drawings are used extensively in catalogs, sales literature, and technical work. They are often used in patent drawings; in piping diagrams; in machine, structural, architectural design, and in furniture design; and for ide-

ation sketching. The sketches for a wooden shelf in Figure 3.32 are examples of axonometric, orthographic, and perspective sketches.

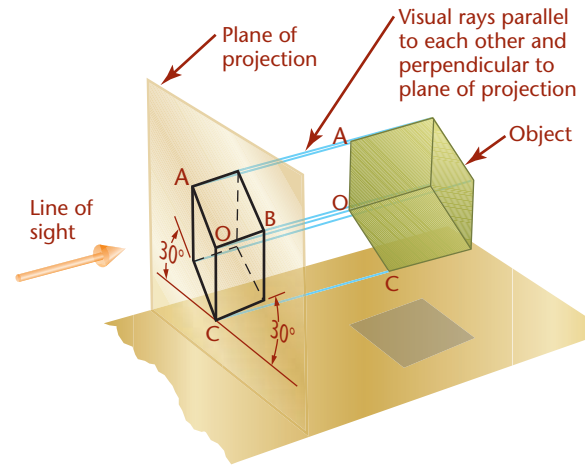
In axonometric and oblique drawings, distant features are not shown proportionately smaller, the way they appear in a photograph or to our vision. Edges that are parallel on the object always appear parallel in an axonometric view. The most common axonometric projection is isometric, which means “equal measure.” When a cube is drawn in isometric, the axes are equally spaced (120° apart). Though not as realistic as perspective drawings, isometric drawings are much easier to draw. CAD software often displays the results of 3D models on the screen as isometric projections. Some CAD software allows you to choose between isometric, axonometric views (see Figure 3.35 for examples), or perspective representation of your 3D models on the 2D computer screen. In sketching, dimetric and trimetric sometimes produce a better view than isometric but take longer to draw and are therefore used less frequently.



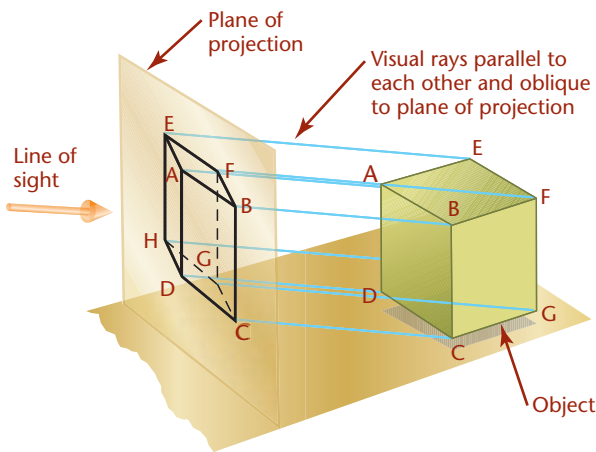
3.32 Sketches for a Wooden Shelf Using Axonometric, Orthographic, and Perspective Drawing Techniques. The axonometric projections in this sketch are drawn in isometric. (Courtesy of Douglas Wintin.)



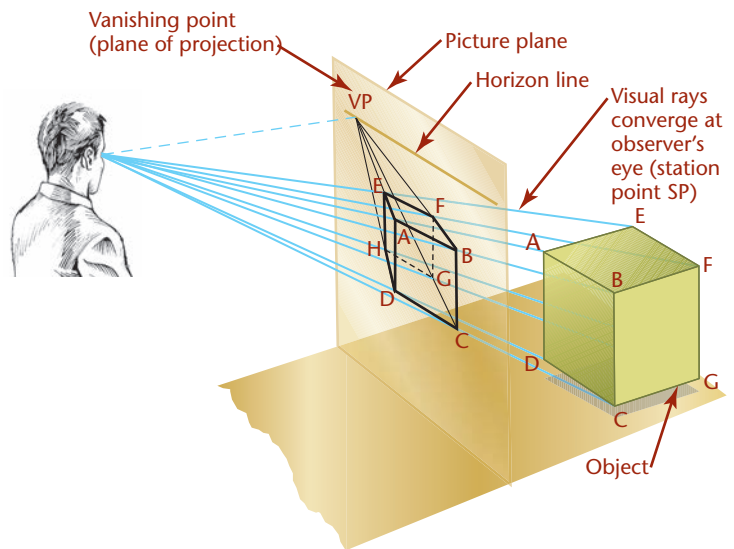
(a) Multiview projection



(b) Axonometric projection (isometric shown)



(c) Oblique projection



(d) Perspective

3.33 Four Types of Projection

3.7 PROJECTION METHODS

The four principal types of projection are shown in Figure 3.33. All except the regular multiview projection (Figure 3.33a) are pictorial types, because they show several sides of the object in a single view. In both **multiview projection** and **axonometric projection**, the visual rays are parallel to each other and perpendicular to the plane of projection. Both are types of **orthographic projection** (Figures 3.33a and 3.33b).

In **oblique projection** (Figure 3.33c), the visual rays are parallel to each other but at an angle other than 90° to the plane of projection.

In perspective (Figure 3.33d), the visual rays extend from the observer's eye, or station point (SP), to all points of the object to form a "cone of rays," so that the portions of the object that are farther away from the observer appear smaller than the closer portions of the object.

3.8 AXONOMETRIC PROJECTION

The feature that distinguishes axonometric projection from multiview projection is the inclined position of the object with respect to the planes of projection. When a surface or edge of the object is not parallel to the plane of projection, it appears foreshortened. When an angle is not parallel to the plane of projection, it appears either smaller or larger than the true angle.

To create an axonometric view, the object is tipped so that principal faces, such as the top, side, and front, show in a single view. This produces a pictorial drawing that is easy to visualize, but because the principal edges and surfaces of the object are inclined to the plane of projection, the lengths of the lines are foreshortened. The angles between surfaces and edges appear either larger or smaller than the true angle. There are an infinite variety of ways that the object may be oriented with respect to the plane of projection.

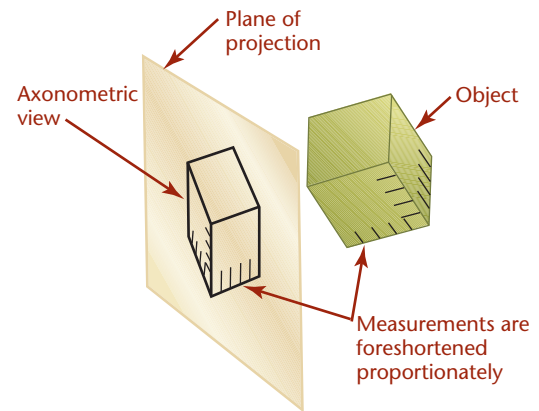
The degree of **foreshortening** of any line depends on its angle to the plane of projection. The greater the angle, the greater the foreshortening. Once the degree of foreshortening is determined for each of the three edges of the cube that meet at one corner, scales can be easily constructed for measuring along these edges or any other edges parallel to them (Figure 3.34).

Use the three edges of the cube that meet at the corner nearest your view as the axonometric axes. Figure 3.35 shows three axonometric projections.

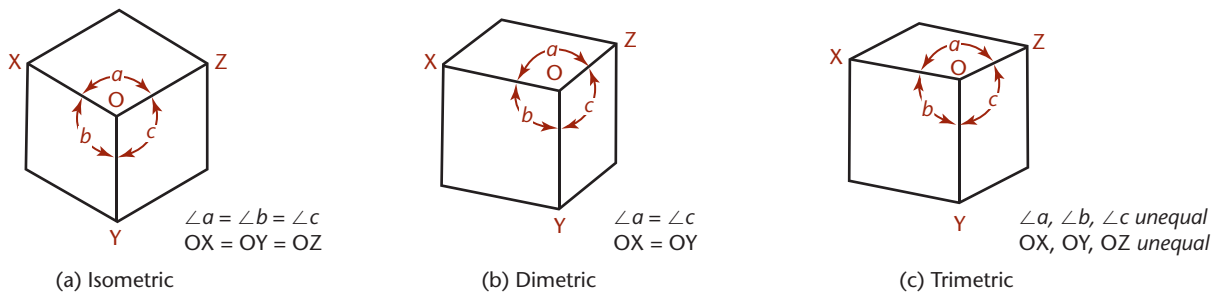
Isometric projection (Figure 3.35a) has equal foreshortening along each of the three axis directions.

Dimetric projection (Figure 3.35b) has equal foreshortening along two axis directions and a different amount of foreshortening along the third axis. This is because it is not tipped an equal amount to all the principal planes of projection.

Trimetric projection (Figure 3.35c) has different foreshortening along all three axis directions. This view is produced by an object that is unequally tipped to all the planes of projection.



3.34 Measurements are foreshortened proportionately based on the amount of incline.



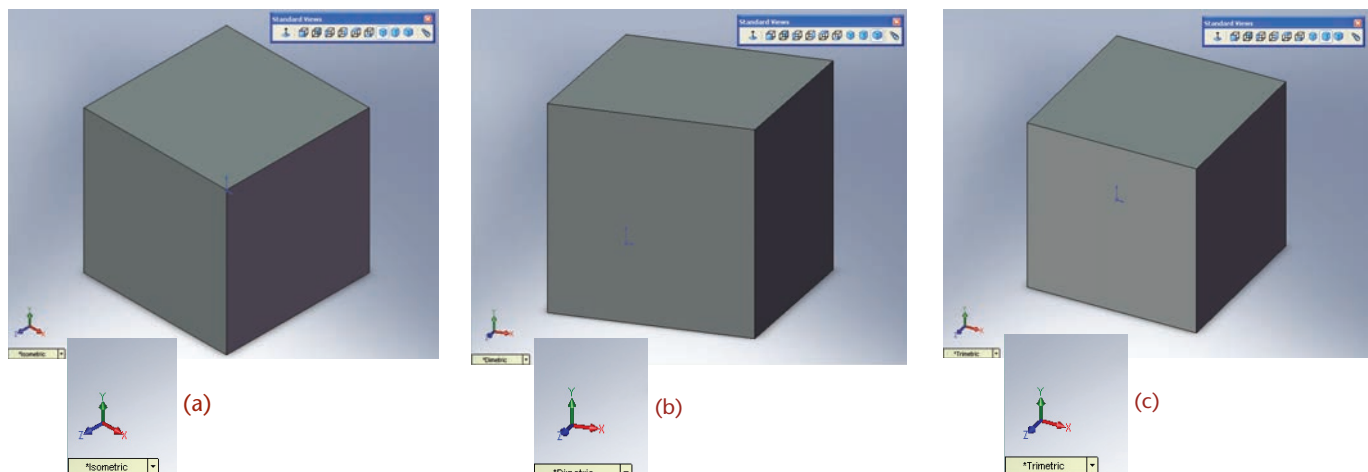
3.35 Axonometric Projections

Axonometric Projections and 3D Models

When you create a 3D CAD model, the object is stored so that vertices, surfaces, and solids are all defined relative to a 3D coordinate system. You can rotate your view of the object to produce a view from any direction. However, your computer screen is a flat surface, like a sheet of paper. The CAD software uses similar projection to produce the view transformations, creating the 2D view of the object on your computer screen.

Most 3D CAD software provides a variety of preset isometric viewing directions to make it easy for you to manipulate the view. Some CAD software also allows for easy perspective viewing on screen.

After rotating the object you may want to return to a preset typical axonometric view like one of the examples shown in Figure 3.36.

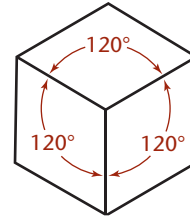


3.36 (a) Isometric View of a 1" Cube Shown in SolidWorks; (b) Dimetric View; (c) Trimetric View

(Images courtesy of ©2016 Dassault Systèmes SolidWorks Corporation.)

3.9 ISOMETRIC PROJECTION

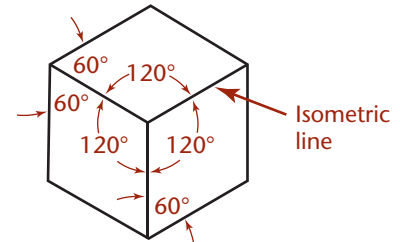
In an isometric projection, all angles between the axonometric axes are equal. To produce an isometric projection, you orient the object so that its principal edges (or axes) make equal angles with the plane of projection and are therefore foreshortened equally. Oriented this way, the edges of a cube are projected so that they all measure the same and make equal angles (of 120°) with each other, as shown in Figure 3.37.



3.37 Isometric Projection

Isometric Axes

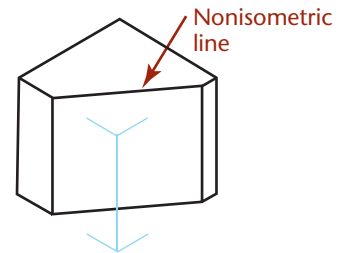
The projections of the edges of a cube make angles of 120° with each other. You can use these as the isometric axes from which to make measurements. Any line parallel to one of these is called an isometric line. The angles in the isometric projection of the cube are either 120° or 60° , and all are projections of 90° angles. In an isometric projection of a cube, the faces of the cube, and any planes parallel to them, are called isometric planes. See Figure 3.38.



3.38 Isometric Axes

Nonisometric Lines

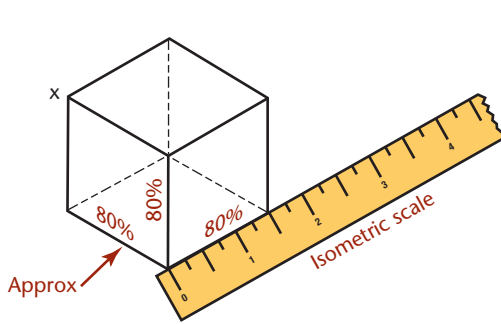
Lines of an isometric drawing that are not parallel to the isometric axes are called nonisometric lines (Figure 3.39). Only lines of an object that are drawn parallel to the isometric axes are equally foreshortened. Nonisometric lines are drawn at other angles and are not equally foreshortened. Therefore the lengths of features along nonisometric lines cannot be measured directly with a scale.



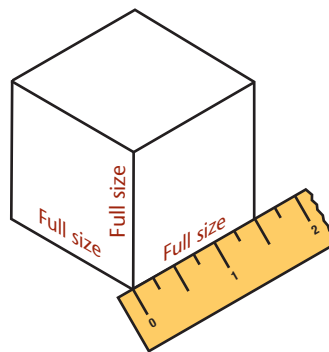
3.39 Nonisometric Edges

Isometric Scales

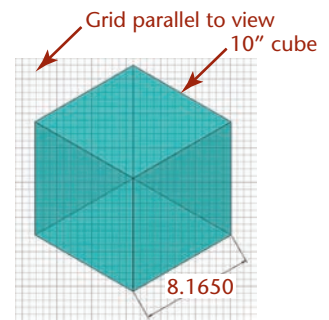
An isometric scale can be used to draw correct isometric projections. All distances in this scale are $\frac{2}{3} \times$ true size, or approximately 80% of true size. Figure 3.40a shows an isometric scale. More commonly, an isometric sketch or drawing is created using a standard scale, as in Figure 3.40b, disregarding the foreshortening that the tipped surfaces would produce in a true projection. Figure 3.40c shows an isometric view of a 10" cube modeled in 3D. When measured parallel to the view, the length appears to be 8.1650 due to the foreshortening (near 80% of the actual size).



(a) Isometric projection



(b) Isometric drawing



(c) CAD model in isometric view

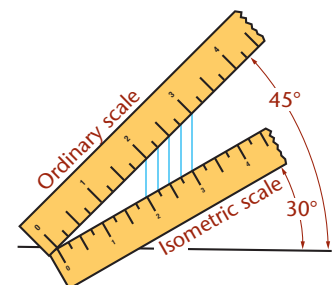
3.40 Isometric and Ordinary Scales

TIP

Making an Isometric Scale

You can make an isometric scale from a strip of paper or cardboard by placing an ordinary scale at 45° to a horizontal line and the paper (isometric) scale at 30° to the horizontal line. To mark the increments on the isometric scale, draw straight lines (perpendicular to the horizontal line) from the division lines on the ordinary scale.

Alternatively, you can approximate an isometric scale. Scaled measurements of $9'' = 1'-0''$, or three-quarter-size scale (or metric equivalent) can be used as an approximation.



3.10 ISOMETRIC DRAWINGS

When you make a drawing using foreshortened measurements, or when the object is actually projected on a plane of projection, it is called an isometric projection (Figure 3.40a). When you make a drawing using the full-length measurements of the actual object, it is an isometric sketch or *isometric drawing* (Figure 3.40b) to indicate that it lacks foreshortening.

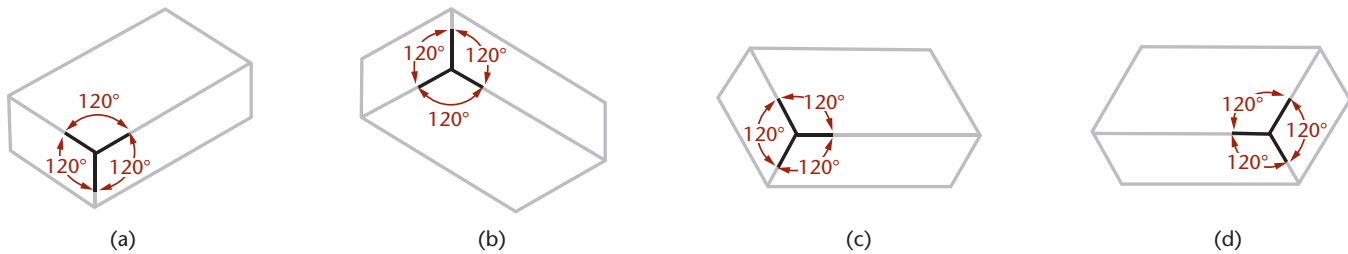
The isometric drawing is about 25% larger than the isometric projection, but the pictorial value is obviously the same in both. Because isometric sketches are quicker (because you can use the actual measurements), they are much more commonly drawn.

Positions of the Isometric Axes

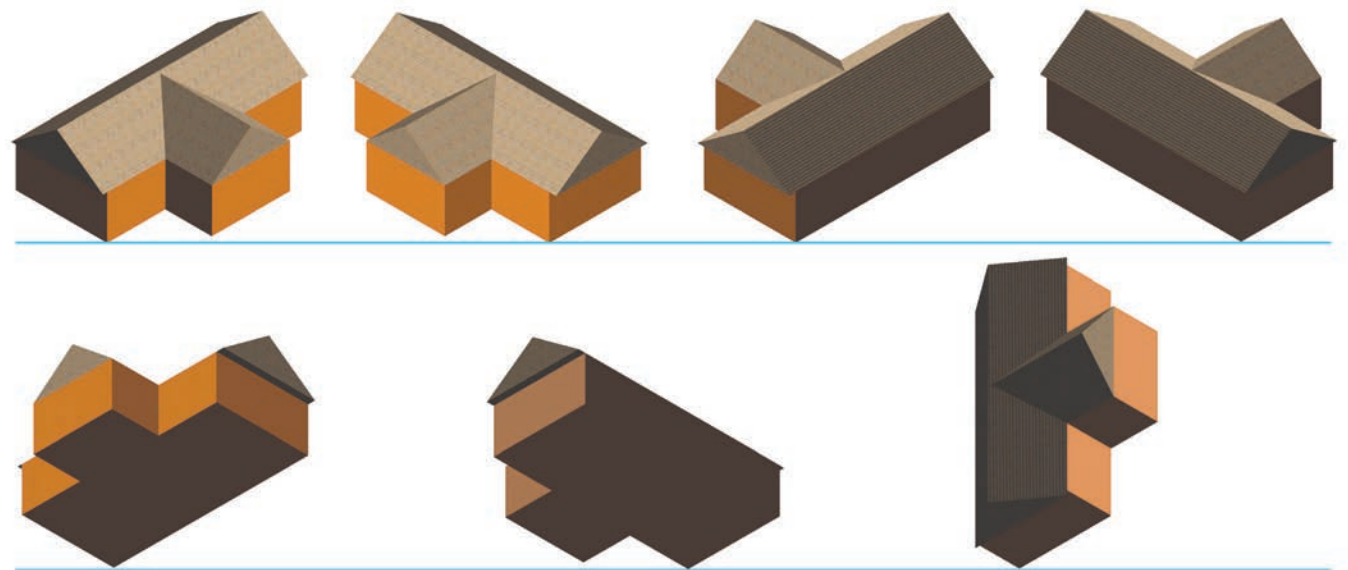
The first step in making an isometric drawing is to decide along which axis direction to show the height, width, and depth, respectively. Figure 3.41 shows four different orientations that you might start with to create an isometric drawing of the block shown. Each is an isometric drawing of the same block, but with a different corner facing your view. These are only a few of many possible orientations.

You may orient the axes in any desired position, but the angle between them must remain 120° . In selecting an orientation for the axes, choose the position from which the object is usually viewed, or determine the position that best describes the shape of the object, or better yet, both.

If the object is a long part, it will look best with the long axis oriented horizontally.



3.41 Positions of Isometric Axes



3.42 Isometric Options. These views of a simple model of a house are all isometric, but some show the features better than others.

TIP

Some CAD software will notify you about the lack of foreshortening in isometric drawings (or allow you to select for it) when you print or save them.